



FPT UNIVERSITY

Capstone Project Document

Online Learning Platform - SkillUp [OLP-SkillUp]

<SEP490_G44>	
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Over the past four months, we have received tremendous help from many people, and we remain deeply thankful for every contribution that has made this project possible.

Definition and Acronyms

Acronym	Definition
SRS	Software requirement specification
SDS	Software design specification
API	Application Programming Interface
AI	Artificial Intelligence
ERD	Entity Relationship Diagram
WBS	Work Breakdown Structure
PM	Project Manager
SDD	Software Design Description
UT	Unit Test
SRS	Software Requirement Specification

I. Project Introduction

1. Overview

1.1 Project Information

- Project name: Online Learning Platform – SkillUp
- Project code: OLP-SkillUp
- Group name: SEP490_G44
- Software type: Website

1.2 Project Team

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2. Product Background

In recent years, the growing demand for flexible and accessible learning solutions has become increasingly evident, especially among students and working professionals in Vietnam. Despite the rise of global e-learning platforms, many remain difficult to access due to high subscription fees, English-based interfaces, and limited interactive support. These limitations have created a gap in the market for a more localised and user-centric learning experience.

The idea for this project originated from feedback gathered through surveys and discussions with potential users—primarily university students, recent graduates, and young professionals seeking skill improvement. They expressed a strong need for an affordable, easy-to-use platform tailored to local preferences and learning habits. Instructors also shared concerns regarding the lack of effective tools for delivering engaging online courses.

In response to these needs, our team recognised the opportunity to develop a platform that better serves local learners and educators. This understanding forms the foundation for proposing the SkillUp online learning platform.

3. Existing Solutions

3.1 Coursera

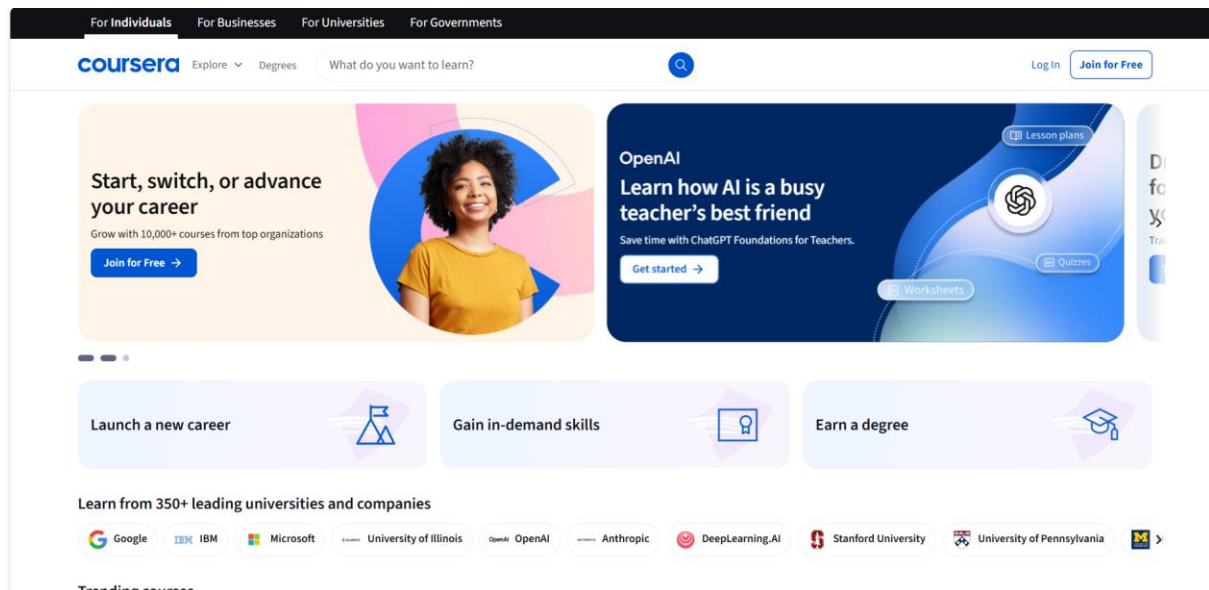


Figure 3.1: Coursera

3.1.1 Description

Coursera, inaugurated in 2012 by Stanford University professors Andrew Ng and Daphne Koller, has established itself as a pioneering force within the massive open online course (MOOC) ecosystem and digital education landscape. The platform operates through strategic collaborative partnerships with over 300 prestigious universities and multinational corporations worldwide, including Stanford University, Yale University, Google, IBM, and Meta. It provides a comprehensively structured hierarchy of educational offerings, encompassing individual skill-development courses, industry-recognized professional certificates, specialized career-focused programs, and fully accredited online degree programs spanning diverse academic disciplines including computer science, business administration, data science, public health, and liberal arts.

Link : <https://www.coursera.org/>

3.1.2 System Actors

The Coursera ecosystem comprises multiple interconnected stakeholders, each fulfilling distinct yet complementary roles within the platform's operational framework:

- **Students/Learners:** End-users who engage with content, complete assessments, and earn verifiable credentials.
- **Instructors/Creators:** Subject matter experts responsible for curriculum design, content creation, and assessment formulation.
- **Partner Organizations:** Universities and corporations that provide high-quality materials and academic credibility.
- **Platform Administrators:** Technical teams managing operations, security, support, and quality control.

3.1.3

Features

Coursera's feature architecture demonstrates sophisticated integration of pedagogical best practices with technological innovation:

- **Multimedia Content:** High-definition videos, readings, and interactive transcripts.

- **Assessments:** Auto-graded quizzes, programming assignments, and peer-reviewed projects.
- **Credentialing:** Blockchain-backed certificates, Specializations, and accredited degrees.
- **Collaboration:** Discussion forums and global learner communities.
- **Accessibility:** Cross-platform support (Web, iOS, Android) with offline capabilities.

3.1.4 Pros

Coursera's competitive advantages manifest across multiple strategic dimensions:

- **Institutional Credibility:** High legitimacy due to backing from top-tier universities and tech giants.
- **Quality Assurance:** Rigorous vetting ensures high-caliber educational content.
- **Broad Scope:** Covers diverse disciplines from Computer Science to Humanities.
- **Career Value:** Credentials are widely recognized by employers.
- **Structured Learning:** Clear roadmaps and specialization tracks reduce decision paralysis.

3.1.5 Cons

Despite significant strengths, Coursera exhibits strategic vulnerabilities that constrain its competitive positioning:

- **Premium Pricing:** High costs for degrees and specialized certifications create barriers for some users.
- **Limited Interaction:** Minimal personalized access to instructors due to the massive scale.
- **Low Completion Rates:** Industry averages of 5-15% suggest challenges in sustaining learner motivation.
- **Western Bias:** Content predominantly reflects Western academic perspectives and contexts.

3.1.6 AI Implementation:

Coursera has strategically integrated artificial intelligence technologies across multiple platform dimensions to enhance scalability, personalization, and pedagogical effectiveness:

- **Automated Grading:** Uses NLP to evaluate text-based assignments and essays.
- **Recommendation Engine:** Suggests courses based on learner behavior and career goals.
- **Adaptive Learning:** Dynamically adjusts content sequencing and difficulty based on performance.
- **Predictive Analytics:** Identifies learners at risk of dropping out to trigger interventions.

3.2 Udemy

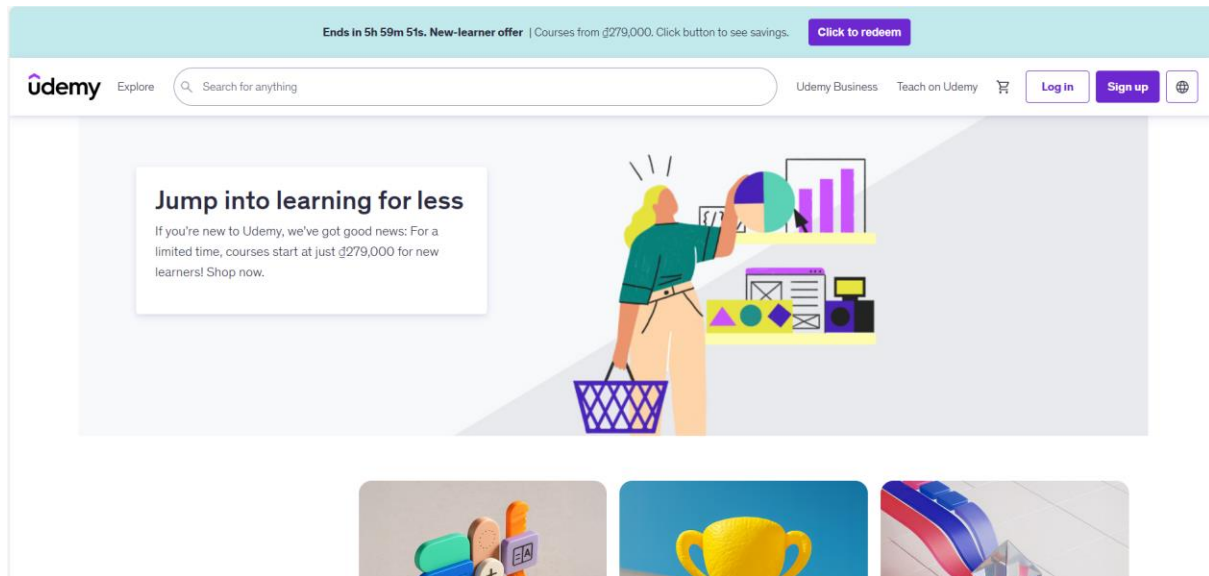


Figure 3.2: Udemy

3.2.1 Description

Udemy, founded in 2010, is a democratized global online education marketplace that allows virtually anyone (instructors) to design, produce, and monetize courses across unlimited subject domains, fundamentally disrupting traditional educational gatekeeping.

Link: <https://www.udemy.com>

3.2.2 System Actors

The Udemy ecosystem comprises diverse stakeholder groups whose interactions constitute the platform's operational dynamics:

- **Learners/Students:** Browse, enroll in self-paced courses, engage with multimedia content, complete assessments, participate in forums, and provide ratings/reviews.
- **Instructors/Teachers/Content Creators:** Conceptualize and produce course content (predominantly video), design supplementary materials, respond to inquiries, monitor analytics, and market their courses.
- **Platform Administrators:** Maintain technical infrastructure, enforce policies, moderate disputes, process transactions, and optimize platform functionality.
- **Corporate Clients/Enterprise Users:** Utilize the Udemy Business offering for employee upskilling and corporate training via volume licensing.

3.2.3 Features

Udemy's feature ecosystem reflects its marketplace philosophy prioritizing diversity, accessibility, and instructor autonomy:

- **Expansive Course Marketplace:** Over 220,000 courses across categories like development, business, design, and niche specializations.

- **Multimedia Instructional Content:** Primarily video lectures enhanced by quizzes, coding exercises, and downloadable resources.
- **Perpetual Access Model:** Learners receive lifetime access upon purchase, including offline viewing capabilities..
- **Cross-Platform Accessibility:** Seamless experience across web and native mobile apps with progress synchronization.
- **Digital Credential System:** Automated certificates of completion (lacking formal accreditation).
- **Instructor Toolkit Suite:** Comprehensive tools for course creation, analytics, and promotional coupon generation.
- **Corporate Learning Solution (Udemy Business):** Curated course collections and administrative tracking for enterprises.
- **Dynamic Pricing and Promotional Ecosystem:** Flexible pricing strategies and frequent, substantial platform-wide sales.

3.2.4 Pros

Udemy's marketplace model generates distinctive competitive advantages across accessibility, diversity, and practical orientation dimensions:

- **Unprecedented Course Diversity and Niche Specialization:** Open model leads to a comprehensive catalog covering specialized topics.
- **Exceptional Cost Accessibility:** Frequent, deep promotional pricing makes education affordable globally.
- **Practical, Application-Oriented Pedagogy:** Content is often created by industry practitioners, emphasizing immediately implementable skills.
- **Flexible Self-Directed Learning:** Self-paced nature accommodates learners with demanding schedules.
- **Rapid Content Currency:** Instructors can quickly create courses responding to emerging trends and technologies.

3.2.5 Cons

Despite marketplace strengths, Udemy's openness generates significant quality assurance, credentialing, and engagement challenges:

- **Substantial Quality Inconsistency:** Minimal vetting leads to significant variance in content accuracy, production quality, and pedagogical effectiveness.
- **Limited Credential Market Recognition:** Certificates lack formal accreditation, limiting value for career advancement or academic credit transfer.
- **Minimal Instructor-Learner Interaction:** synchronous nature limits opportunities for personalized feedback or mentorship.
- **Course Discovery and Selection Challenges:** Overwhelming volume and quality variance make selection difficult.
- **Content Currency Maintenance Issues:** Many courses become outdated as instructors lack systematic obligations to update them.

3.2.6 AI Implementation:

Udemy has demonstrated notably limited integration of advanced artificial intelligence technologies compared to competitors, with AI implementation remaining rudimentary relative to platform scale:

- **Basic Recommendation Systems:** Machine learning generates personalized course suggestions based on basic patterns and popularity.
- **Search Relevance Optimization:** Natural language processing enhances search to match user queries with relevant course content semantically.

Notable Absence of Advanced AI Features: Lacks sophisticated AI capabilities seen in competitors, such as adaptive learning paths, predictive analytics, or intelligent tutoring systems.

4. Business Opportunity

Vietnam's EdTech market is projected to reach significant valuation by 2029, fueled by a digital economy demanding continuous upskilling. The shift from 'degree-based' to 'skill-based' hiring creates an urgent need for short-term, high-impact courses.

SkillUp captures this opportunity through a hybrid value proposition:

- For Learners: A frictionless path from discovery to mastery with localized support and payments.
- For Instructors: A 'Business-in-a-Box' solution. Instead of building their own websites, instructors use SkillUp's infrastructure (hosting, payment, LMS) to monetize their knowledge instantly.
- Scalability: The platform model allows for exponential content growth without proportional increases in operational costs, ensuring long-term sustainability.

The online learning market in Vietnam is rapidly growing, fueled by widespread Internet adoption, the demand for flexible learning, and a strong cultural emphasis on continuous skill development. According to the [Vietnam Edtech Market Summary, Competitive Analysis and Forecast to 2029](#), the sector is poised for significant expansion, driven by government initiatives to digitize education and a rising middle class willing to invest in quality learning resources. Projections indicate that the number of users in the online learning platform market alone could reach approximately 11.8 million by 2029, highlighting a massive addressable market.

Despite this growth, a significant gap remains. Both students and working professionals are actively seeking a user-friendly platform that is not only affordable but also provides practical, applicable knowledge tailored to the local context. SkillUp is uniquely positioned to bridge this gap. By offering intuitive tools for instructors to easily create and monetize courses, and a localized interface for learners, SkillUp fosters a vibrant, interactive learning community. Beyond serving individual learners, the platform has the potential to capture the B2B segment by expanding into corporate training, helping organizations improve workforce quality at optimized costs in this booming digital economy.

5. Software Product Vision

SkillUp is an online learning platform designed to help Vietnamese learners easily access high-quality knowledge, develop practical skills, and expand career opportunities. The product delivers a personalized, flexible, and effective learning experience for individuals, while supporting instructors

and organizations in managing and measuring training programs. The vision of SkillUp is to become the leading digital learning ecosystem in Vietnam, where advanced technology goes hand in hand with human development, contributing to a dynamic and sustainable learning community.

6. Project Scope & Limitations

No.	ID	Feature	Description
1	FE-01	Common	Login: Users can log in using email/password or Google OAuth. The system validates credentials and generates JWT tokens for authenticated sessions..
			Register: New users can register by providing email, password, and personal information. Email verification via OTP is required to activate the account.
			Change Password: Authenticated users can change their password by providing their current password and a new password.
			Forgot Password: Users who forget their password can request a reset link via email. The system sends a secure token to reset the password.
			Notification: Users receive real-time notifications about course updates, payment confirmations, and system announcements via SignalR hub.
			Profile Management: Users view their personal information, can edit, update information or change passwords
2	FE-02	Student's Features	View Course: view course details including curriculum, lecturer info, ratings, and preview lessons before enrollment.
			Learning Course: enrolled courses with structured sections and lessons. Video playback with progress tracking , and lesson completion status. The system provides instant grading with multiple-choice and multiple-answer questions.Results show correct/incorrect answers and score percentage.
			Learning progress tracking: system automatically tracks student progress across sections , lessons , and quizzes.
			Forum & Community: Students can create forum posts, comment on posts, like/unlike comments, and engage in course-related discussions.
			AI chat in lesson : Students can chat with an AI assistant while watching lessons to ask questions, clarify doubts, and get instant responses related to course content.
			Shopping cart & payment: add multiple courses to cart, apply vouchers for discounts, and complete purchases via PayOS payment gateway.
3	FE-03	Lecturer's Feature	Apply as Lecturer : apply to become a lecturer by submitting CV (PDF) and degree certificates (images). Applications are reviewed by System Moderators.
			Course Management : Lecturers can create new courses with details, edit course information, and manage course status.
			Quiz Management : Lecturers create quizzes within sections, assign questions from question bank.
			Student Analytics : Lecturers view enrolled students, track their progress, quiz scores, completion rates, and student engagement metrics per course.

			Question Management : Lecturers create question pools with multiple-choice or multiple-answer questions, add answer options, mark correct answers, and attach images if needed. Questions are reusable across quizzes. Revenue and income report: Lecturers access revenue reports showing total earnings, course sales breakdown, and financial analytics with charts.
4	FE-04	System Moderator's Features	Lecturer Application Review: System Moderators review lecturer applications including CV and degree certificates. They can accept applications (converting Student to Lecturer role and setting account status to Active) or reject with reasons via email notification. User Account Management: Moderators view all user accounts across roles, search/filter users, view user details, manage account status (Active/Pending/Banned), and monitor user activities for security purposes. Ticket Management: System Moderators handle support tickets from lecturers and students. They can view ticket details, provide solutions, escalate issues, mark tickets as resolved, and track resolution time. AI suggests ticket titles for faster categorization. System Moderator Dashboard: Central dashboard displaying platform statistics: total users by role, pending lecturer applications, unresolved tickets, recent user activities, and system health metrics.
5	FE-05	Content Moderator's Features	Course Approval: Content Moderators review course submissions from lecturers. They can approve courses (making them public), reject with reasons, or request modifications. Quality checks include content accuracy and appropriateness. News Management: Moderators create news articles using TinyMCE editor with rich formatting, upload images, publish/unpublish news, edit existing articles, and manage news visibility on platform. Forum Post Moderation: Moderators monitor forum posts and comments for inappropriate content, spam, or violations. They can delete posts, ban users temporarily, and enforce community guidelines. Category Management: Moderators create, edit, or delete course categories and subcategories. Categories organize courses for easier browsing and help students find relevant content. Report Comment Management: Moderators review reported comments from students/lecturers. They investigate reports, take action (delete comment, warn user), and resolve report status. Content Moderator Dashboard: Central dashboard providing an overview of platform content and moderation status. It displays key content metrics, tracks course approval status (Public, Pending Approval, Hidden, Reported), and highlights pending tasks, such as unprocessed reports. It also shows recent publishing activity. Banner Management: Moderators upload promotional banners for homepage, set banner priority/order, configure display duration, and manage banner visibility for marketing campaigns.
6	FE-06	Admin's Features	Moderator Management: Admins create Content Moderator and System Moderator accounts, assign permissions, manage moderator roles, deactivate accounts, and send account credentials via email. Financial Reports: Admins generate lecturer salary reports based on course sales, calculate platform commission, export financial data, track payment transactions, and manage revenue distribution

Platform Notifications: Admins broadcast system-wide notifications to all users or specific roles (e.g., maintenance announcements, policy updates, promotional messages). Notifications support rich text formatting.

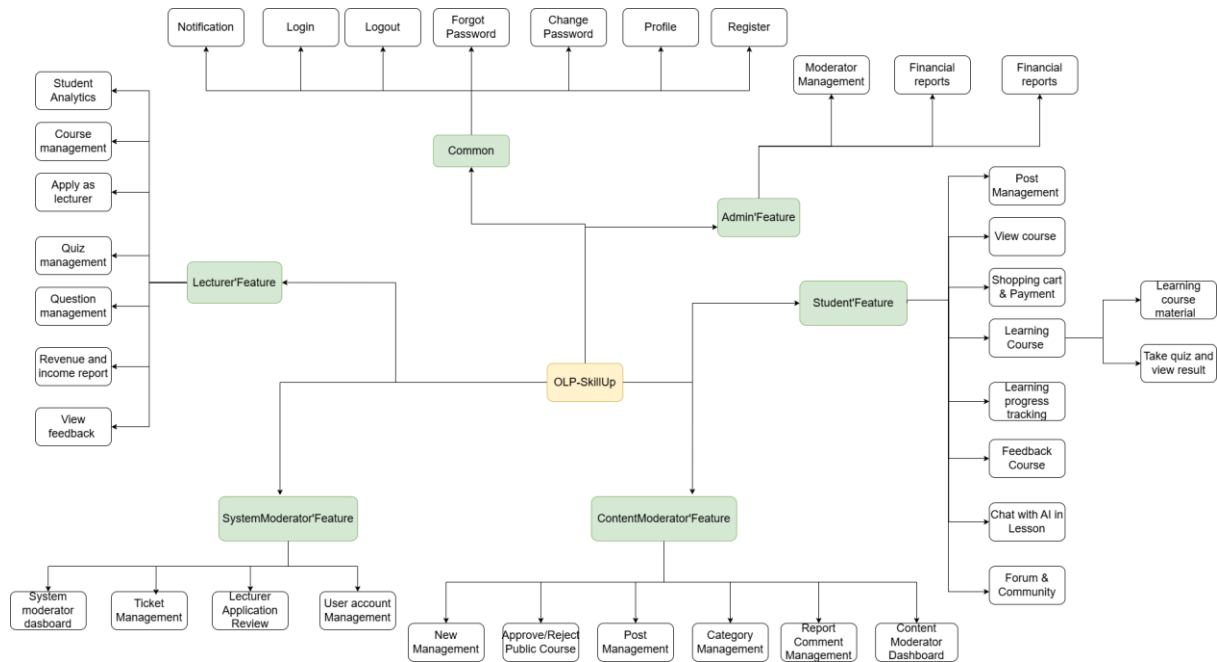


Figure 5.1: [OLP-SkillUp Feature](#)

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

#	Work Package	Est. Effort (pds)	Deadline
1	Feature 1: Common Feature	13	
1.1	Login	2	Week 1
1.2	Logout	1	Week 1
1.3	Change Password	2	Week 1
1.4	Forgot Password	2	Week 1
1.5	Register	2	Week 1
1.6	Homepage	2	Week 1
1.7	Search engine course at homepage	2	Week 1
2.	Feature 2: Profile Feature	4	
2.1	View profile	2	Week 1

2.2	Update profile	2	Week 2
3	Feature 3: New Feature	4	
3.1	Create news	1	Week 2
3.2	View news	1	Week 2
3.3	Update news	1	Week 2
3.4	Delete news	1	Week 2
4	Feature 4: Ticket Feature	6	
4.1	Create ticket	1	Week 2
4.2	View detail ticket	1	Week 2
4.3	Update ticket	1	Week 2
4.4	View list ticket	1	Week 2
4.5	Resolve ticket	2	Week 2
5	Feature 5: Forum Feature	15	
5.1	View my post list	2	Week 3
5.2	View post list user	2	Week 3
5.3	Create post	1	Week 3
5.4	Update post	2	Week 3
5.5	Delete post	1	Week 3
5.6	Comment post	1	Week 3
5.7	Resolve comment	1	Week 3
5.8	Like comment	1	Week 3
5.9	Report comment	1	Week 3
5.10	View post list all	1	Week 3
5.11	Manage Forum	2	Week 3
6	Feature 6: Category Feature	7	
6.1	Create category	2	Week 4
6.2	Update category	2	Week 4
6.3	Delete category	1	Week 4
6.4	View category	2	Week 4
7	Feature 7: Forum Category Feature	4	

7.1	Create forum category	1	Week 4
7.2	Update forum category	1	Week 4
7.3	Delete forum category	1	Week 4
7.4	View forum category	1	Week 4
8	Feature 8: SubCategory Feature	4	
8.1	Create subcategory	1	Week 4
8.2	Update subcategory	1	Week 4
8.3	Delete subcategory	1	Week 4
8.4	View subcategory	1	Week 4
9	Feature 9: Notify Feature	5	
9.1	View notify userId	2	Week 5
9.2	View notify detail	1	Week 5
9.3	Send notify system	2	Week 5
10	Feature 10: Course Feature	25	
10.1	View the list of all courses by subcategory	2	Week 5
10.2	View detail course	4	Week 5
10.3	View the list of courses enrolled in by students	3	Week 5
10.4	View the list of all courses for the Moderator	1	Week 5
10.5	View my course list for the Lecturer	1	Week 5
10.6	Create course	3	Week 5
10.7	Update course	3	Week 5
10.8	Delete course	2	Week 5
10.9	Ban/unban course	2	Week 6
10.10	Report course	1	Week 6
10.11	Approve/Reject the course	2	Week 12
10.12	Publish course	1	Week 12
11	Feature 11: Voucher Feature	7	
11.1	View voucher	2	Week 6
11.2	Create voucher	2	Week 6
11.3	Update voucher	2	Week 6

11.4	Disable voucher	1	Week 6
12	Feature 12: Lesson Feature	35	
12.1	View Lesson	3	Week 6
12.2	Create Lesson	4	Week 7
12.3	Update Lesson	4	Week 7
12.4	Delete Lesson	2	Week 7
12.5	Create comment lesson	2	Week 10
12.6	Update comment lesson	2	Week 10
12.7	Delete comment lesson	2	Week 10
12.8	View comment lesson	2	Week 11
12.9	Like comment lesson	2	Week 11
12.10	Resolve comment lesson	2	Week 11
12.11	Lesson-based AI Chat	10	Week 7
13	Feature 12: Cart Feature	6	
13.1	View Cart	1	Week 7
13.2	Add to cart	2	Week 7
13.3	Remove item from cart	1	Week 7
13.4	Enter and apply voucher	2	Week 8
14	Feature 14: Question Bank Feature	15	
14.1	Create a question in the question bank	3	Week 8
14.2	View question bank	1	Week 8
14.3	Update question bank	4	Week 8
14.4	Delete the question from the question bank	2	Week 8
14.5	Add multiple questions from Excel to the question bank	5	Week 8
15	Feature 15: Quiz Feature	27	
15.1	Create quiz	3	Week 8
15.2	View quiz	2	Week 8
15.3	Update quiz	4	Week 8
15.4	Delete quiz	2	Week 8

15.5	Student do quiz	6	Week 12
15.6	View result detail quiz	2	Week 12
15.7	Add question to quiz	4	Week 9
15.8	Update question in quiz	4	Week 9
16	Feature 16: Section Feature	4	
16.1	Create section	1	Week 9
16.2	View section	1	Week 9
16.3	Update section	1	Week 9
16.4	Delete section	1	Week 9
17	Feature 16: Lecturer Application Feature	7	
17.1	Apply lecturer Application	3	Week1
17.2	Manage lecturer application	2	Week1
17.3	View lecturer application	2	Week1
18	Feature 18: Question Feature	20	
18.1	Create question	6	Week1
18.2	View question	4	Week1
18.3	Update question	6	Week1
18.4	Delete question	4	Week1
19	Feature 18: Answer Feature	7	
19.1	Create answer	2	Week1
19.2	View answer	1	Week1
19.3	Update answer	2	Week1
19.4	Delete answer	2	Week1
19	Feature 19: Rating Feature	2	
19.1	Create rating	1	Week1
19.2	View rating	1	Week1
20	Feature 20: Student Progress Feature	3	
20.1	View learning progress	2	Week 10
20.2	Resume Learning	1	Week 10
21	Feature 21: Payment Feature	4	

21.1	Buy course now	2	Week 13
21.2	Buy course in cart	2	Week 13
22	Feature 22: Transaction Feature	6	
22.1	View transaction	2	Week 11
22.2	Remove transaction	2	Week 11
22.3	Create transaction	2	Week 11
23	Feature 23: Banner Feature	9	
23.1	Create banner	2	Week 13
23.2	View banner	2	Week 13
23.3	Update banner	2	Week 13
23.4	Hidden banner	2	Week 13
23.5	Visible banner	1	Week 13
24	Feature 24: User Feature	2	
24.1	User management	2	Week 9
25	Feature 25: Lecturer Feature	4	
25.1	View lecturer	2	Week 10
25.2	Update lecturer	2	Week 10
26	Feature 26: Admin Feature	4	
26.1	Moderator management	2	Week 9
26.2	Lecturer Payroll Report	2	Week 11

#	Work Package	Est. Effort (pds)	Deadline
1	Report 1: Project Introduction	3	Week 14
2	Report 2: Project Management Plan	3	Week 14
3	Report 3: Software Requirement Specification	6	Week 14
4	Report 4: Software Design Specification	6	Week 14
5	Report 5: Test Documentation	3	Week 14
6	Report 5.1: Unit Test	6	Week 14
7	Report 5.2: Integration Test	5	Week 14
8	Report 5.3: System Test	5	Week 14
9	Report 6: Software User Guides	3	Week 14
10	Report 7: Final Project Report	5	Week 14
11	Meeting & review	30	Week 14
12	Project Management Overhead	30	Week 14

Total Estimated Effort (person-days)

354 person-day

1.2 Project Objectives

#	Metric	Unit	Planned	Actual	Notes / References
1	Effort Usage	Person-day	350	354	Need to analyze more for the actual effort distribution (as tracked in the Project Tracking, sheet Scope)
2	Review Defects	No of defects	40	55	Actually found more defects (55/40), indicating a strong and effective review process.
3	Unit Test	No of defects	100	80	312 test cases (test functions) successfully passed. The target was to find 100 bugs during unit testing, 80 have been fixed, evidence is described in the following file:
4	Integration Test	No of defects	25	18	130 test cases successfully passed. Low integration defects (18/25) combined with high test case pass rate.
5	System Test	No of defects	15	10	145 test cases successfully passed. System defects are low (10/15) compared to the plan.

6	Acceptance Test	No of defects	5	2	Very low defects found by the end-user (2/5), confirming high output quality.
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1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	Lack of specific customers	The system may not meet real learning or teaching needs, reducing usability and adoption	Medium	Conduct early surveys with students, lecturers, and moderators; validate requirements through prototypes and sprint reviews
2	Requirement changes during development	Delays schedule, increases workload, causes rework	Medium	Freeze requirements per increment, apply change-control review, and document all updates clearly
3	Limit team availability due to school schedules	Tasks may not be completed on time, affecting sprint commitments	High	Plan workload based on real weekly capacity, prioritize essential features, and redistribute tasks early
4	Poor communication between team members	Misunderstandings in feature behavior, delays in integration, lower quality	Medium	Hold meetings every evening , provide daily reports, and

				maintain clear documentation
5	Dependency on third-party services(payment APIs, email sender, storage, QR system, Genemi AI)	Payment or notification features may fail if third parties have downtime	Medium	Choose reliable providers; implement fallback messages; retry mechanisms and error logging
6	Performance issues due to large course data, many users, or heavy queries	Slow loading pages, affecting learning experience and user satisfaction	Low	Optimize database queries; load test high-traffic features (Course, Forum)

2. Management Approach

2.1 Project Process

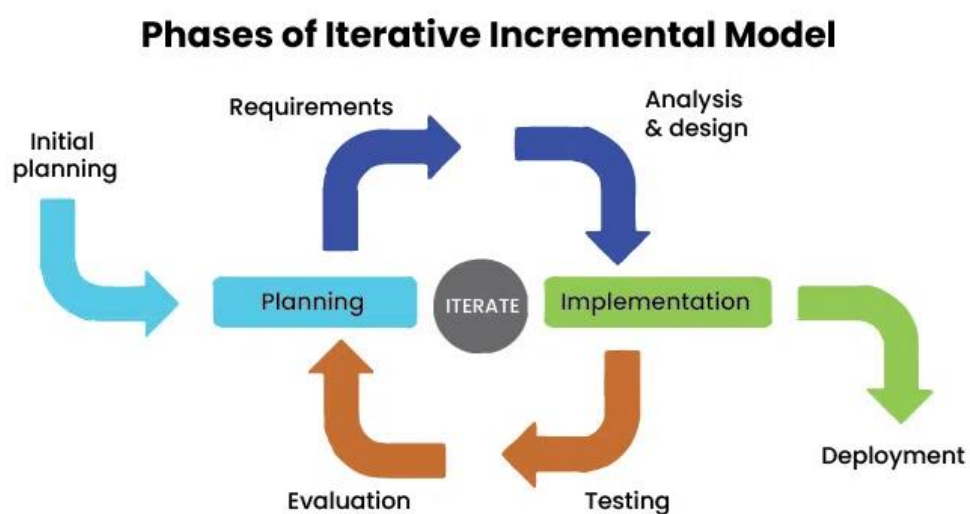


Figure 2.1 - The process of developing a software project

After evaluating user roles, customer expectations, team availability, and the modular nature of the system requirements, the team selected the Iterative–Incremental model as the most suitable methodology. The system serves students, lecturers, and moderators, all requiring a stable, intuitive platform that evolves continuously. With customers expecting updates every 2–3 weeks and the team having varied schedules, the Iterative–Incremental model ensures flexibility and predictable delivery.

The project is divided into three increments across the 14-week timeline. Increment 1 (Weeks 1–4) delivers the foundational system, including authentication, homepage and shared UI, user profile, news module, ticket management, basic forum interactions, and category structures, ensuring an early functional version for feedback. Increment 2 (Weeks 5–9) expands the system into a complete learning ecosystem by implementing notifications, course modules with CRUD and role-specific behaviors, voucher and cart system, lesson and section management, quiz and question bank, and user/moderator management. Increment 3 (Weeks 10–13) provides all advanced and commercial features, including learning progress tracking, direct and cart-based course purchasing flows, reporting for courses and lessons, quiz-taking and result review, lesson interactions (comment, reply, like, report, resolve), transaction history, administrative functions (salary, feature toggles, course approval/publishing), search engine, drag-and-drop course structure, lecturer dashboard, lecturer profiles, and enrolled-student views.

Week 14 is dedicated to system testing, integration validation, bug fixing, performance checks, and deployment preparation. By delivering fully testable modules at each increment, the team ensures rapid feedback, early defect detection, and steady progress toward project goals. This Iterative–Incremental approach minimizes risk, maximizes efficiency, and fits the constraints and capabilities of the development team while ensuring all objectives are achieved within the 14-week schedule.

2.2 Quality Management

To ensure the quality of the Automated Essay Scoring System and reach the project quality objectives, the team will apply a comprehensive approach including three main stages: Defect Prevention, Reviewing, and Testing.

2.2.1. Defect Prevention (Coding Standards)

Approach: All team members must strictly follow the defined Coding Conventions (naming rules, comment styles, folder structure) for both Frontend and Backend.

Goal: To maintain code consistency and minimize syntax or logic errors during the development phase.

2.2.2. Reviewing (Peer Reviews)

Document Review: Requirements and design documents (SRS, SDS) are reviewed by the Team Leader and Instructor before implementation to ensure clarity.

Code Review: Before merging any code into the main branch (e.g., on GitHub/GitLab), a Pull Request (PR) must be created and reviewed by at least one other team member. This helps detect hidden bugs and optimize code logic.

2.2.3. Testing Strategy The project conducts testing using 3 levels:

Unit Testing: Performed by Developers during coding. We verify individual functions/methods to ensure they work correctly. (Tools: JUnit, NUnit, or Jest).

Integration Testing: Performed by Developers when combining modules (e.g., connecting the Essay Scoring Algorithm with the Database).

System Testing: Performed by the Tester. The entire application is tested as a complete system to verify functionality against the SRS requirements.

2.2.4. Defect Tracking

All defects found are logged into a tracking sheet (e.g., Excel) with priority levels (Low, Medium, High, Critical).

Process: Log Bug -> Assign to Dev -> Fix -> Retest -> Close.

2.3 Training Plan

Training Area	Participants	When, Duration
Git & GitHub Workflow(Clone, Branch, Commit, Pull Request)	All team members	Week 1(1 day workshop)
Backend Core(C#, .NET, RESTful API structure)	BinhTB,QuatBT, HienVD, HaiTT	Week 1 - 2(1 week)
Database & ORM(SQL Server, Entity Framework Core)	BinhTB,QuatBT, HienVD, HaiTT	Week 2(1 week)
System Integration(Connecting React with .NET API)	BinhTB,QuatBT, HienVD, HaiTT,TuanCM	Week 3(3 days)
Frontend Development(ReactJS, Hooks, Tailwind CSS)	BinhTB,QuatBT, HaiTT,TuanCM	Week 1 - 2(2 weeks)
Software Testing(Unit Test techniques, Bug logging)	All team members	Week 4(1 week)

3. Project Deliverables

#	Deliverable	Due Date	Notes
1	Project Plan	07/09/2025	Product Background, Software Product Vision, Project Scope and Limitation.
2	Report 1 – Project Introduction	07/09/2025	Product Background, Existing Systems, Business Opportunity, Product Vision, Project Scope & Limitations
3	Report 2 – Project Management Plan	12/09/2025	WBS, Project Process, Plan and Schedule, Project Organization, Project Communication, and Configuration Management
4	Report 3 – Software Requirement Specification	15/09/2025	Business Rules, Use Case Diagram & Use Case Description, Functional Requirements, and Non-Functional Requirements
5	Report 4 - Software Design Document	10/10/2025	
6	Report 5 - Software Testing	25/10/2025	Write testing documents and synthesize test cases.

4. Responsibility Assignments

Work Package	H E1 76 38 2	H E 1 7 3 2 4 4	H E1 8 6 6 2 1	H E 1 7 2 2 3 8	H E 1 6 3 3 4 4
Feature 1: Common Feature	R, A	R	I	C	I
Feature 2: Profile Feature	A	R	I	C	I
Feature 3: News Feature	A	I	R	R, C	I
Feature 4: Ticket Feature	R	I	R	C	I
Feature 5: Forum Feature	A	I	I	R	R
Feature 6: Category Feature	A	I	R	C	R
Feature 7: Forum Category Feature	A	I	R	C	R
Feature 8: SubCategory Feature	A	I	R	C	R
Feature 9: Notification Feature	A	I	R	C	R
Feature 10: Course Feature	R,A	R	I	R, C	I
Feature 11: Voucher Feature	R,A	I	R	C	I
Feature 12: Lesson Feature	R,A	I	I	R, C	I
Feature 13: Cart Feature	R,A	I	I	C	I
Feature 14: Question Bank Feature	A	C	R	C	I
Feature 15: Quiz Feature	A	R	I	R, C	I
Feature 16: Section Feature	A	I	I	R, C	R
Feature 17: Lecturer Application Feature	R	R	I	C	I
Feature 18: Question Feature	A	R	R	R, C	I
Feature 19: Answer Feature	A	R	R	R, C	I
Feature 20: Rating Feature	A	I	I	R, C	R
Feature 21: Student Progress Feature	A	R	I	R, C	I
Feature 22: Payment Feature	R,A	I	I	C	I
Feature 23: Transaction Feature	R,A	I	R	C	I
Feature 24: Banner Feature	A	I	R	R, C	I
Feature 25: User Feature	A	R	I	C	I
Feature 26: Lecturer Feature	A	R	I	C	R

Work Package	H E1 76 38 2	H E 1 7 3 2 4 4	H E1 8 6 6 2 1	H E 1 7 2 2 3 8	H E 1 6 3 3 4 4
Report 1: Project Introduction	R, A	R	R	R,C	R
Report 2: Project Management Plan	R, A	R	R	R,C	R
Report 3: Software Requirement Specification	R, A	R	R	R,C	R
Report 4: Software Design Specification	R, A	R	R	I,C	R
Report 5: Test Documentation	R, A	R	R	R,C	R
Report 5.1: Unit Test	R, A	R	R	I,C	R
Report 5.2: Integration Test	I, A	I	I	R,C	I
Report 5.3: System Test	I, A	I	I	R,C	I
Report 6: Software User Guides	I, A	I	I	R,C	I
Report 7: Final Project Report	R, A	R	R	R,C	R
Meeting & review	R, A	R	R	R,C	R
Project Management Overhead	R, A	R	R	R,C	R

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Daily Meeting	All team members	Report tasks that have been done. Discuss issues, resolve problems. Define the next task	21:00 every week	Online (Google Meet)
Quick Messaging - Group chat	All team members	When you need to have a quick casual talk or schedule a meeting with a supervisor or among team members.	Every day	Online (Zalo)
Weekly meeting with the supervisor	Team Members, Supervisor	Report the project's progress, log issue, define next week plan	21:00 on Sunday, every week	Offline,Online (Google Meet)

6. Configuration Management

6.1 Document Management

Regarding documents, we will store documents related to this project as well as documents related to meetings in Google Drive, Zalo. At the same time, some important documents will also have a copy on GitHub.

6.2 Source Code Management

Regarding source code management, we chose GitHub as the place to store project versions because GitHub is a popular and easy-to-use tool.

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	ReactJS(FrontEnd), .NET(BackEnd), Python(Backend)
Database	SQL Server, Qdrant
IDEs/Editors	Visual Studio Code (FrontEnd), Visual Studio 2022 (BackEnd)
Diagramming	DrawIO, Visual Paradigm, Lucidchart
Documentation	Google Docs/Sheets/Canva/Powerpoint
Version Control	GitLocal/Github (Source Codes), Google Drive (Documents)
Deployment server	Vercel (Frontend), Vietnix Server (Backend)
Project management	Github, Zalo, Google Drive

III. Software Requirement Specification

1. Requirement Overview

1.1 Context Diagram

The SkillUp Online Learning Platform is a comprehensive web-based educational system that modernizes and streamlines the online learning experience for students, lecturers, and educational institutions. This system replaces traditional fragmented learning management approaches by providing an integrated platform that supports course creation and management, interactive learning, real-time collaboration, secure payment processing, and intelligent content moderation. The context diagram below illustrates the external entities and system interfaces for release 1.0.

The primary external entities interacting with the SkillUp system include:

Students: Learners who browse, purchase, and consume educational content, participate in discussions, complete quizzes, and provide course ratings and feedback.

Lecturers: Content creators who develop courses, upload learning materials, manage course structure, interact with students, and track their teaching performance.

Content Moderators: Staff members responsible for reviewing and moderating user-generated content, managing forum posts, handling content reports, and ensuring community guidelines compliance.

System Moderators: Administrators who manage lecturer applications, monitor system-wide activities, handle support tickets, and oversee platform operations.

System Administrators: High-level administrators who manage platform configuration, moderators accounts.

Payment Gateway (PayOS): External payment processing service that handles secure transaction authorization, payment verification, and refund processing for course purchases.

Cloud Storage Service (Cloudinary): Third-party media storage and delivery platform for hosting course videos, images, and other multimedia content with CDN support.

Email Service Provider: External SMTP service for sending transactional emails including registration confirmations, password resets, notifications, and system alerts.

AI Services (Gemini/Ollama): External artificial intelligence APIs for subtitle generation, content analysis, and RAG (Retrieval-Augmented Generation) capabilities for enhanced learning support.

Vector Database (Qdrant): External vector search service for implementing semantic search and intelligent content recommendation features.

The system is expected to evolve over several releases, with future enhancements including integration with additional payment methods (credit/debit card processors, digital wallets), third-party authentication providers (Google, Facebook OAuth), video conferencing platforms for live classes, mobile applications (iOS and Android), advanced analytics and reporting dashboards, AI-powered personalized learning paths, and integration with external learning content providers and certification authorities. AI-powered personalized learning paths, and integration with external learning content providers and certification authorities.

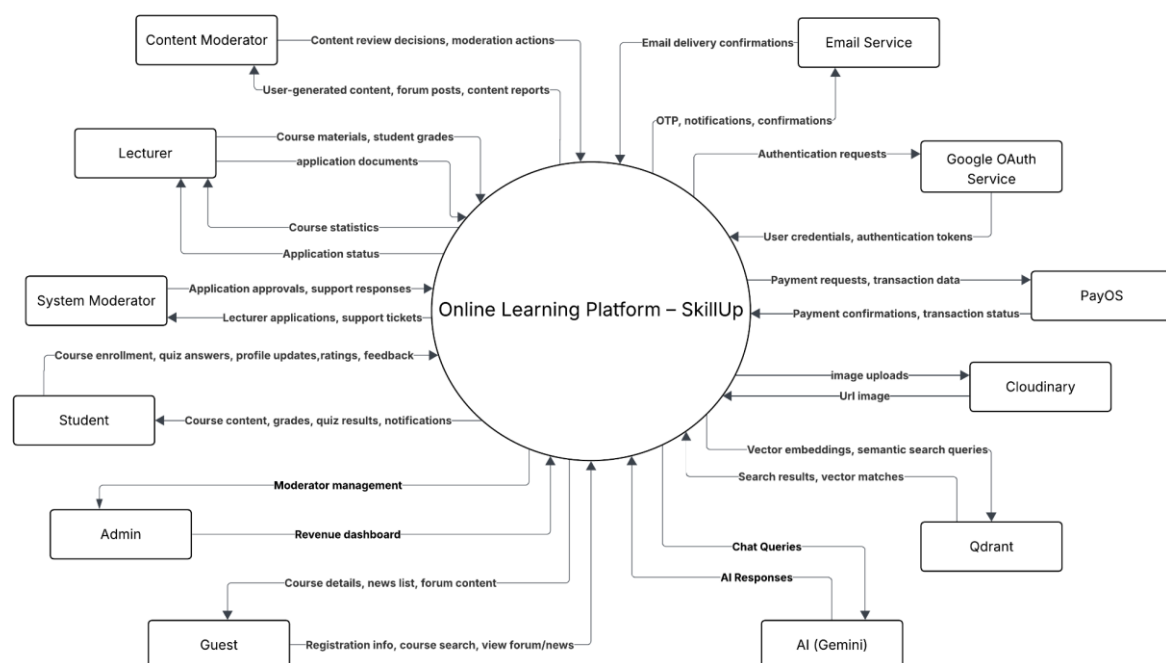


Figure 1.1: Context Diagram

1.2 User Requirements

1.2.1 Actors

#	Actor	Description
1	Administrator	An administrator who has access to the system dashboard, with the ability to monitor overall activities, view key metrics, and manage Moderators by adding, updating, or removing their roles and permissions.
2	Guest	A non-registered visitor who can browse the homepage, view available courses, add courses to the shopping cart, and read news updates, but cannot enroll or access premium features without creating an account.
3	Student	A registered user who can enroll in courses, manage their profile, access customer support, recharge their wallet, provide feedback, and navigate their personalized student dashboard.
4	System Moderator	Moderators have content monitoring and moderation rights. They can manage student and lecturer accounts, manage and approve user's tickets, assess lecturer applications, manage their own profile, and view statistics in the System Moderator Dashboard.
5	Content Moderator	A Content Moderator oversees posts, comments, news, categories, and promotional images to ensure all content follows guidelines and keeps the platform safe and organized.
6	Lecturer	A lecturer is a person who creates and manages course content. Can create/edit/delete courses, add lectures (video, text, quiz, slide), manage quiz, view student list, receive revenue payments, and track sales performance via Lecturer Dashboard.

1.2.2 Use Cases(UC)

ID	Use Case	Feature	Use Case Description
01	Login	Common Feature	Login allows users to easily access the course library, track their learning progress, and take advantage of features that support skill development on SkillUp.
02	Register	Common Feature	Register allows new users to create an account on SkillUp by providing personal information. Upon successful registration, users receive a verification email to activate their account and gain access to platform features.
02	Forgot Password	Common Feature	Forgot Password allows users who cannot remember their login credentials to initiate a password recovery process. Users enter their registered email address, and the system sends a password reset link to verify their identity.
03	Reset Password	Common Feature	Reset Password allows users to create a new password after clicking the verification link received via email. Users enter and confirm a new password that meets security requirements, updating their account credentials.
04	Change Password	Common Feature	Change Password allows authenticated users to update their current password for security purposes. Users must provide their current password for verification, then enter and confirm a new password meeting system requirements.
05	View Lecturer's Courses	Course Feature	The Lecturer views a comprehensive list of all courses they have created, along with their current status, recent activity, and quick actions.
06	Create New Course	Course Feature	The Lecturer initializes a new course shell by providing basic metadata. This creates a "Draft" course in the system.
07	Update Course & Manage General Settings	Course Feature	The Lecturer updates the core metadata, image, price, and coupon information for an existing course. This Use Case covers the actions within the Course Editor sidebar tabs.
08	Manage Sections	Course Feature	The Lecturer structures the course content by creating, modifying, and deleting sections (chapters) within the "Khung chương trình" tab.
09	Manage Lessons	Course Feature	The Lecturer populates sections with learning units, choosing between a standard Bài học (Video/Text) or a Quiz format, and managing supplementary material.
10	Submit the course for publication	Course Feature	The Lecturer submits the finalized course for review by a Content Moderator once all core prerequisites are met.

11	Ban/unban course as Moderator Content	Course Feature	Content Moderator manages courses in the system, including viewing course list, approved/rejected pending courses, and ban/unban published courses.
12	Approve/Rejected Course as Moderator Content	Course Feature	Content Moderator manages courses in the system, including viewing course list, approved/rejected pending courses, and ban/unban published courses.
13	View section	Section Feature	The "View Section Detail" use case allows an enrolled Student to view the contents of a specific chapter (section) inside the learning interface. It displays the list of lessons, current progress within that chapter, and allows the student to navigate to specific lessons.
14	View lesson	Lesson Feature	The "View Lesson Content" use case allows an enrolled Student to view the specific content of a lesson (Video, Article, or Quiz). The student can read or watch the material, mark the lesson as completed, and navigate to the next or previous lesson.
15	Take quiz	Quiz Feature	The "Take Quiz" use case allows an enrolled Student to attempt a quiz assigned in a course. The student can view questions, select answers, navigate between questions, save progress, and submit the quiz for grading.
16	View detail result	Quiz Feature	The "View Quiz Result Detail" use case allows a Student to review their performance after submitting a quiz. The system displays an overall summary (score, pass/fail status) and a detailed breakdown of each question, showing the user's selected answers versus the correct answers with specific feedback.
17	Add cart item	Cart Feature	The "Add Course to Cart" use case allows a logged-in Student to select a specific course and add it to their shopping cart database. This allows the student to collect multiple courses before proceeding to a single checkout process.
18	Remove cart item	Cart Feature	The "Remove Item from Cart" use case allows a logged-in Student to remove a specific course from their shopping cart if they decide not to purchase it. The system updates the cart total immediately after removal.
19	Apply voucher	Cart Feature	The "Apply Voucher" use case allows a Student to enter a promotional code during the shopping cart review process. If the code is valid, the system

			calculates the discount amount and deducts it from the total price.
20	Create Lecturer Application	Lecturer Application	The "Submit Lecturer Application" use case allows a newly registered Lecturer to complete their profile by submitting professional details (experience, CV, degrees). This step is mandatory to activate the teaching privileges (create courses).
21	View detail Lecturer Application	Lecturer Application	The "View My Application Detail" use case allows a Lecturer (who has submitted a registration form) to review the details of their application. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and access the "Edit" function if the application is still pending or was rejected.
22	Manage Lecturer Application	Lecturer Application	The "Manage Lecturer Application" use case allows a System Moderator to review the details of lecturer applications. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and choose to reject or accept the application.
23	Buy course	Shopping cart & payment feature	The "Buy Course" use case allows a logged-in Student to purchase courses. There are two entry points: Direct Purchase (Mua ngay): Buying a single course directly from the detail page. Cart Checkout (Thanh toán giỏ hàng): Buying multiple courses saved in the cart. Upon success, the system enrolls the student in the course(s) and records the transaction.
24	View history transaction	Shopping cart & payment feature	The "View Transaction History" use case allows a logged-in Student to view a chronological list of their financial activities within the platform. This includes wallet top-ups, course purchases, and refunds (if any). It helps users track their spending and verify payment statuses.
25	Feedback course	Shopping cart & payment feature	The "View Transaction History" use case allows a logged-in Student to view a chronological list of their financial activities within the platform. This includes wallet top-ups, course purchases, and refunds (if any). It helps users track their spending and verify payment statuses.

26	View Profile	Profile Management Feature	The "View Lecturer Profile" use case allows a logged-in Lecturer to view their account details. This includes Public Information (visible to students like Title, Profession, Bio) and Private Information (Financial details for payouts like Bank Account).
27	Update Profile	Profile Management Feature	The "Update Lecturer Profile" use case allows a logged-in Lecturer to modify their personal, professional, and financial details. Note that profile picture updates are not supported in this function; users can only update text-based information.
28	Manage Quiz	Quiz Feature	Allows Lecturers to create new quizzes , update existing test information (title , quiz time , passing score) , or delete quizzes in a specific section to assess student knowledge.
29	Add question to quiz	Quiz Feature	Allows Lecturers to add questions to a specific quiz to build the content. Lecturers can either create new questions manually or select existing questions from the Question Bank
30	Manage Question	Question Feature	Allows Lecturer to view a list of existing questions in a Quiz, make edits to question content, or delete questions from the Quiz.
31	Manage Question Bank	Question Feature	Allows Lecturers to manage a shared question bank for each Section of the course. Functions include: viewing the list, creating new questions , editing, deleting, and importing in bulk from Excel files.
32	Manage Answer	Answer Feature	Allows Lecturers to manage answer options for a specific question. Includes adding answers, editing content, deleting answers, uploading answer illustrations, and configuring correct answers.
33	View Enrolled Students	Answer Feature	Allows Lecturers to view a list of all students enrolled in their courses. Lecturers can search and filter the list by specific course to track learning progress.
34	Manage Profile	User Feature	The "View Profile" use case allows a student & lecturer to view their personal information and account details on the SkillUp platform
35	Manage Ticket for Student and Lecturer	Ticket Feature	The "Manage Ticket" feature allows a student or lecturer to create, view and update their support ticket, which will be sent to a Moderator for consideration.

36	Manage Ticket for System Moderator	Ticket Feature	The "Manage Ticket" feature allows a System Moderator to view and handle all Tickets submitted by the Students and Lecturers.
37	Manage News for Content Moderator	News Feature	Allows a moderator to create, edit, publish, and delete news articles that will be displayed to students and other users on the platform.
38	View News	News Feature	Allows students and guests to view a list of published news articles and click on one to read its full content.
39	View List Post	Post Feature	Content Moderator is authenticated and inside “Quản lý bài đăng”.
40	Manage voucher	Voucher Feature	The Lecturer is authenticated and is on the course update screen, at the "Phiếu giảm giá" step. The Lecturer can view the list of vouchers for the course, create new vouchers, edit existing ones, or delete them.
41	Create Post	Forum Feature	Allows authenticated users to create new discussion posts in forum categories
42	Edit/Delete Post	Forum Feature	Allows post authors to edit or delete their own posts.
43	View Post Detail	Forum Feature	Allows users to view the full content of a forum post, including all comments, likes, and replies in a threaded conversation format.
44	Manage Forum Categories	Forum Feature	Allows Content Moderators to create, edit, or delete forum categories to organize discussions
45	Report Post	Report Feature	Allows users to report inappropriate forum posts that violate community guidelines.
46	Manage Post Reports	Report Feature	Allows Content Moderators to view, investigate, and take action on reported posts.
47	Comment on Lesson	Comment Feature	Allows enrolled Students to post questions or comments on specific lessons within a course.
48	Comment on Post	Comment Feature	Allows authenticated users to reply to forum posts , creating threaded discussions.
49	Edit/Delete Comment	Comment Feature	Allows comment authors to edit or delete their own comments on lessons or posts.
50	Like Comment on Lesson	Comment Feature	Allows enrolled Students to like helpful comments on lessons
51	Like Comment on Post	Comment Feature	Allows users to like comments on forum posts, indicating agreement or appreciation.

52	Manage Comment Reports	Comment Feature	Allows Content Moderators to review reported comments, view context, and take action
53	View Courses by SubCategory	Course Feature	Allows users to filter and view courses within a specific subcategory
54	View Course Detail	Course Feature	Displays comprehensive course information including description, curriculum outline, instructor profile, ratings, reviews, preview videos, and enrollment button.
56	Report Course	Report Feature	Allows enrolled students to report issues with courses such as outdated content, misleading information, copyright violations, or poor quality.
57	Manage Course Reports	Report Feature	Allows Content Moderators to review course reports, investigate claims, contact instructors for corrections, and take action
58	View Notifications	Notification Feature	Allows authenticated users to view real-time notifications about course updates, comment replies, ticket responses, application status, purchases, and system announcements.
59	Comment in Lesson	Comment Feature	Allows enrolled students to use real-time chat during lesson viewing to ask questions and interact with other students currently watching the same lesson.
60	Auto-Generate Subtitles	AI Feature	Allows Lecturers to automatically generate subtitles for uploaded videos using AI services.

1.2.3 Use Case Diagrams

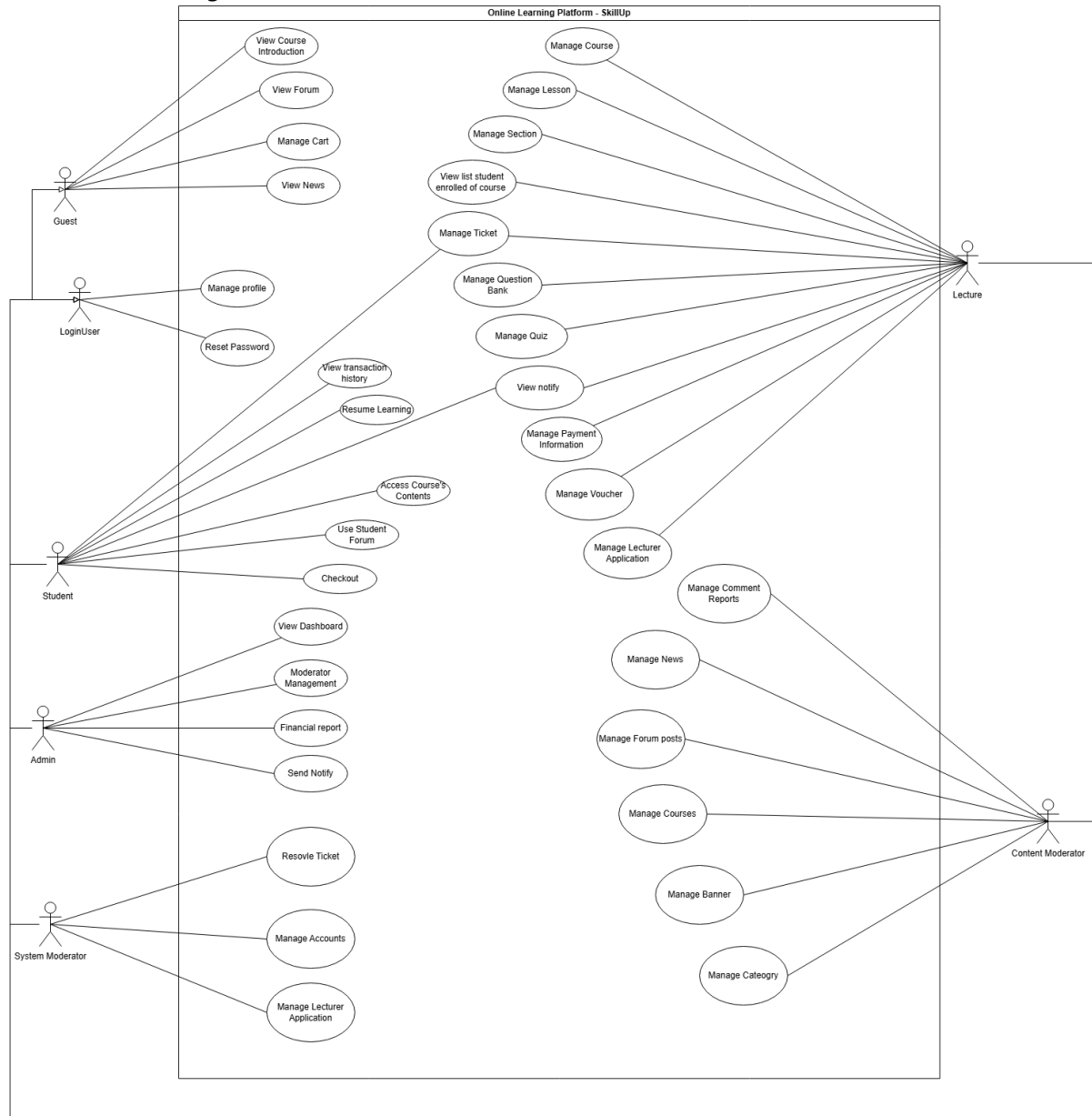


Figure 1.3.3 - Overview use case diagram

1.2.3.1 UCs for Logged In User

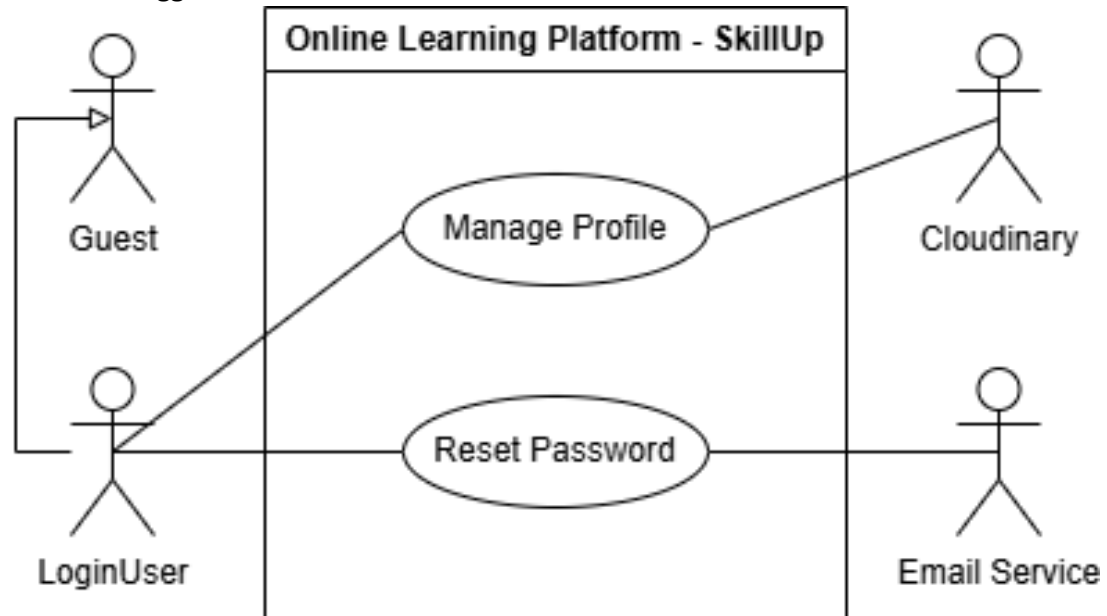


Figure 1.3.3.1 - use case logged in user diagram

1.3.3.2 UCs for Guest

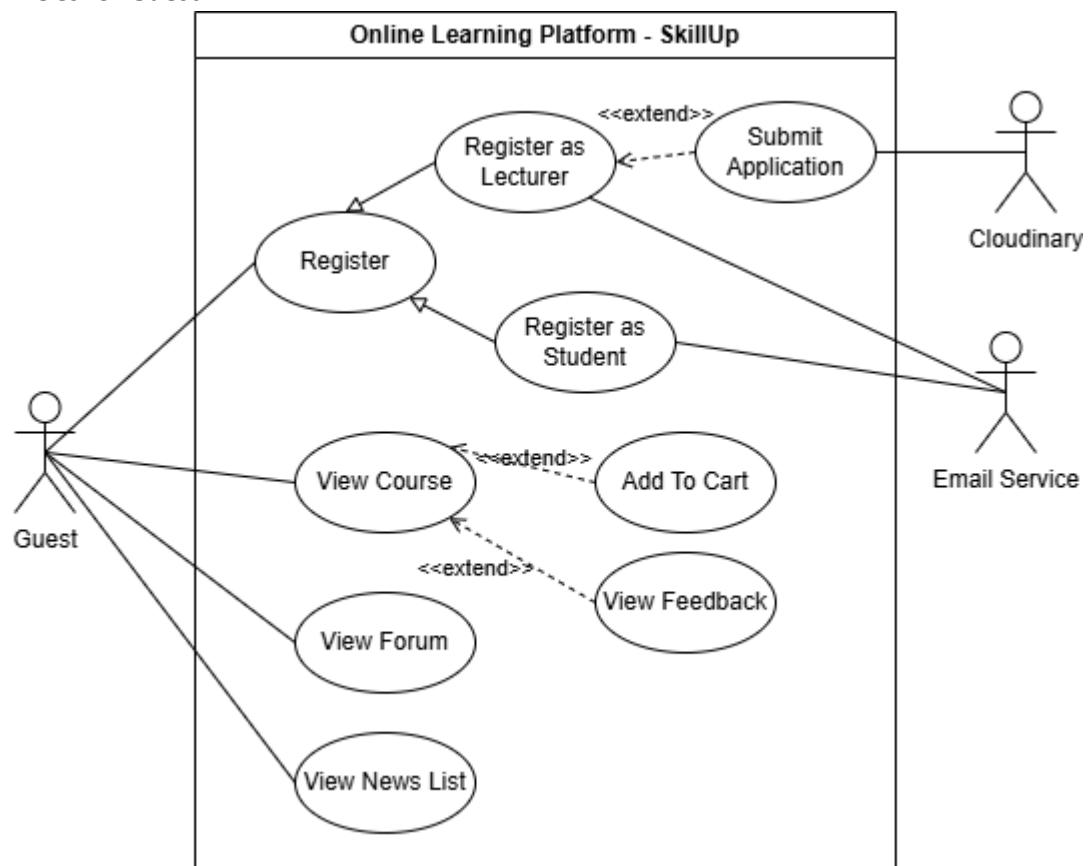


Figure 1.3.3.2 - use case guest diagram

1.3.3.3 UCs for Student

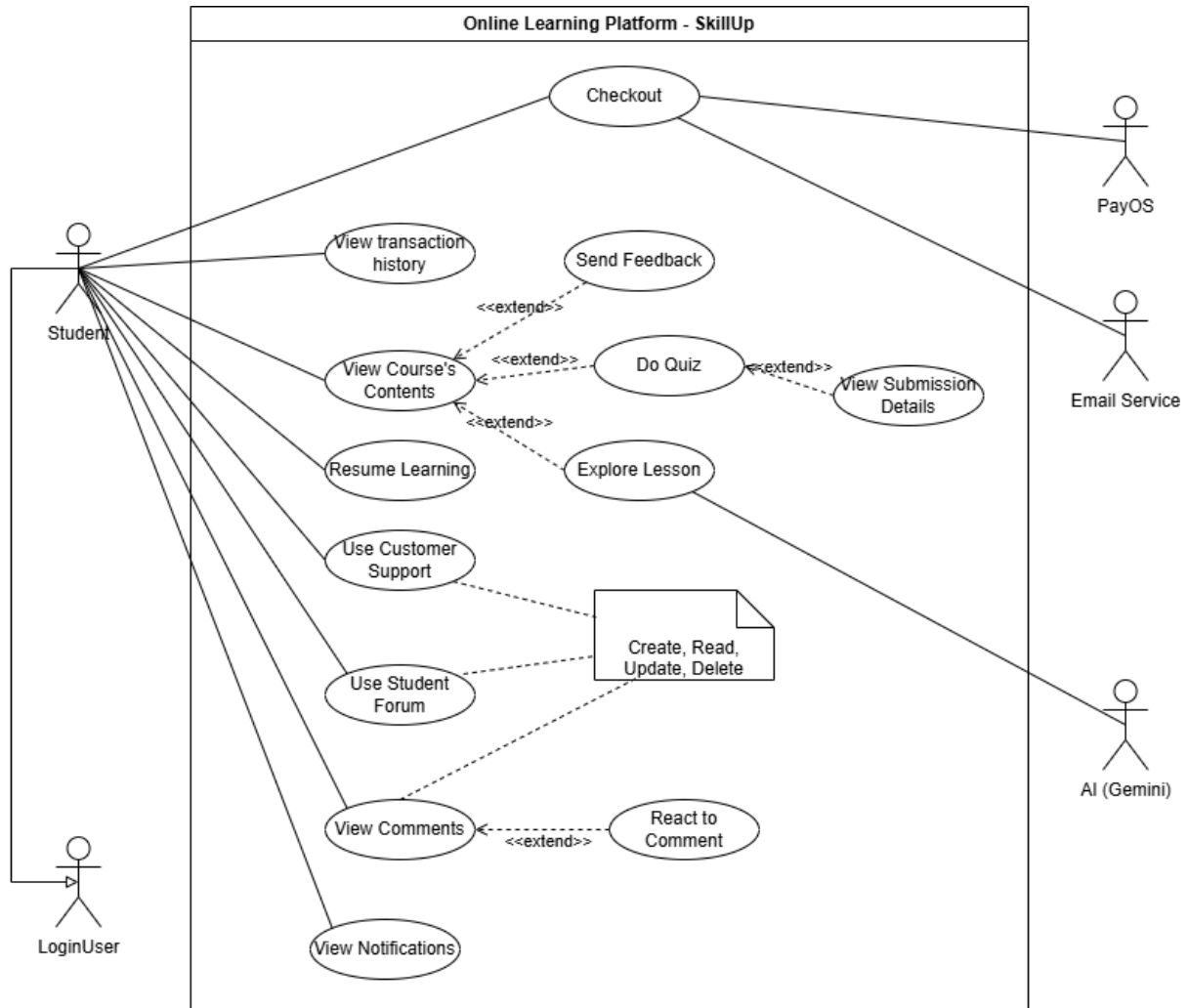


Figure 1.3.3.3 - use case student diagram

1.3.3.4 UCs for Lecturer

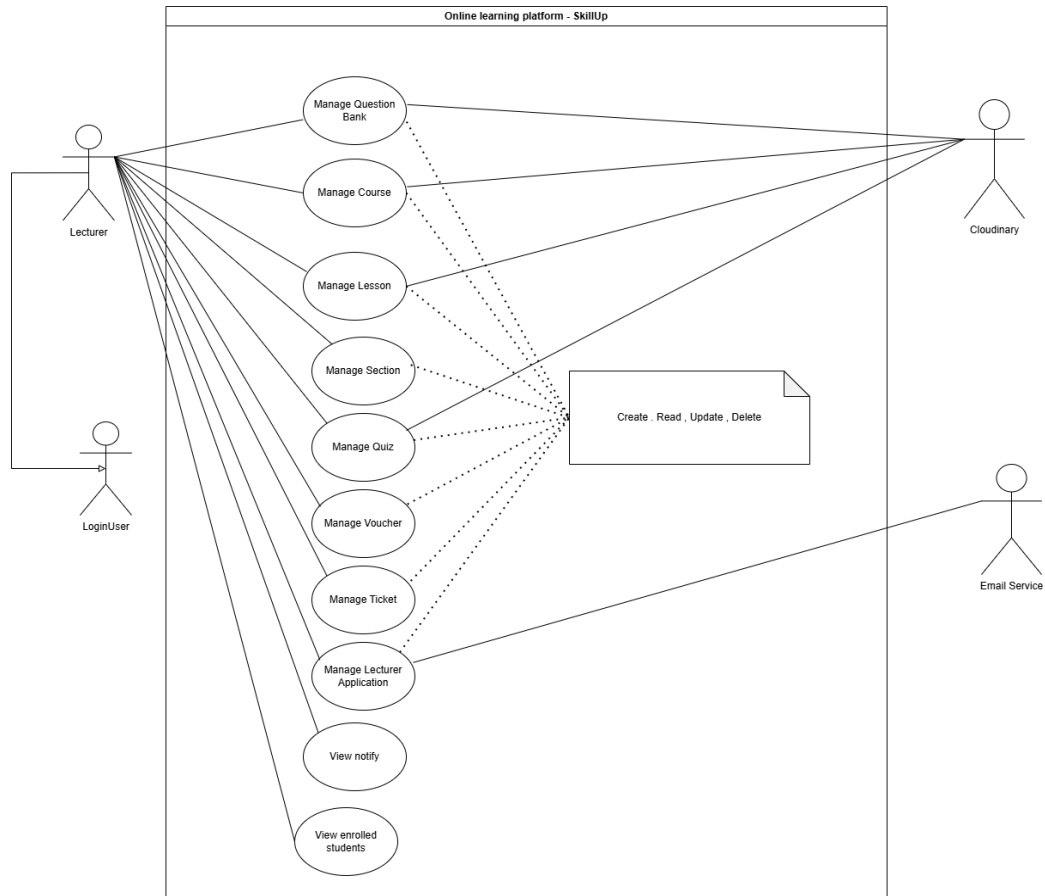


Figure 1.3.3.4 - use case lecturer diagram

1.3.3.5 UCs for Content Moderator

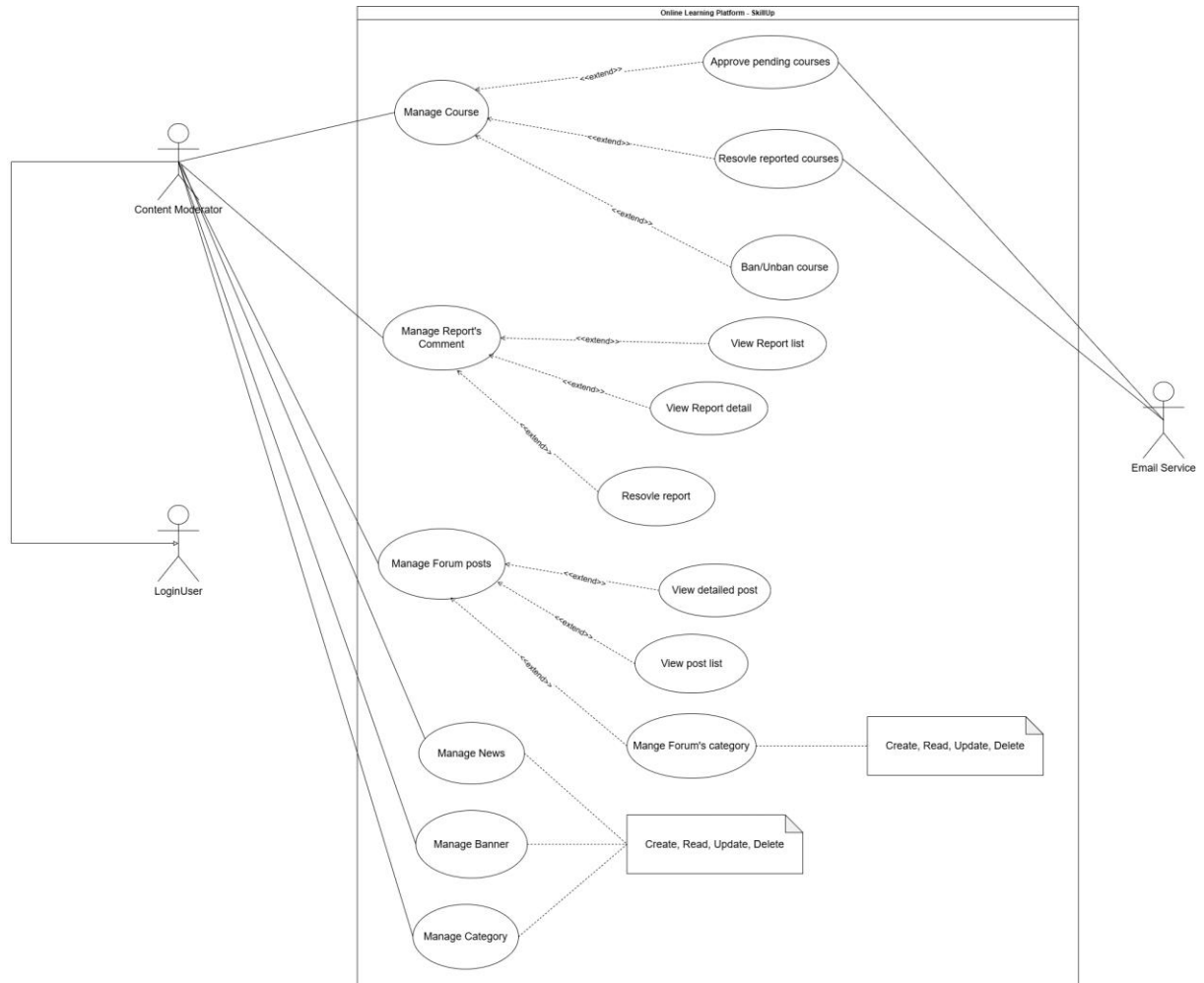


Figure 1.3.3.5 - use case content moderator diagram

1.3.3.6 UCs for System Moderator



Figure 1.3.3.6 - use case system moderator diagram

1.3.3.7 UCs for Admin

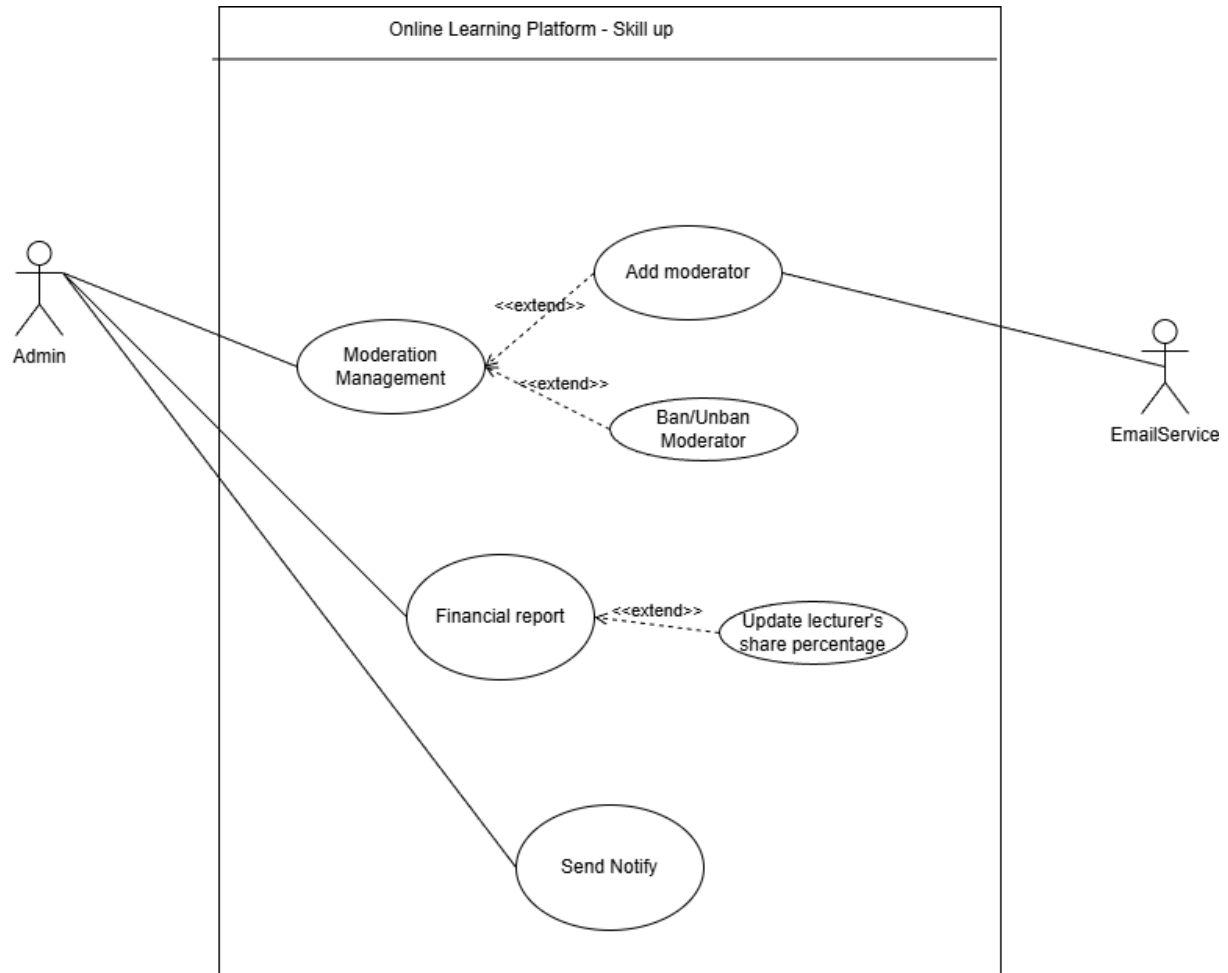


Figure 1.3.3.7 - use case admin diagram

1.3 System Functionalities

1.3.1 Screens Flow

1.3.1.1 Guest Screens Flow

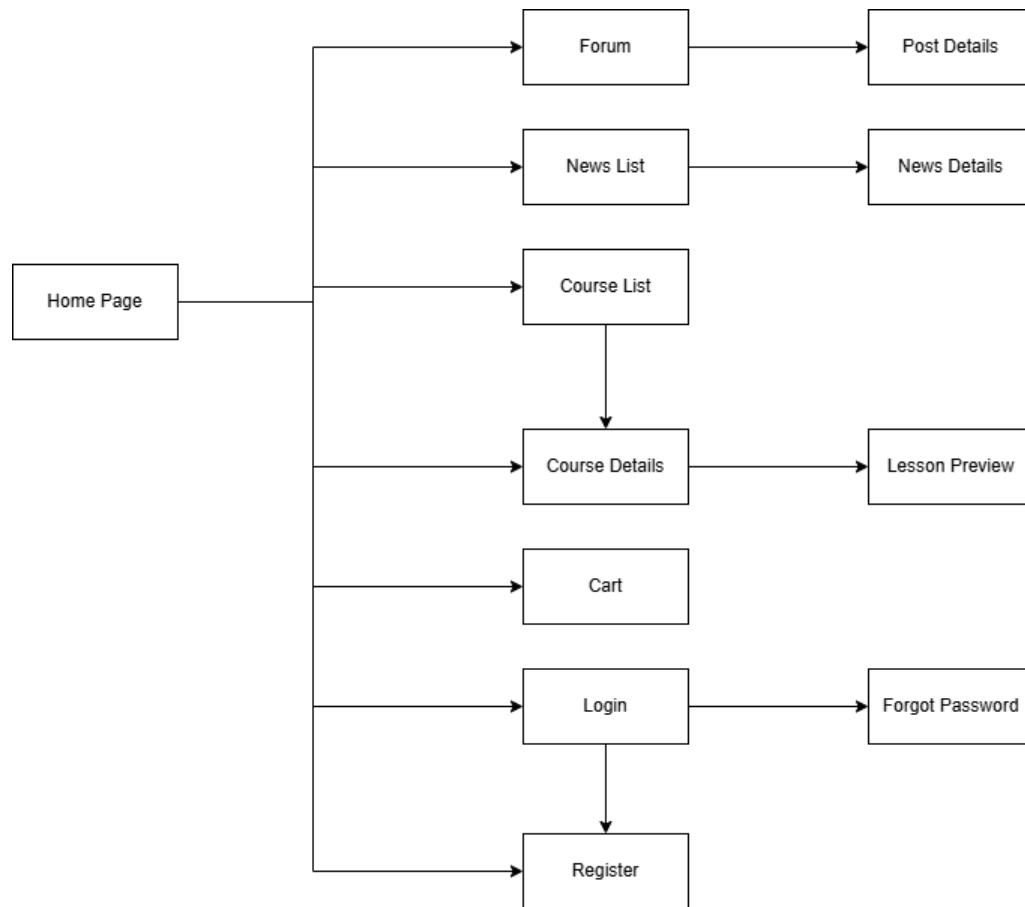


Figure 1.3.1.1 - Guest Screens Flow

1.3.1.2 Content Moderator Screens Flow

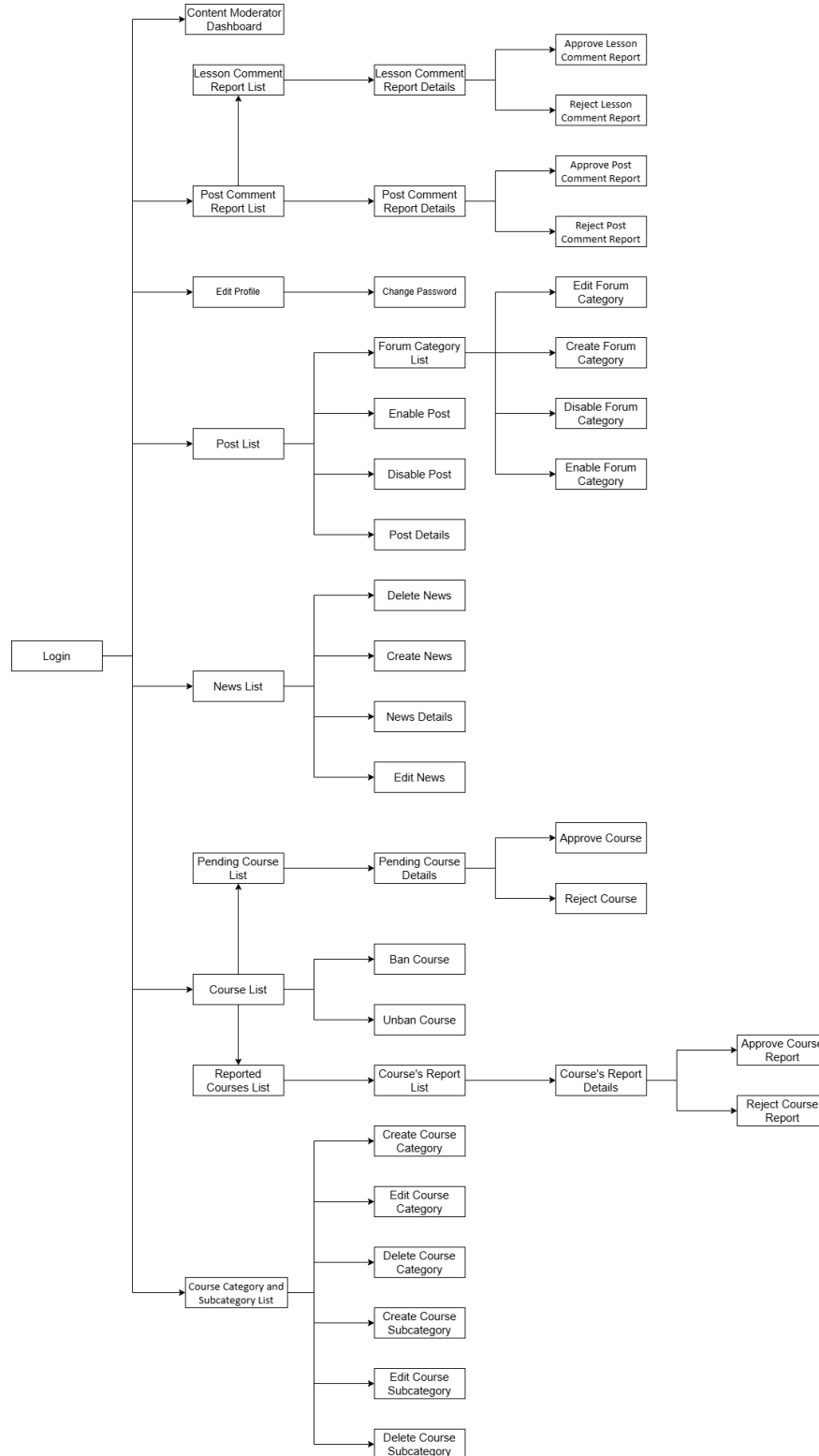


Figure 1.3.1.2 - Content moderator Screens Flow

1.3.1.3 Lecturer Screens Flow

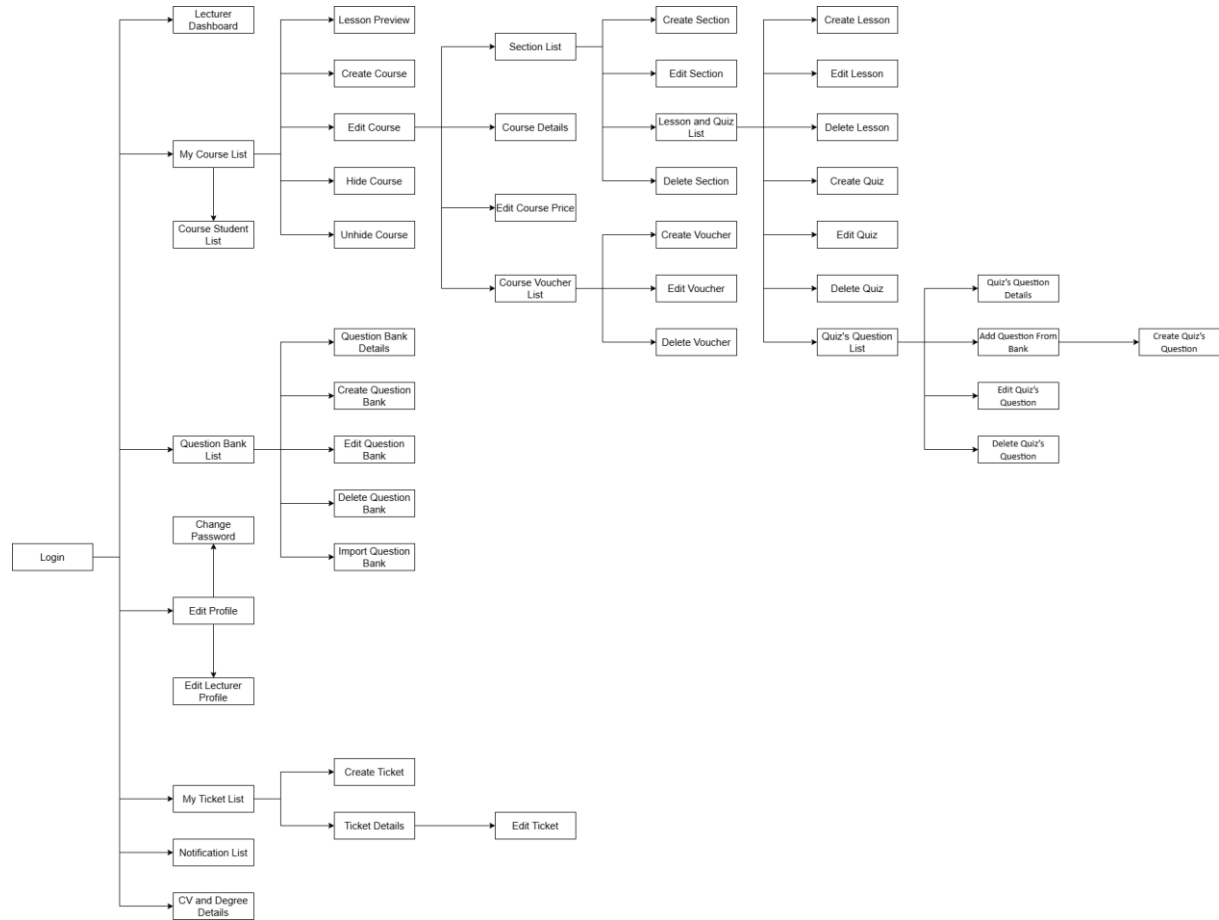


Figure 1.3.1.3 - Lecturer Screens Flow

1.3.1.4 Admin Screens Flow

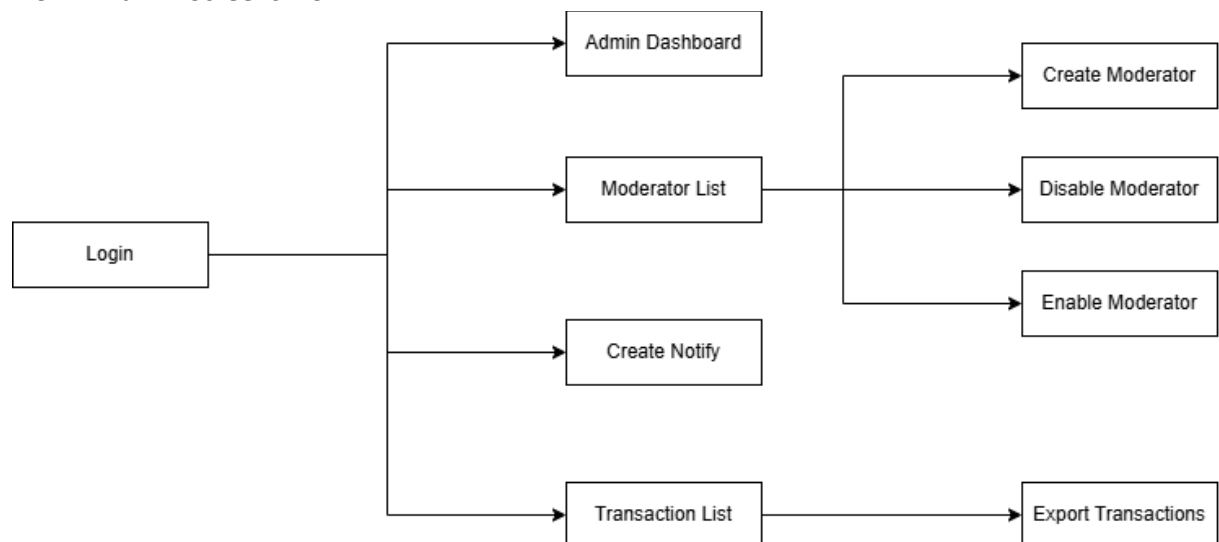


Figure 1.3.1.4 - Admin Screens Flow

1.3.1.5 System Moderator Screens Flow

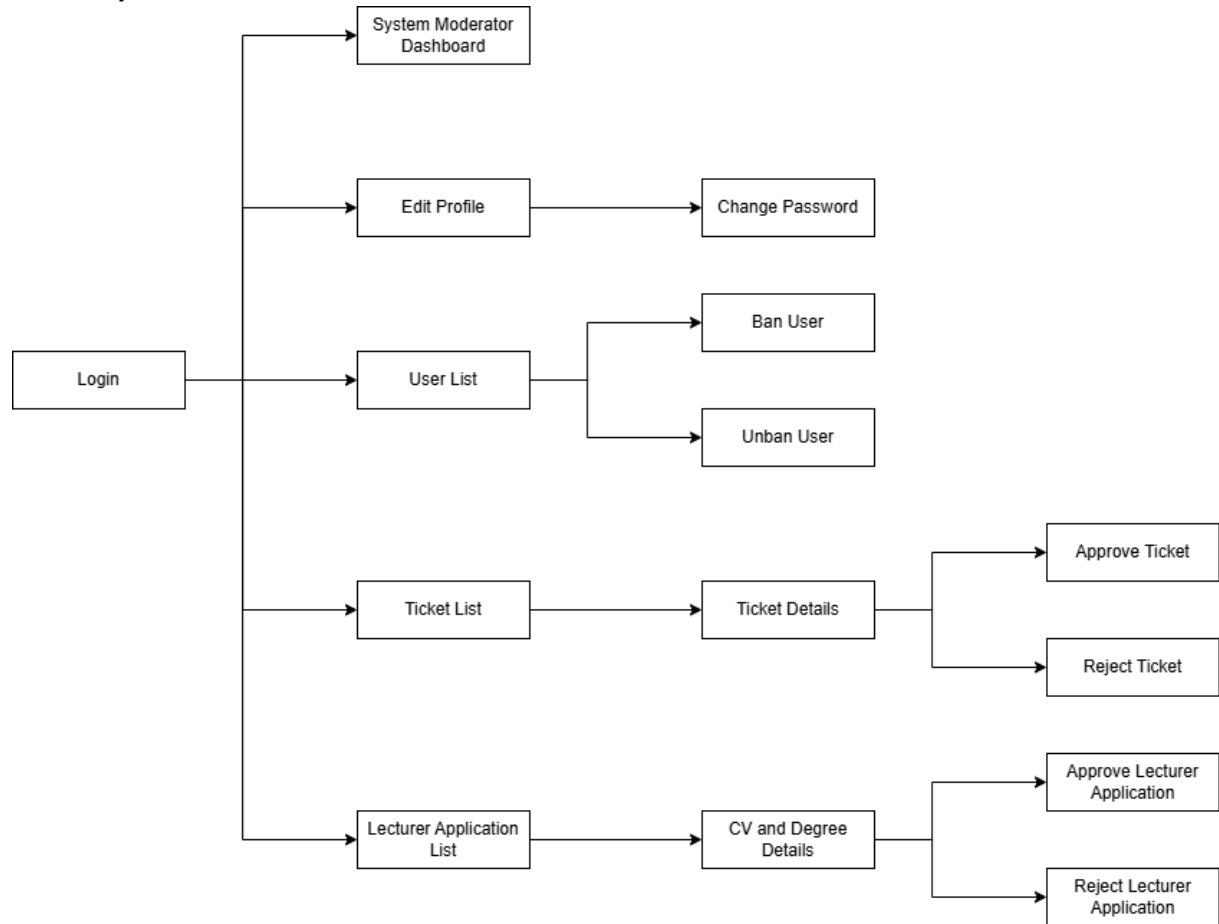


Figure 1.3.1.5 - System moderator Screens Flow

1.3.1.6 Student Screens Flow

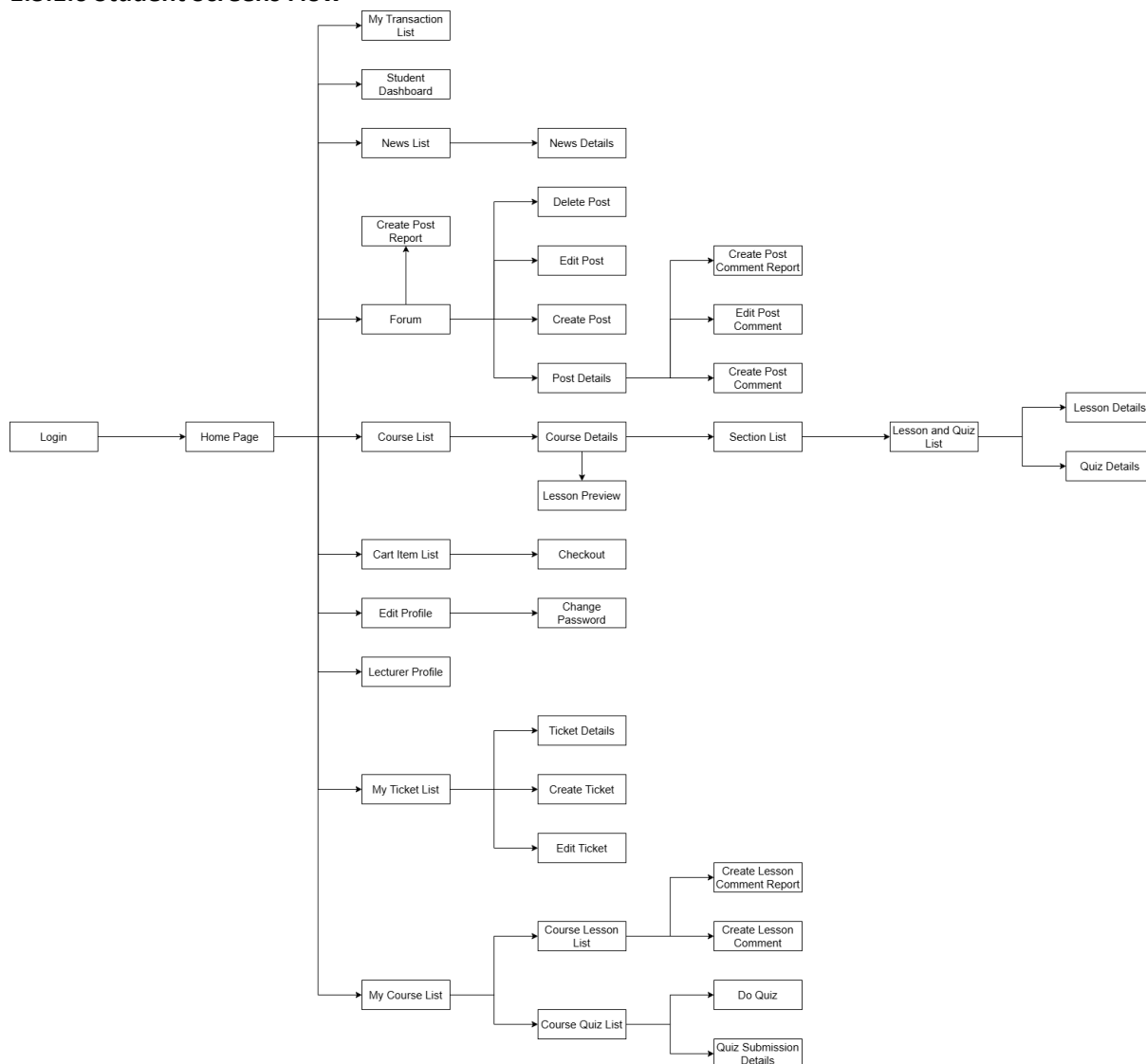


Figure 1.3.1.6 - Student Screens Flow

1.3.2 Screen Authorization

The following table illustrates the authorization structure for the system's screens. Each screen may be accessible by one or more user roles, with an "x" marking where access is permitted. This helps clarify role restrictions and ensures that users only interact with functions relevant to their assigned permissions.

Screen	Guest	Admin	System Moderator	Content Moderator	Lecturer	Student
Home Page	x					x
Login	x	x	x	x	x	x
Forgot Password	x					
Register	x					
Forum	x					x
Post Details	x			x		x
News List	x			x		x

News Details	x			x		x
Course List	x			x		x
Ban Course				x		
Unban Course				x		
Pending Course List				x		
Pending Course Details				x		
Course Details	x				x	x
Lesson Preview	x				x	x
Cart Item List	x					x
Content Moderator Dashboard				x		
Create News				x		
Edit News				x		
Delete News				x		
Approve Course				x		
Reject Course				x		
Reported Courses List				x		
Reported Courses Details				x		
Course's Report List				x		
Course's Report Details				x		
Approve Course Report				x		
Reject Course Report				x		
Post List				x		
Disable Post				x		
Enable Post				x		
Forum Category List				x		
Create Forum Category				x		
Edit Forum Category				x		
Disable Forum Category				x		
Enable Forum Category				x		
Post Comment Report List				x		
Post Comment Report Details				x		
Approve Post Comment Report				x		
Reject Post Comment Report				x		
Lesson Comment Report List				x		

Lesson Comment Report Details				x		
Approve Lesson Comment Report				x		
Reject Lesson Comment Report				x		
Course Category and Subcategory List				x		
Create Course Category				x		
Edit Course Category				x		
Delete Course Category				x		
Create Course Subcategory				x		
Edit Course Subcategory				x		
Delete Course Subcategory				x		
Lecturer Dashboard					x	
Hide Course					x	
Unhide Course					x	
Create Course					x	
Edit Course					x	
Create Section					x	
Section List					x	x
Edit Section					x	
Delete Section					x	
Lesson and Quiz list					x	x
Lesson Details						x
Create Lesson					x	
Edit Lesson					x	
Delete Lesson					x	
Quiz Details						x
Create Quiz					x	
Edit Quiz					x	
Delete Quiz					x	
Course Voucher List					x	
Edit Voucher					x	
Create Voucher					x	
Delete Voucher					x	
Edit Course Price					x	
Question Bank List					x	
Question Bank Details					x	
Create Question Bank					x	

Edit Question Bank					x	
Delete Question Bank					x	
Import Question Bank					x	
Quiz's Question List					x	
Quiz's Question Details					x	
Create Quiz's Question					x	
Edit Quiz's Question					x	
Delete Quiz's Question					x	
Add Question From Bank					x	
Edit Profile			x	x	x	x
Lecturer Profile						x
Edit Lecturer Profile					x	
Change Password			x	x	x	x
System Moderator Dashboard			x			
Course Student List					x	
User List			x			
Ban User			x			
Unban User			x			
Ticket List			x			
Create Ticket					x	x
My Ticket List					x	x
Ticket Details			x		x	x
Edit Ticket					x	x
Approve Ticket			x			
Reject Ticket			x			
Lecturer Application List			x			
CV and Degree Details			x		x	
Approve Lecturer Application			x			
Reject Lecturer Application			x			
Admin Dashboard		x				
Moderator List		x				
Create Moderator		x				
Disable Moderator		x				
Enable Moderator		x				
Create Notify		x				
Student Dashboard						x
Create Post						x
Edit Post						x
Delete Post						x
Create Post Report						x

Create Post Comment						x
Edit Post Comment						x
Create Post Comment Report						x
My Course List					x	x
Create Lesson Comment						x
Delete Lesson Comment						x
Like Lesson Comment						x
Create Lesson Comment Report						x
Create Quiz Submission						x
Quiz Submission Details						x
Transaction List		x				
Export Transactions		x				
My Transaction List						x
Notification List					x	x

1.3.3 Non-UI Functions

#	Feature	System Function	Description
1	Email	Send Email	Send email with contents that are defined within workflow
2	Authentication	Authentication	Used to authentication users within system
3	AI (Gemini)	AI system	Supporting user experience(e.g recommend course, giving hint quiz,..)

1.4 Entity Relationship Diagram

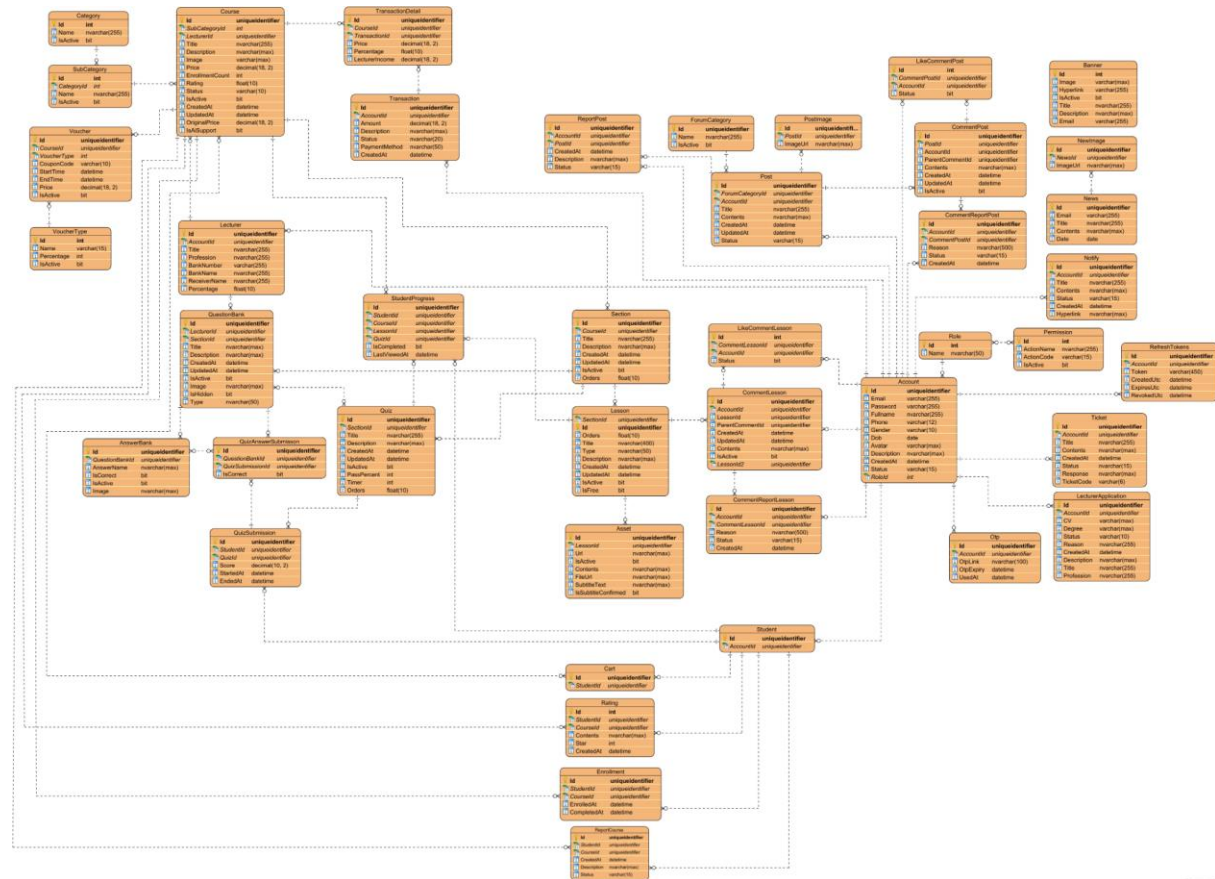


Figure 1.5.1 - Overview entity relationship diagram

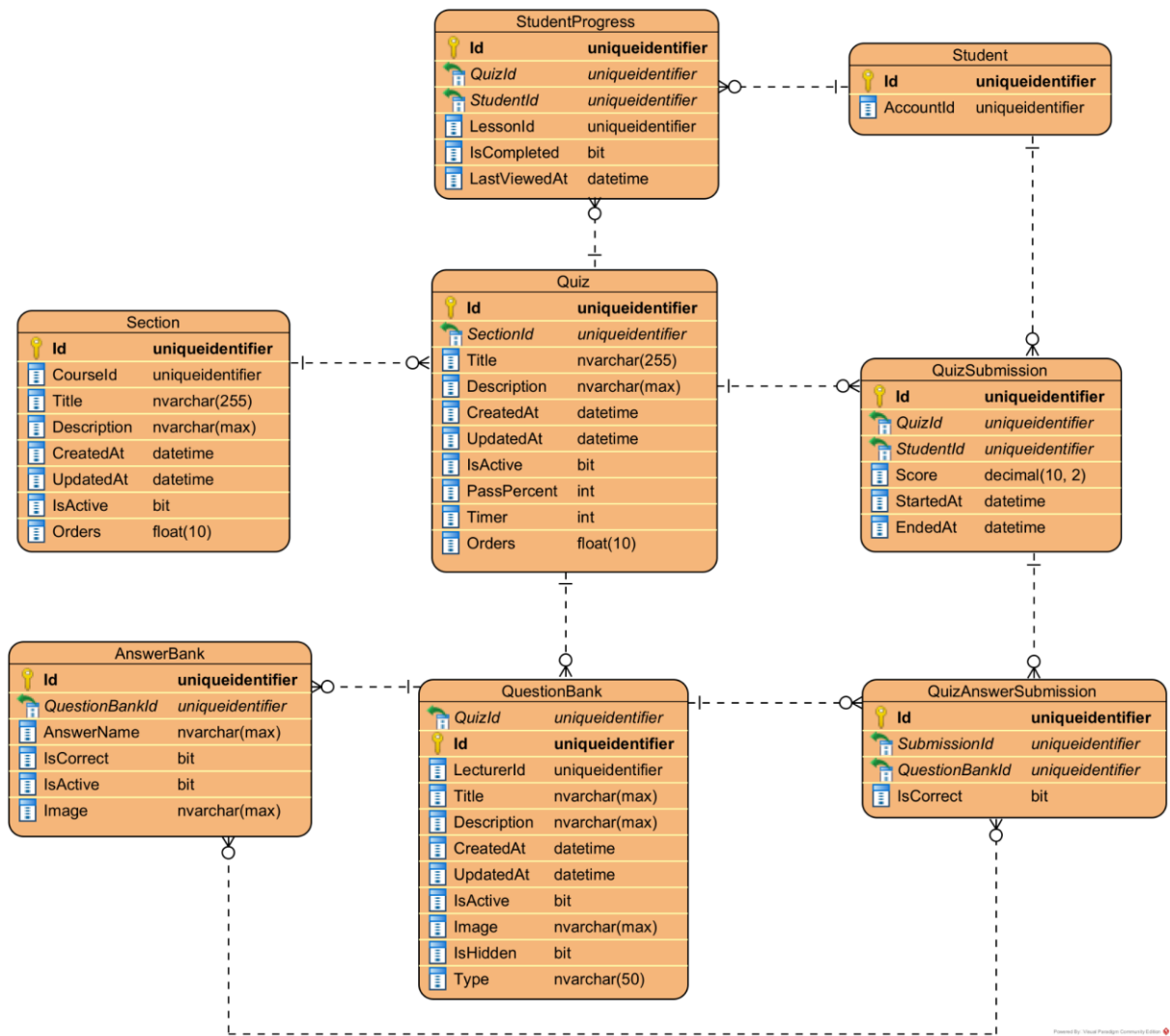


Figure 1.5.2 - Quiz entity relationship diagram

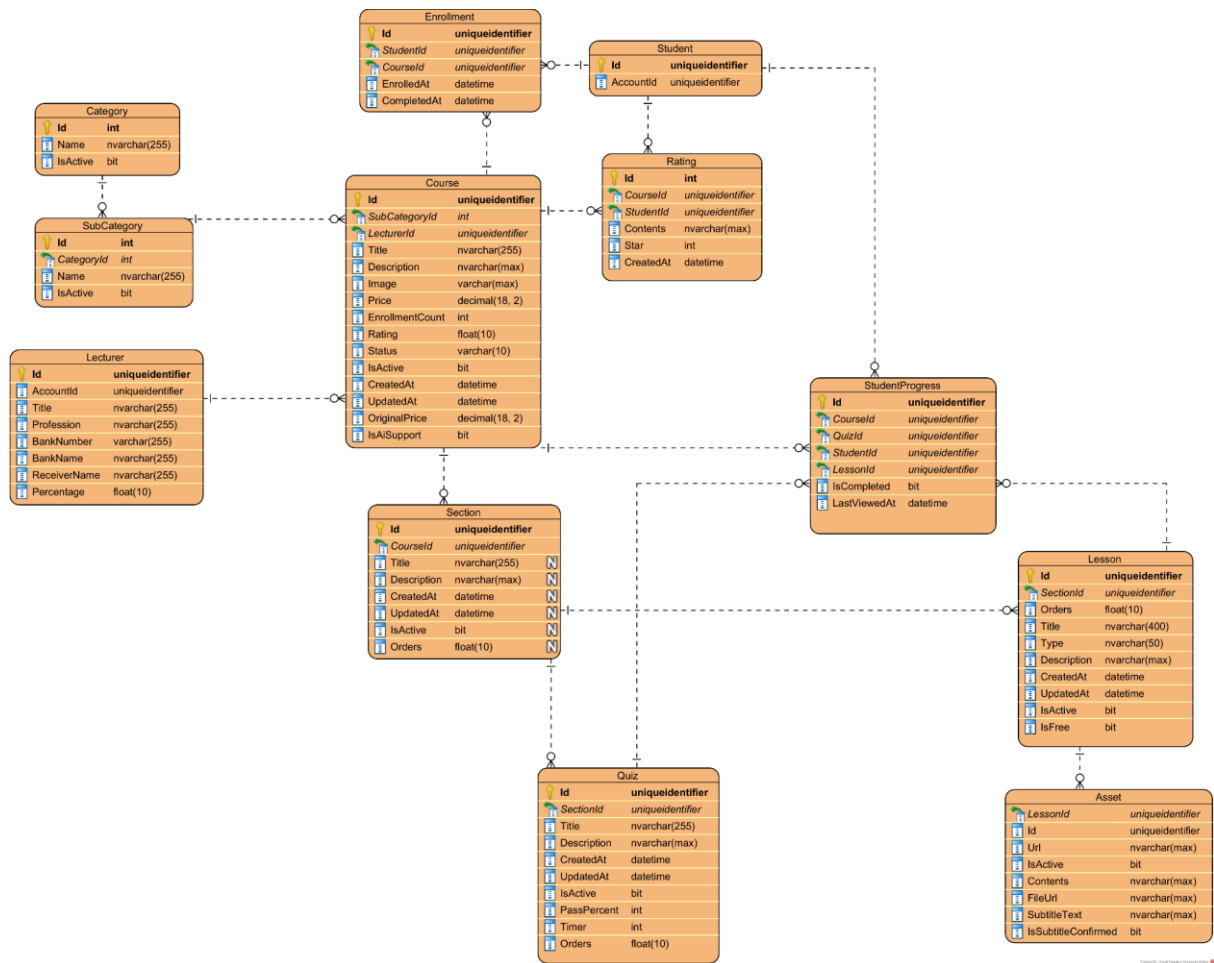


Figure 1.5.3 - Course relationship diagram



Figure 1.5.4 - LecturerApplication relationship diagram

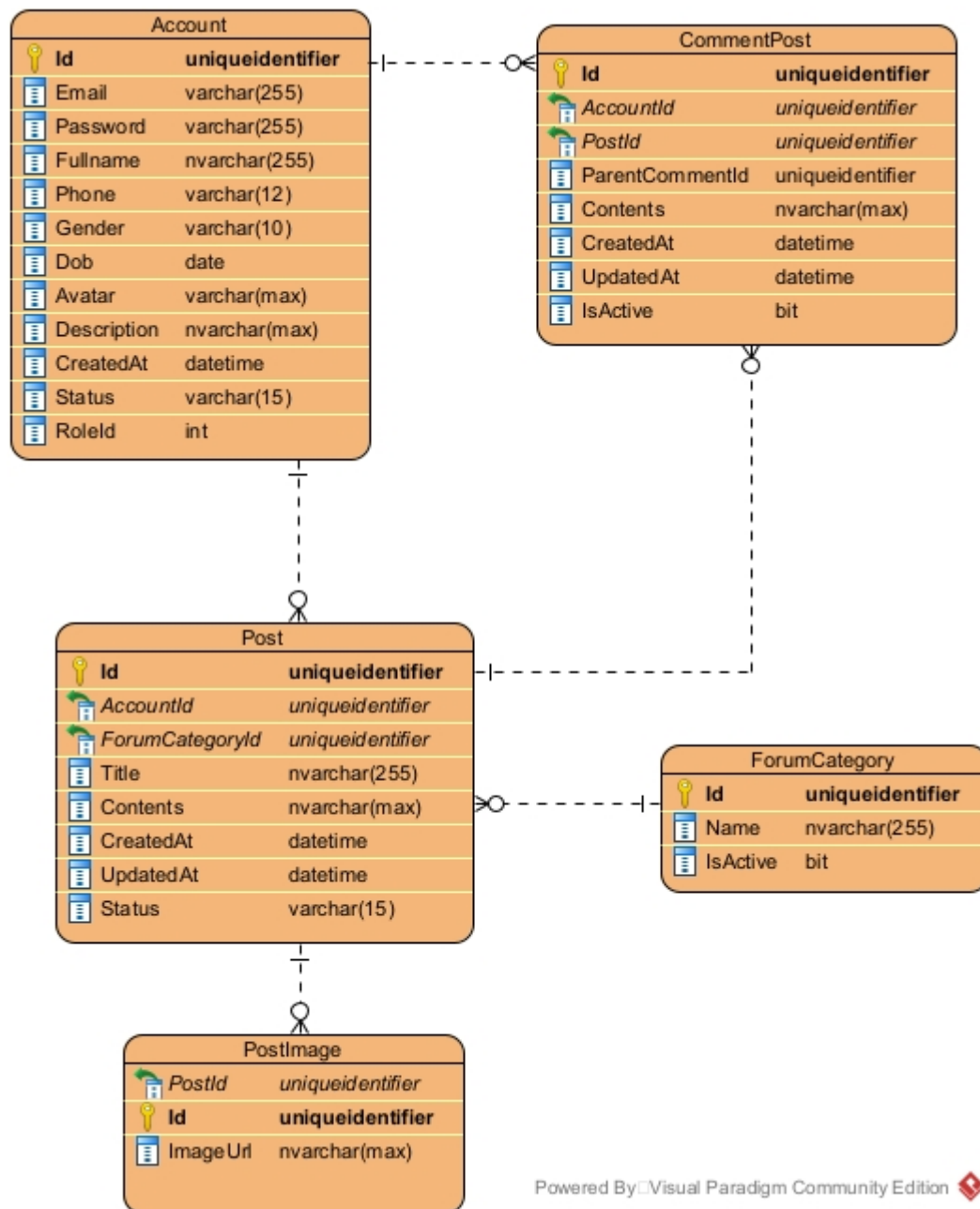


Figure 1.5.5 - Forum relationship diagram

Entities Description

#	Entity	Description
1	Category	Defines high-level classifications for organizing courses and posts.
2	SubCategory	Represents sub-classifications belonging to a specific category.
3	Voucher	Represents discount vouchers that can be applied to course purchases
4	VoucherType	Defines types of vouchers, including discount rules and conditions.

#	Entity	Description
5	Course	Represents an online course with details such as title, price, and status.
6	TransactionDetail	Stores detailed information of each course within a transaction.
7	Transaction	Records payment transactions made by users for course enrollments.
8	Lecturer	Represents instructors who create and manage courses.
9	QuestionBank	Stores questions used for quizzes and assessments..
10	StudentProgress	Tracks a student's learning progress within enrolled courses.
11	AnswerBank	Stores possible answers for questions in the question bank.
12	QuizAnswerSubmission	Records answers submitted by students for quiz questions.
13	Quiz	Represents a quiz associated with a course or lesson..
14	QuizSubmission	Stores quiz submission results of students, including scores and status.
15	ReportPost	Records reports submitted by users for inappropriate posts.
16	ForumCategory	Defines categories used to organize discussion forum posts..
17	PostImage	Stores images associated with forum posts.
18	Post	Represents discussion posts created by users in forums.
19	Section	Divides a course into structured sections or modules.
20	LikeCommentLesson	Stores likes on comments related to lessons.
21	Lesson	Represents individual lessons within a course section.
22	CommentLesson	Stores user comments on lessons.
23	Asset	Stores learning materials such as videos, documents, or files.
24	CommentReportLesson	Records reports submitted for inappropriate lesson comments.
25	Cart	Stores courses added to a user's cart before checkout.
26	Rating	Stores user ratings and reviews for courses.
27	Enrollment	Records student enrollments in courses.
28	ReportCourse	Stores reports submitted by users for inappropriate courses.
29	Student	Represents learners enrolled in courses on the platform.
30	LikeCommentPost	Stores likes on comments related to forum posts.
31	CommentPost	Stores user comments on forum posts.
32	CommentReportPost	Records reports submitted for inappropriate post comments.
33	Banner	Stores banner images displayed on the platform.
34	NewImage	Stores images associated with news or announcements.
35	News	Represents news or announcements published on the platform.
36	Notify	Stores system notifications sent to users.
37	Role	Defines user roles such as Admin, Lecturer, or Student.
38	Permission	Defines access permissions assigned to roles.
39	RefreshTokens	Stores refresh tokens used for authentication.
40	Account	Stores user account information and authentication data.
41	Ticket	Represents support tickets submitted by users.
42	Otp	Stores one-time passwords used for verification processes.
43	LecturerApplication	Stores applications submitted by users to become lecturers.

2. Use case specifications

Below are the main use case specifications, for a more detailed description, please refer to the following content: 2. Use case specification in Report 3.

2.1 Common

2.1.1 Register

Primary Actors	Guest	Secondary Actors	Email Service, Google OAuth Service
Description	Register allows new users to create an account in the system. After registration, users receive a verification email to activate their account.		
Preconditions	• The user does not have an account in the system • The system is operating normally • A stable internet connection is available • The email address is not already registered in the system		
Postconditions	• New account is created • Verification email is sent • User can login after email verification • For Lecturer: Account pending approval, activated after moderator acceptance		
Normal Sequence/Flow	Register: 1. User navigates to Register page 2. System shows the User Registration screen with tab selector (Học viên/Giảng viên) 3. User selects role type (Học viên/Giảng viên) 4. User types in registration details ("Email", "Mật khẩu", "Xác nhận mật khẩu", "Họ và tên") 5. User clicks the "Đăng ký ngay" button 6. System validates input data (email format, password match, password length ≥ 6 characters, all fields required) 7. System checks if email already exists 8. System creates a new account with "Inactive" status 9. System generates verification token and sends verification email 10. System displays success message: "Đăng ký thành công! Vui lòng kiểm tra email để xác thực tài khoản." (MSG05) 11. System redirects user to login page after 2 seconds		
Alternative Sequences/Flows	Step 6.1. Invalid password format: 1. System displays "Mật khẩu phải có ít nhất 6 ký tự" 2. Return to step 4 6.2. Password confirmation doesn't match: 1. System displays "Mật khẩu không khớp" 2. Return to step 4 6.3. Required fields missing: 1. System highlights missing fields 2. Return to step 4		
Exception	E1_Email Service Failure		

	● Trigger: Occurs at Normal Flow Step 9 (System attempts to send verification email). ● Sequence: 1. System creates the account in the database but fails to connect to the Email Service (SMTP timeout or service down). 2. System logs the error. 3. System displays a warning message: "Tài khoản đã được tạo nhưng không thể gửi email xác thực lúc này. Vui lòng liên hệ quản trị viên hoặc thử lại chức năng 'Gửi lại email' sau." (MSG-ERR-EMAIL). 4. Postcondition: Account is created with status "Inactive"; User is on the Login page or a specific notification page. E2_System/Database Error ● Trigger: Occurs at Normal Flow Step 7 or 8. ● Sequence: 1. System encounters a database write error or connection timeout. 2. System displays message: "Đã có lỗi xảy ra trong quá trình xử lý. Vui lòng thử lại sau." (MSG-SYS-01). 3. Postcondition: Account is not created; User remains on the Registration screen with their data preserved in the fields. E3_Google Registration Error ● Trigger: Occurs if the user uses the Google OAuth Service (implied by Secondary Actors list, though not detailed in Normal Flow). ● Sequence: 1. User attempts to register via Google. 2. Google returns an error or the email is already linked to a manual account without support for merging. 3. System displays message: "Không thể đăng ký bằng Google. Vui lòng sử dụng đăng ký thủ công." (MSG-SYS-02). 4. Postcondition: User remains on the Registration screen.
--	--

2.2 Course Feature

2.2.1 Create New Course

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer initializes a new course shell by providing basic metadata. This creates a "Draft" course in the system.		
Preconditions	● The user must be logged in as a Lecturer.		

	• The Lecturer account must be active and verified. • The Lecturer must have at least one active course and have created sections. • The lecturer has navigated to the "Khóa học" (Courses) menu item.
Postconditions	• A new course ID is generated. • The course status is set to "Nháp" (Draft). • The Lecturer is redirected to the Course Editor interface.
Normal Sequence/Flow	1. Lecturer clicks the yellow "+ Tạo khóa học mới" (Create New Course) button on the dashboard. 2. The system displays the creation form. 3. Lecturer enters the Tên khóa học (Course Name) and Mô tả khóa học (Course Description). 4. Lecturer selects a Danh mục (Category) and Danh mục con (Sub-category). 5. Lecturer clicks the "Tạo khóa học" (Create) button. 6. System saves the data and redirects the Lecturer to the Course Editor's "Trang giới thiệu" tab.
Alternative Sequences/Flows	• A1. Missing Information: If the Lecturer attempts to proceed without filling mandatory fields, the System displays an error message next to the required fields. • A2. Cancel Creation: Lecturer clicks "X" on the modal. The process is aborted.
Exception	E1_System/Database Save Error • Trigger: Occurs at Normal Flow Step 5 (System attempts to save the new course). • Sequence: 1. System validates data successfully but fails to insert the record into the database (e.g., deadlock, storage full, service timeout). 2. System logs the critical error. 3. System displays an error message on the modal: "Không thể tạo khóa học lúc này. Vui lòng thử lại sau." (MSG-SYS-01). 4. Postcondition: User remains on the creation modal; Data entered is preserved so the user doesn't have to re-type it. E2_Network Connection Lost • Trigger: Occurs at Normal Flow Step 5. • Sequence: 1. Lecturer clicks "Tạo khóa học". 2. Client detects no internet connection or the request times out. 3. System displays a toast/alert: "Kết nối mạng không ổn định. Vui lòng kiểm tra lại đường truyền." (MSG-NET-01). 4. Postcondition: User remains on the creation modal. E3_Session Expired (Time-out)

	● Trigger: Occurs at Normal Flow Step 5. ● Sequence: 1. Lecturer spends a long time filling out the description. 2. The session/JWT token expires in the background. 3. Lecturer clicks "Tạo khóa học". 4. System rejects the request with a 401 Unauthorized error. 5. System displays a modal: "Phiên làm việc đã hết hạn. Vui lòng đăng nhập lại để tiếp tục." 6. Postcondition: User is redirected to the Login page. (Ideally, the system should autosave the draft to local storage to prevent data loss, but in a standard exception flow, the data is often lost).
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2.2.2 Update Course & Manage General Settings

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer updates the core metadata, image, price, and coupon information for an existing course. This Use Case covers the actions within the Course Editor sidebar tabs.		
Preconditions	● The course exists in the database. ● Lecturer is inside the Course Editor mode (e.g., by clicking the Edit icon)		
Postconditions	● Course details, price, and coupon lists are updated in the database.		
Normal Sequence/Flow	1. Lecturer views the Course Editor interface. 2. Lecturer navigates using the sidebar links: "Trang giới thiệu", "Giá khóa học", or "Phiếu giảm giá". 3. Lecturer modifies the fields on the active tab (e.g., updates the course description, sets a price value, or creates a new coupon). 4. The lecturer clicks the "Lưu" (Save) button (implied action for saving the form). 5. System confirms the update and ensures the completion status checkmarks are updated in the "Tổng quan tiến độ" box.		
Alternative Sequences/Flows	● A1. Unpublish Course: The Lecturer returns to the Dashboard and chooses to "Ẩn khóa học". If the course was published, the System confirms the action and sets the status to Unpublished. Students who previously bought the course retain access. ● A2. Permanent Delete (Draft only): If the course status is "Nháp" (Draft), the System allows permanent deletion upon confirmation.		
Exception	E1_Invalid Input Data (Validation Error) ● Trigger: Occurs at Normal Flow Step 4 (When clicking "Save"). ● Sequence: 1. System validates the input data on the server side.		

	2. System detects a violation of business rules (e.g., "Price" is negative, "Coupon Expiry Date" is in the past, or "Start Date" is after "End Date"). 3. System halts the save process. 4. System highlights the specific field causing the error. 5. System displays an error message: "Dữ liệu không hợp lệ. Vui lòng kiểm tra lại các trường được đánh dấu đỏ." (MSG-VAL-01). 6. Postcondition: User remains on the current tab; Data is not saved. E2_Media Upload Failure ● Trigger: Occurs at Normal Flow Step 3 (When uploading a Course Image/Thumbnail). ● Sequence: 1. Lecturer selects an image and attempts to upload. 2. System/Storage Service rejects the file (File size too large, unsupported format, or storage server timeout). 3. System displays an error message: "Tải ảnh thất bại. Vui lòng chọn ảnh .JPG/.PNG dưới 5MB." (MSG-ERR-UPLOAD). 4. System reverts the image preview to the previous state. 5. Postcondition: User remains on the "Trang giới thiệu" tab.
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2.2.3 Manage Sections

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer structures the course content by creating, modifying, and deleting sections (chapters) within the " Khung chương trình " tab.		
Preconditions	● Lecturer selects the "Khung chương trình" sidebar tab		
Postconditions	● The course structure is updated with new, modified, or deleted sections.		
Normal Sequence/Flow	1. Lecturer clicks the "+ Chương mới" (New Chapter) button. 2. The system displays the input form. 3. The lecturer inputs the mandatory "Tên chương" (Chapter Title). 4. (Optional) Lecturer inputs the "Mô tả chương". 5. The lecturer clicks "Lưu" (Save). 6. The new chapter is rendered as a collapsible item in the curriculum list.		
Alternative Sequences/Flows	● A1. Edit Section: Lecturer clicks the Pencil icon on a section row, modifies the title/description, and saves. ● A2. Delete Section: Lecturer clicks the Trash Bin icon. The system prompts a warning before deletion.		
Exception	E1_Validation Error (Missing/Invalid Title) ● Trigger: Occurs at Normal Flow Step 5 (Lecturer clicks "Save").		

	- Sequence: - System validates the input. - System detects that "Tên chương" is empty or exceeds the character limit (e.g., > 255 chars). - System displays an inline error message: "Tên chương là bắt buộc và không quá 255 ký tự." (MSG-VAL-02). - Postcondition: User remains in the input form mode; Section is not created.
--	--

2.2.4 Manage Lessons

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer populates sections with learning units, choosing between a standard Bài học (Video/Text) or a Quiz format, and managing supplementary material.		
Preconditions	- At least one Section exists. - The lecturer has clicked the option to add content inside a section.		
Postconditions	- A new Lesson or Quiz is added to the section. - Uploaded media (video, PDF) is stored.		
Normal Sequence/Flow	- Lecturer selects the "Bài học" card from the "Chọn loại nội dung" section. - Lecturer inputs "Tên bài học" and "Mô tả bài học". - The lecturer selects "Video" from the type dropdown. - The lecturer clicks "Chọn video" to upload the media source file. (Optional) Lecturer checks "Miễn phí" for a free preview lesson. (Optional) Lecturer uploads "Tài liệu PDF". - Lecturer clicks "Thêm bài học" (Add Lesson).		
Alternative Sequences/Flows	A1. Create Text Lesson: In Step 3, Lecturer selects "Văn bản" (Text). A Rich Text Editor is displayed for inputting documentation. Steps 5-7 are followed. A2. Create Quiz (Use case 2.4.1): In Step 1, Lecturer selects the "Quiz" card. The system redirects the user to the quiz creation interface. A3. Edit/Delete Lesson: Lecturer clicks the Pencil or Trash Bin icon on an existing lesson item.		
Exception	E1_Invalid File Format or Size (Upload Error) - Trigger: Occurs at Normal Flow Step 4 or 6 (User selects a file). - Sequence: - Lecturer selects a file that violates system rules (e.g., Video > 2GB, Format is .exe instead of .mp4, or PDF > 50MB). - System immediately validates the file metadata before uploading. - System rejects the file.		

	4. System displays an error message: "File không hợp lệ. Vui lòng chọn Video MP4 dưới 2GB hoặc PDF dưới 50MB." (MSG-ERR-FILE). 5. Postcondition: User remains on the form; No file is attached. E2_Upload Interrupted (Network Failure) ● Trigger: Occurs during Normal Flow Step 4 (While the progress bar is filling). ● Sequence: 1. The upload starts, but the internet connection is lost or unstable during the transfer. 2. The upload process stalls or fails. 3. System displays an error: "Quá trình tải lên bị gián đoạn. Vui lòng kiểm tra mạng và thử lại." (MSG-NET-UPLOAD). 4. System provides a "Resume" (Thử lại) button if supported, or resets the upload field. 5. Postcondition: User remains on the form. E3_Missing Core Content (Validation) ● Trigger: Occurs at Normal Flow Step 7 (Clicking "Add Lesson"). ● Sequence: 1. Lecturer clicks "Thêm bài học". 2. System detects that the "Lesson Type" is Video but no video file has been uploaded, or Type is Text but the editor is empty. 3. System blocks the creation. 4. System highlights the missing area. 5. System displays message: "Nội dung bài học không được để trống." (MSG-VAL-CONTENT). 6. Postcondition: User remains on the form.
--	---

2.2.5 Submit Course for Publish

Primary Actors	Lecturer	Secondary Actors	Content Moderator
Description	The Lecturer submits the finalized course for review by a Content Moderator once all core prerequisites are met.		
Preconditions	● The "Tổng quan tiến độ" (Progress Overview) must show 3/3 bước completed. ● All tabs must be marked green as completed.		
Postconditions	● Course Status changes from "Nháp" (Draft) to "Chờ duyệt" (Pending). ● The course is locked against major edits by the Lecturer.		
Normal Sequence/Flow	1. Lecturer verifies that the progress box confirms "Tất cả đã hoàn thành! Có thể đề xuất khóa học". 2. Lecturer clicks the active green button "Đề xuất khóa học" (Submit Course).		

	- The system performs final validation checks on course integrity. - The system updates the course status and displays a confirmation message.
Alternative Sequences/Flows	A1. Prerequisites Not Met: If any section of the course (Intro, Curriculum, or Price) is incomplete, the "Đề xuất khóa học" button remains disabled/grayed out, and the Lecturer cannot proceed.
Exception	E1_Backend Validation Failure (Data Sync Error) - Trigger: Occurs at Normal Flow Step 3 (System performs final validation). - Sequence: - The User Interface indicates "100% Complete," but the Backend detects missing critical data (e.g., a video file was deleted in another tab, or a text description is null). - System rejects the submission request. - System reloads the "Progress Overview" box to reflect the actual missing items (e.g., turning a green checkmark back to red). - System displays an error message: "Dữ liệu không đồng bộ. Vui lòng hoàn tất các mục còn thiếu được đánh dấu đỏ." (MSG-ERR-SYNC). - Postcondition: Course remains "Draft"; User stays on the page to fix errors. E2_Minimum Content Policy Violation - Trigger: Occurs at Normal Flow Step 3. - Sequence: - System checks the course against specific quality policies (e.g., Total video duration must be > 30 minutes, or Course must have at least 5 lessons). - System detects the course is too short or empty despite technically having sections. - System displays a policy warning: "Khóa học chưa đạt tiêu chuẩn tối thiểu (VD: Tổng thời lượng > 30 phút). Vui lòng bổ sung nội dung." (MSG-POL-01). - Postcondition: Course remains "Draft". E3_Account Restricted/Banned - Trigger: Occurs at Normal Flow Step 2. - Sequence: - Lecturer clicks "Submit". - System checks the Lecturer's current account status. - System detects the account has been flagged, banned, or suspended by an Admin. - System denies the action. - System displays a modal: "Tài khoản của bạn đang bị hạn chế quyền. Vui lòng liên hệ hỗ trợ." (MSG-AUTH-BAN).

	6. Postcondition: Course remains "Draft"; User might be logged out depending on severity. E4_System/Network Error ● Trigger: Occurs at Normal Flow Step 4. ● Sequence: 1. System attempts to update the status in the database but fails. 2. System displays message: "Không thể gửi yêu cầu lúc này. Vui lòng thử lại sau." (MSG-SYS-01). 3. Postcondition: Course remains "Draft" and unlocked.
--	--

2.2.6 Approve/Rejected Course as Moderator Content

Primary Actors	Content Moderator	Secondary Actors	Email Service
Description	Content Moderator manages courses in the system, including viewing course list, approved/rejected pending courses, and ban/unban published courses.		
Preconditions	● Content Moderator has logged into the system ● Content Moderator has "Content Moderator" permission (RoleId = 3) ● Email services are operational ● At least one course exists in the system (for viewing list)		
Postconditions	● Course status is updated (Pending → Public/Draft, or IsActive changed) ● Email notification is sent to lecturer (when approved/rejected) ● Course list is displayed with updated information		
Normal Sequence/Flow	Approve course: 1. Content Moderator navigates to "Quản lý Khóa học" page 2. System displays a list of courses filtered "Đã công khai" 3. Content Moderator select "Đang chờ duyệt" 4. System shows only courses with status "Đang chờ duyệt" (status="pending") 5. Content Moderator clicks "Duyệt" button 6. System shows approval confirmation dialog 7. Content moderator fills in "Duyệt khoá học" confirmation reason and click "Xác nhận duyệt" button 8. System validates: - Course exists - Course status is "Pending" - Moderator has permission (RoleId = 3) 9. System sends approval email to lecturer 10. System displays success message: "Khóa học đã được duyệt" 11. System refreshes course list		
Alternative Sequences/Flows	Rejected course: 1. Content Moderator navigates to "Quản lý Khóa học" page 2. System displays a list of courses filtered "Đã công khai"		

	3. Content Moderator select "Đang chờ duyệt" 4. System shows only courses with status "Đang chờ duyệt" (status="pending") 5. Content Moderator clicks "Từ chối" button 6. System shows approval confirmation dialog 7. Content moderator fills in "Từ chối khoá học" confirmation reason and click "Xác nhận từ chối" button 8. System validates: - Course exists - Course status is "Pending" - Moderator has permission (RoleId = 3) 9. System sends rejected email to lecturer 10. System displays success message: "Từ chối khóa học thành công" 11. System refreshes course list
Exception	E1_Email Service Failure ● Trigger: Occurs at Normal Flow Step 9 or Alternative Flow "Rejected" Step 9. ● Sequence: 1. The System successfully updates the course status in the database (to Public or Draft). 2. The System attempts to send the notification email to the Lecturer but fails (SMTP timeout, service down, or invalid email address). 3. System logs the email error. 4. System displays a warning message: "Trạng thái khóa học đã được cập nhật, nhưng không thể gửi email thông báo đến giảng viên." (MSG-WARN-EMAIL). 5. Postcondition: Course status is updated (Success), but the Moderator remains on the list view with the warning. E2_Course Status Conflict (Concurrency Error) ● Trigger: Occurs at Normal Flow Step 7 or Alternative Flow Step 7 (Clicking "Xác nhận"). ● Sequence: 1. Moderator A attempts to process the course. 2. System checks the current status of the course in the database. 3. System detects that the status is no longer "Pending" (e.g., The Lecturer cancelled the submission, or Moderator B already approved/rejected it 5 seconds ago). 4. System prevents the update. 5. System displays an error message: "Trạng thái khóa học không hợp lệ hoặc đã được xử lý bởi người khác." (MSG-ERR-STATUS). 6. System auto-refreshes the course list. 7. Postcondition: User returns to the updated Course List. E3_Missing Rejection Reason (Validation)

	● Trigger: Occurs at Alternative Flow "Rejected" Step 7. ● Sequence: 1. Moderator clicks "Xác nhận từ chối". 2. System validates that the "Lý do" (Reason) field is empty. (Rejection usually requires a reason, whereas Approval might not). 3. System displays an inline error: "Vui lòng nhập lý do từ chối để gửi cho giảng viên." (MSG-VAL-REASON). 4. Postcondition: User remains in the Dialog box. E4_Database Transaction Failure ● Trigger: Occurs at Normal Flow Step 9. ● Sequence: 1. System attempts to write the status change to the database but fails due to a connection error or deadlock. 2. System rolls back the transaction (status remains "Pending"). 3. System displays an error message: "Đã có lỗi hệ thống. Vui lòng thử lại sau." (MSG-SYS-01). 4. Postcondition: User remains in the Dialog box; Course status is unchanged.
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2.2.7 View detail course

Primary Actors	Student	Secondary Actors	None
Description	The "View Detail Course" use case allows a logged-in Student to view the full information of a specific course, including the curriculum, instructor details, ratings, and pricing, to decide whether to enroll or start learning.		
Preconditions	● The course must exist in the system database. ● The course status must be "Published" (Active). ● The user has navigated to the SkillUp website.		
Postconditions	● The course details are displayed successfully. The system identifies if the user has already enrolled to show the appropriate action button (Buy vs. Go to Course).		
Normal Sequence/Flow	1. The Student clicks on a course thumbnail or title from the Homepage or Course List. 2. The system verifies the Student's login session. 3. The system retrieves the course data from the database. 4. The system checks the Student's enrollment status for this specific course. 5. The system displays the Course Detail Page with:		

	● Main Content: Course Title, Description, Curriculum, Instructor, Reviews. ● Sidebar Actions: ○ Case A (Not Enrolled): Displays "Mua ngay" (Buy Now) and "Thêm vào giỏ" (Add to Cart). ○ Case B (Enrolled): Displays "Vào học" (Go to Course). 6. The Student views the course information.
Alternative Sequences/Flows	Unauthenticated Access (User Not Logged In): At step 1, if a user tries to access the Course Detail page without logging in: 1. The system detects no active session. 2. The system displays a message: "Vui lòng đăng nhập để xem chi tiết khóa học." 3. The system redirects the user to the Login Page. 4. After successful login, the system redirects the user back to the Course Detail page. Course Not Found: At step 3, if the course ID is invalid or the course is disabled: 1. The system displays an error message: "Khóa học không tồn tại." 2. The system redirects the Student to the Course List. Session Expired: During step 6, if the Student's session expires while viewing the page and they try to click "Buy Now" or "Go to Course": 1. The system displays a session timeout message. 2. The system redirects the Student to the Login page.
Exception	E1_System/Database Retrieval Error ● Trigger: Occurs at Normal Flow Step 3 (System retrieves course data). ● Sequence: 1. The system attempts to query the database for course details (Curriculum, Reviews, Instructor). 2. The request times out or encounters a server-side error (500 Internal Server Error). 3. System displays a generic error page: "Hệ thống đang bảo trì hoặc gặp sự cố. Vui lòng quay lại sau." (MSG-ERR-LOAD). 4. System provides a "Quay lại trang chủ" (Back to Home) button. 5. Postcondition: User is not on the Course Detail page; No data is shown. E2_Course Status Changed (Race Condition) ● Trigger: Occurs at Normal Flow Step 3. ● Sequence:

	1. User clicks the course card from the list (where it looked "Active"). 2. In the milliseconds between the list load and the click, an Admin/Lecturer changed the status to "Draft", "Hidden", or "Banned". 3. System retrieves the course but sees Status != Published. 4. System denies access. 5. System displays a modal/page: "Khóa học này hiện không khả dụng hoặc đã bị ẩn." (MSG-ERR-ACCESS). 6. System redirects to Course List. 7. Postcondition: User is returned to the Course List. E3_External Media Failure (Partial Load) ● Trigger: Occurs at Normal Flow Step 5 (Displaying content). ● Sequence: 1. The page text loads successfully, but the Promo Video (hosted on YouTube/Vimeo/Cloudinary) fails to load due to a third-party API error. 2. System detects the media failure. 3. System replaces the video player with a placeholder image or a "Video unavailable" icon. 4. System displays a toast notification: "Không thể tải video giới thiệu. Vui lòng kiểm tra kết nối mạng." (MSG-WARN-MEDIA). 5. Postcondition: The user can still view the text details and buy the course, but the rich media is missing (Graceful Degradation). E4_Enrollment Check Failure ● Trigger: Occurs at Normal Flow Step 4. ● Sequence: 1. Course data loads, but the service checking "Did this user buy this?" fails. 2. System defaults to the "Safe" state (Assuming Not Enrolled) to prevent unauthorized access. 3. System displays the "Mua ngay" button. 4. If the user (who actually owns the course) clicks "Mua ngay", the system catches the duplicate order at the Checkout stage (Next Use Case). 5. Postcondition: Page loads with "Buy" buttons.
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2.2.8 View course

Primary Actors	Student, Guest	Secondary Actors	None
Description	The "View Course List by Category" use case allows a Student or Guest to browse a specific category of courses (e.g., "Học IT"). Users can view sub-categories, filter courses by rating, sort results, and select a specific course to view its details.		
Preconditions	• The system database must contain active courses and categories. • The "Học IT" category (and others) must be defined in the system. • The user has accessed the SkillUp website.		
Postconditions	• A list of courses matching the selected category and filters is displayed. • The user can proceed to the "View Detail Course" use case by clicking a course card.		
Normal Sequence/Flow	1. The Student clicks on a Category tab (e.g., "Học IT") from the main navigation or Homepage. 2. The system retrieves the category information, available sub-categories (e.g., C#, C++), and the list of associated courses. 3. The system displays the Course List Page with: • Header: Category Title ("Học IT") and Description. • Sub-navigation: List of sub-categories (Lập trình game, Kinh tế, C#, C++). • Sidebar Filter: "Đánh giá" (Rating) with checkboxes for star ratings (3 to 5 stars). • Sorting Tool: A dropdown to sort courses (e.g., "Nổi bật nhất"). • Course Grid: A list of course cards showing the Thumbnail, Title, Instructor Name, Rating, and Price (or "Miễn phí"). 4. The Student views the list of courses. 5. Optional: The Student clicks on a specific sub-category (e.g., "C#"). • The system refreshes the grid to show only courses with the "C#" tag. 6. Optional: The Student selects a rating filter (e.g., "4 sao trở lên"). • The system filters the list to show only courses with an average rating of 4.0+. 7. The Student clicks on a specific course card. 8. The system redirects the Student to the Course Detail Page.		
Alternative Sequences/Flows	No Courses Found: • If the selected category or applied filter yields no results (e.g., selecting "C++" combined with "5 stars" where none exist): 1. The system displays an empty state message: "Không tìm thấy khóa học nào phù hợp." (No matching courses found).		

	2. The system allows the user to clear filters or select a different category. Pagination (Many Courses): • If the number of courses exceeds the page limit (e.g., more than 12 courses): 1. The system displays pagination controls at the bottom of the grid. 2. The Student clicks "Next" or a page number. 3. The system loads the next set of courses without reloading the entire page. Database Connection Error: • At step 2, if the system fails to retrieve course data: 1. The system displays an error message: "Đã xảy ra lỗi khi tải danh sách khóa học. Vui lòng thử lại sau." 2. The system provides a "Reload" button.
Exception	E1_Invalid Input Data (Validation Error) • Trigger: Occurs at Normal Flow Step 4 (When clicking "Save"). • Sequence: 1. System validates the input data on the server side. 2. System detects a violation of business rules (e.g., "Price" is negative, "Coupon Expiry Date" is in the past, or "Start Date" is after "End Date"). 3. System halts the save process. 4. System highlights the specific field causing the error. 5. System displays an error message: "Dữ liệu không hợp lệ. Vui lòng kiểm tra lại các trường được đánh dấu đỏ." (MSG-VAL-01). 6. Postcondition: User remains on the current tab; Data is not saved. E2_Media Upload Failure • Trigger: Occurs at Normal Flow Step 3 (When uploading a Course Image/Thumbnail). • Sequence: 1. Lecturer selects an image and attempts to upload. 2. System/Storage Service rejects the file (File size too large, unsupported format, or storage server timeout). 3. System displays an error message: "Tải ảnh thất bại. Vui lòng chọn ảnh .JPG/.PNG dưới 5MB." (MSG-ERR-UPLOAD). 4. System reverts the image preview to the previous state.

	5. Postcondition: User remains on the "Trang giới thiệu" tab.
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2.2.9 View my course

Primary Actors	Student	Secondary Actors	None
Description	The "View My Courses" use case allows a logged-in Student to access their personal learning dashboard. Here, they can track their learning statistics (total courses, hours spent), search through their enrolled courses, resume learning, and view recent activities or learning tips.		
Preconditions	- The User must be logged in as a Student. - The User navigates to the "Khóa học của tôi".		
Postconditions	The dashboard is displayed with up-to-date statistics and course list. The student can navigate to a specific course or return to the main control panel.		
Normal Sequence/Flow	- The Student accesses the My Courses page. - The system retrieves the Student's enrollment data and learning history from the database. - The system calculates and displays Learning Statistics: - Total enrolled courses (Khóa học đã đăng ký). - Active courses (Đang học). - Completed courses (Đã hoàn thành). - Total learning hours (Tổng giờ học). - The system displays the Course List section with a search bar. - The system displays the Sidebar Widgets: "Hoạt động gần đây" (Recent Activity) and "Mẹo học tập" (Learning Tips). - The Student views their progress or searches for a specific course to continue learning.		
Alternative Sequences/Flows	No Courses Found: At step 3 or after filtering, if no courses match the criteria: - The system displays an empty state message: "Không tìm thấy khóa học nào phù hợp." - The system keeps the structure visible so the user can clear filters or choose another category. Database Connection Error: At step 2, if the system cannot retrieve course data: - The system displays an error message: "Đã xảy ra lỗi khi tải danh sách. Vui lòng thử lại sau."		

Exception	E1_System/Database Retrieval Error (Global Failure) ● Trigger: Occurs at Normal Flow Step 2 (System attempts to retrieve enrollment and history data). ● Sequence: 1. The system attempts to query the database but fails due to a connection timeout or server error (500). 2. System logs the error. 3. System displays a "Global Error" state: "Đã xảy ra lỗi khi tải dữ liệu. Vui lòng tải lại trang." (MSG-ERR-LOAD). 4. System provides a "Tải lại" (Reload) button. 5. Postcondition: User remains on the My Courses page (empty state); No data is displayed. E2_Session Expired (Unauthorized Access) ● Trigger: Occurs at Normal Flow Step 2. ● Sequence: 1. System validates the user's session token while fetching data. 2. System detects the token is expired or invalid. 3. System interrupts the data loading. 4. System displays a modal: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-AUTH-01). 5. Postcondition: System redirects the user to the Login page.
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2.3 Lesson Feature

2.3.1 View lesson

Primary Actors	Student	Secondary Actors	None
Description	The "View Lesson Content" use case allows an enrolled Student to view the specific content of a lesson (Video, Article, or Quiz). The student can read or watch the material, mark the lesson as completed, and navigate to the next or previous lesson.		
Preconditions	● The Student is logged in and enrolled in the course. ● The Student has navigated to the Learning Interface.		
Postconditions	● The lesson content is displayed. If the student clicks "Mark as completed", their progress is updated in the database.		

Normal Sequence/Flow	1. The Student clicks on a specific lesson from the Sidebar Curriculum List or clicks "Start Learning" from the dashboard. 2. The system retrieves the lesson details (Title, Type, Content URL/Body). 3. The system displays the Learning Screen with: ● Sidebar: List of lessons in the current chapter, highlighting the active lesson. ● Main Content: The video player (if video) or text content (if article). ● Action Area: "Mark as completed" button and Navigation buttons (Previous/Next). 4. The Student consumes the content (reads text or watches video). 5. The Student clicks "Đánh dấu hoàn thành" (Mark as completed). 6. The system updates the progress status to 100% for this lesson. 7. The Student clicks "Tiếp theo" (Next) to move to the next lesson.
Alternative Sequences/Flows	Navigate via Sidebar: At step 4, the Student clicks a different lesson in the left sidebar: 1. The system immediately loads the selected lesson content. Lesson Locked (Sequential Learning): If the course requires sequential learning and the previous lesson is not done: 1. The system keeps the next lesson locked (grayed out). 2. The Student cannot access it until the current lesson is marked complete.
Exception	E1_Media Resource Unavailable (404/Broken Link) ● Trigger: Occurs at Normal Flow Step 2 or 3 (System attempts to load the video URL). ● Sequence: 1. System retrieves the lesson metadata, but the video hosting service (e.g., YouTube/Vimeo) returns an error (Video deleted, Private video, or Service Down). 2. System loads the page structure but replaces the video player with an error placeholder. 3. System displays a message: "Video bài học không khả dụng. Vui lòng báo cáo với hỗ trợ viên." (MSG-ERR-MEDIA). 4. Postcondition: User can read the description/text but cannot watch the video. E2_Progress Save Failure (Network/Sync Error) ● Trigger: Occurs at Normal Flow Step 5 (Student clicks "Mark as completed").

	● Sequence: 1. Student clicks the button. 2. Client attempts to send the API request to update progress. 3. Request fails due to internet disconnection or server timeout. 4. System displays a toast notification: "Không thể lưu tiến độ. Vui lòng kiểm tra kết nối mạng và thử lại." (MSG-ERR-SAVE). 5. The button state reverts to "Mark as completed" (not checked). 6. Postcondition: Lesson remains "Incomplete" in the database. E3_Access Denied (Direct URL Access) ● Trigger: Occurs at Normal Flow Step 2. ● Sequence: 1. Student attempts to access a specific lesson ID directly via the browser URL (bypassing the UI). 2. System checks permissions. 3. System detects that the user is Enrolled but has not completed the prerequisite previous lesson (in a Sequential Course). 4. System redirects the user to the latest unlocked lesson. 5. System displays a warning: "Bạn cần hoàn thành bài học trước để truy cập nội dung này." (MSG-LOCK-01). 6. Postcondition: User is redirected to the correct safe lesson. E4_Concurrent Deletion (Lesson Not Found) ● Trigger: Occurs at Normal Flow Step 1 or 7. ● Sequence: 1. Student clicks "Next Lesson". 2. In the time between the student loading the course and clicking next, the Lecturer deleted that specific lesson. 3. System returns a "404 Not Found" error. 4. System displays a modal: "Bài học này không còn tồn tại." 5. System refreshes the Curriculum Sidebar to show the updated structure. 6. Postcondition: User remains on the current lesson or is moved to the Section view.
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2.4 Quiz Feature

2.4.1 Take quiz

Primary Actors	Student	Secondary Actors	None
Description	The "Take Quiz" use case allows an enrolled Student to attempt a quiz assigned in a course. The student can view questions, select answers, navigate between questions, save progress, and submit the quiz for grading.		
Preconditions	• The Student is logged in and enrolled in the course. • The quiz is available (within the allowed time frame or attempts). • The Student has clicked "Start Quiz" (Làm bài).		
Postconditions	• User answers are saved to the database. The system calculates the score (for auto-graded quizzes) and updates the student's progress.		
Normal Sequence/Flow	1. The Student initiates the quiz. 2. The system retrieves the quiz questions and configuration (time limit, passing grade). 3. The system displays the Quiz Interface containing: • Left Sidebar: Question navigation grid (Question Map), Timer (if applicable), and Action buttons (Submit, Save). • Main Content: The current question text and answer options. 4. The Student reads the question (e.g., "API là viết tắt của gì?"). 5. The Student selects one or more options (e.g., "Application Program Interface"). 6. The Student clicks "Câu sau" (Next) to move to the next question. 7. The Student repeats steps 4-6 for all questions. 8. The Student clicks "Nộp bài" (Submit). 9. The system asks for confirmation. 10. The Student confirms. 11. The system grades the quiz and displays the Result Screen.		
Alternative Sequences/Flows	Jump to Question: At step 6, instead of clicking "Next", the Student clicks a specific number (e.g., "5") in the Sidebar Question Grid. 1. The system immediately displays Question 5. Save Draft (Lưu bài làm): During the attempt, the Student clicks "Lưu bài làm". 1. The system saves the current selected answers without submitting/grading. 2. The Student can resume the quiz later (if the time limit allows). Time Limit Exceeded: If the quiz has a timer and it reaches 00:00: 1. The system automatically submits the current answers.		

	2. The system redirects the Student to the Result Screen.
Exception	E1_Submission Network Failure (Data Rescue) ● Trigger: Occurs at Normal Flow Step 9 (Student confirms submission). ● Sequence: 1. Student clicks "Submit". 2. Client attempts to send the answer payload to the server. 3. Connection fails (Internet lost or Server timeout). 4. System displays a blocking alert: "Mất kết nối mạng. Không thể nộp bài!" (MSG-NET-SUBMIT). 5. System caches the final answers locally (in LocalStorage/SessionStorage) to prevent data loss. 6. System changes the "Submit" button to "Retry" (Thử lại). 7. Postcondition: User remains on the Quiz interface with all answers preserved; Quiz is not submitted yet.

2.4.2 View detail result

Primary Actors	Student	Secondary Actors	None
Description	The "View Quiz Result Detail" use case allows a Student to review their performance after submitting a quiz. The system displays an overall summary (score, pass/fail status) and a detailed breakdown of each question, showing the user's selected answers versus the correct answers with specific feedback.		
Preconditions	● The Student has submitted a quiz. ● The quiz grading process is complete. ● The Student is on the Quiz Result page.		
Postconditions	● The student understands their score and which questions they answered correctly or incorrectly.		
Normal Sequence/Flow	1. The system calculates the score immediately after submission. 2. The system displays the Quiz Result Screen containing two main areas: ● Left Sidebar (Summary): ○ Overall Score Percentage (e.g., 22%). ○ Pass/Fail Status (e.g., "Chưa vượt qua" - Not Passed). ○ Statistics: Correct answers, Wrong answers, Total questions. ○ Question Map: A grid of numbers colored by result (Green = Correct, Red = Wrong). ● Right Panel (Detail View): ○ Displays the first question (or selected question) details.		

	○ Shows the Question Text. ○ Shows Answer Options with visual feedback (Correct, Wrong, Missed). 3. The Student reviews the summary statistics. 4. The Student clicks on a specific question number in the Question Map or uses "Câu sau" (Next) buttons. 5. The system updates the Right Panel to show the selected question. 6. The Student reviews the explanation for the answer. 7. The Student clicks "Đóng" (Close) or "Làm lại" (Retry) (if attempts allowed).
Alternative Sequences/Flows	Retry Quiz: If the quiz allows multiple attempts: 1. The Student clicks the "Làm lại" (Retry) button (if available). 2. The system resets the attempt and redirects to the Take Quiz screen.
Exception	● E1_Session Expired Trigger: Occurs at Precondition validation (Student's session expires). Sequence: 1. Student's session expires (JWT token expired). 2. Student attempts to access Quiz Result page. 3. System validates session token. 4. System detects expired authentication token. 5. System prevents page access. 6. System displays toast: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-SESSION-EXPIRED). 7. System redirects to login page. 8. After re-authentication, the system redirects back to Quiz Result page (if quiz ID preserved). Postcondition: Result NOT displayed; Student must re-authenticate; Redirected to login; Returns to results after login.

2.4.3 Add question to quiz

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturers to add questions to a specific quiz to build the content. Lecturers can either create new questions manually or select existing questions from the Question Bank.		
Preconditions	• The user must be logged in as a Lecturer. • A Quiz must already exist within a Section. • The Lecturer is viewing the details/content of that specific Quiz.		
Postconditions	• New questions are successfully linked to the Quiz. • The total score/number of questions of the Quiz is updated.		
Normal Sequence/Flow	1. The Lecturer navigates to the “Khung chương trình” page of a specific course. 2. The Lecturer clicks on an existing Quiz item to view details. 3. The System displays the Quiz interface with a list of current questions (if any). 4. The Lecturer clicks the “Thêm câu hỏi” button. 5. The system transfers to the Question Bank interface. 6. The lecturer searches and selects questions to add. 7. The lecturer clicks the “Thêm câu hỏi” button to confirm. 8. The system links the selected questions to the Quiz, returns to the "Khung chương trình" page and notifies that the question has been successfully added.		
Alternative Sequences/Flows	A1: Create new question : ○ At step 5 (Question Bank interface), the Lecturer clicks the "Tạo mới" button. ○ The System displays the "Tạo câu hỏi" form including: ■ Question Type (Single/Multiple Choice). ■ Question Text/Content. ■ Answers/Options (and mark the correct one). ○ The Lecturer fills in the required information and clicks "Tạo câu hỏi". ○ The System validates the data, creates the new question, adds it to the Question Bank, and links it to the current Quiz. ○ The flow merges back to Step 8.		
Exception	E1: Validation error ○ In A1, if the Lecturer leaves mandatory fields blank (e.g., no question text, no correct answer selected) ○ The System display the error and prevents saving.		

2.5 Lecturer Application Feature

2.5.1 Create Lecturer Application

Primary Actors	Lecturer (New/Pending Approval)	Secondary Actors	moderator system
Description	The "Submit Lecturer Application" use case allows a newly registered Lecturer to complete their profile by submitting professional details (experience, CV, degrees). This step is mandatory to activate the teaching privileges (create courses).		
Preconditions	- The User has registered a Lecturer account successfully. - The User is logged into the Lecturer Dashboard. - The User's account status is "Pending" or "Not Verified".		
Postconditions	The application data (CV, Description) is saved. The application status updates to "Đang chờ duyệt" (Pending Approval). The User sees the submission confirmation screen.		
Normal Sequence/Flow	- Registration Phase: - Guest selects "Lecturer" tab on Register screen . - Guest fills basic info (Email, Name, Password) and clicks Register. - System creates account with Role = Lecturer. - Application Phase: - User logs in and is directed to the Lecturer Dashboard. - User navigates to the "Giảng dạy" (Teaching) or "Đơn ứng tuyển" (Applications) section. - System displays the Application Form. - User fills in "Mô tả / Kinh nghiệm" (Description/Experience). - User uploads "Ảnh bằng cấp / CV" (Credentials). - User clicks "Nộp đơn" (Submit). - Completion Phase: - System validates files and text content. - System saves the application record. - System displays the Success Screen With message "Đơn ứng tuyển đã được gửi!".		
Alternative Sequences/Flows	Re-submission (If Rejected): If a previous application was rejected: - User views the rejection reason on the dashboard. - User clicks "Nộp đơn mới" (Submit New) or "Chỉnh sửa" (Edit). - User updates the info and resubmits.		
Exception	E1_Unauthorized Access (Non-Lecturer Role)		

	● Trigger: Occurs at Precondition validation (User attempts to access application form). ● Sequence: 1. User (Student, Moderator, or Admin) attempts to navigate to lecturer application submission page. 2. System validates user role. 3. System detects that logged-in user does not have 'Lecturer' role. 4. System prevents access to application form. 5. System displays error message: "Bạn không có quyền truy cập trang này. Chỉ tài khoản giảng viên mới có thể nộp đơn ứng tuyển." (MSG-UNAUTHORIZED-ACCESS). 6. System redirects user to appropriate dashboard. 7. System may log unauthorized access attempt. Postcondition: Unauthorized user cannot access application form; User redirected; Access attempt logged; Security maintained. E2_Missing_Required_Files ● Trigger: User does not upload CV or degree files ● System: Display error "Vui lòng tải Cv lên" or "Vui lòng tải bằng cấp lên" ● Result: Application submission blocked E3_Invalid_File_Format ● Trigger: CV is not PDF or degree is not image (JPG/PNG) ● System: Display "CV chỉ chấp nhận dạng PDF" or "Chỉ ảnh JPG / PNG được chấp nhận" ● Result: File rejected, must reselect E4_File_Too_Large ● Trigger: File size > 5MB ● System: Display "File không vượt quá 5 MB" ● Result: File rejected E5_Already_Has_Pending_Application ● Trigger: User already has application with Status = "Pending" ● System: Block new submission, redirect to view existing application ● Result: Must wait for current application to be approved/rejected E6_Unauthenticated ● Trigger: Token expired or not logged in ● System: Return 401 "Unauthorized" ● Result: Must login again E7_Upload_Service_Failed ● Trigger: Cloudinary error (network issue, quota exceeded, server down) ● System: Return 500 Internal Server Error ● Result: Application not saved, retry required E8_Database_Error ● Trigger: Error during SaveChanges() to database ● System: Return "Không thể nộp CV , làm ơn thử lại" ● Result: Application not saved, must resubmit
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2.5.2 View detail Lecturer Application

Primary Actors	Lecturer	Secondary Actors	none
Description	The "View My Application Detail" use case allows a Lecturer (who has submitted a registration form) to review the details of their application. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and access the "Edit" function if the application is still pending or was rejected.		
Preconditions	• The User is logged in as a Lecturer. • The User has already submitted an application. • The User navigates to the "Teaching" or "My Application" section in the Dashboard.		
Postconditions	• The application details and current status are displayed. The User knows whether they need to update information or wait for approval.		
Normal Sequence/Flow	1. The User accesses the Lecturer Dashboard. 2. The system retrieves the latest application record for this user. 3. The system displays the Application Detail Screen containing: • Status Badge: e.g., "Đang chờ duyệt" (Pending) or "Từ chối" (Rejected). • Description: The text previously entered by the user. • Attachments: List of uploaded images/files (CV, Degrees). • Actions: "Chỉnh sửa" (Edit) button (if applicable). 4. The User reviews the information to ensure accuracy. 5. <i>Optional</i>: The User clicks on an attachment to preview it. 6. <i>Optional</i>: If the status allows, the User clicks "Chỉnh sửa" to update details.		
Alternative Sequences/Flows	Application Rejected: At step 3, if the status is "Rejected": 1. The system displays the status in Red. 2. The system displays the Rejection Reason provided by the Admin (e.g., "Ảnh bằng cấp bị mờ"). 3. The system enables the "Chỉnh sửa" or "Nộp đơn mới" button. Application Approved: At step 3, if the status is "Approved": 1. The system displays the status in Green. 2. The fields become Read-only (cannot be edited). 3. The system displays a message: "Bạn đã trở thành giảng viên chính thức."		

Exception	E1_No Application Found ● Trigger: Occurs at Normal Flow Step 2 (System retrieves latest application record). ● Sequence: 1. Lecturer navigates to "My Application" section in Dashboard. 2. System attempts to retrieve application record for logged-in user. 3. System detects that no application exists (user never submitted application). 4. System displays empty state screen: a. Empty state illustration (e.g., document icon) b. Message: "Bạn chưa nộp đơn ứng tuyển nào." (MSG-NO-APPLICATION) c. Subtitle: "Hãy nộp đơn ứng tuyển để trở thành giảng viên chính thức." d. Call-to-action button: "Nộp đơn ứng tuyển" (Submit Application) 5. Application detail view not displayed. Postcondition: Empty state displayed; User informed no application exists; Clear guidance to submit new application; Good onboarding experience. E2_Unauthorized_Access ● Trigger: User not logged in or token expired ● Sequence: 1. User attempts to access application details 2. System checks userId from token 3. System detects userId == null 4. System returns Code 401 - "Unauthorized" ● Postcondition: Access denied, redirect to login required E3_Non_Lecturer_Role ● Trigger: Logged-in user does not have Lecturer role (RoleId ≠ 4) ● Sequence: 1. User navigates to application details page 2. System validates account.RoleId 3. System detects RoleId ≠ 4 4. System throws UnauthorizedAccessException: "Bạn không có quyền truy cập." Postcondition: Access denied, error displayed
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	E4_Application_Not_Found_By_ID ● Trigger: Specific application ID does not exist in database ● Sequence: 1. User clicks on specific application link 2. System queries GetApplicationByIdAsync(applicationId) 3. System returns null (application not found) 4. System returns Code 404 - "Không tìm thấy CV" Postcondition: Detail view not displayed, "Not found" message shown E5_Server_Error Trigger: Database error, network failure, or unexpected exception Sequence: 1. User requests application details 2. System encounters exception during retrieval 3. System catches Exception 4. System returns Code 500 + exception message Postcondition: Error screen displayed, user prompted to retry
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2.5.3 Manage Lecturer Application

Primary Actors	System Moderator	Secondary Actors	none
Description	The "Manage Lecturer Application" use case allows a System Moderator to review the details of lecturer applications. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and choose to reject or accept the application.		
Preconditions	● The moderator logged into the system with an account. ● The account Role must be "System Moderator". ● The account status must be "Active". ● The Moderator clicked on the "Quản lý ứng tuyển" tab on the side menu and navigate to the "Quản lý đơn ứng tuyển của giảng viên" page.		

Postconditions	● View Application List: The system successfully displays the list of all Lecturer's applications. ● Approve Application: The application status is updated successfully to "Đã chấp nhận", and an email is sent afterward. ● Reject Application: The application status is updated successfully to "Đã từ chối", and an email is sent afterward.
Normal Sequence/Flow	View Application List: 1. The moderator accesses the "Quản lý đơn ứng tuyển của giảng viên" page. 2. The system successfully display the list of all Lecturer's applications, consist of following informations: ● Họ tên (Full name) ● Email ● Tiêu đề (Title) ● Chức danh (Profession) ● Mô tả (Short Description) ● Trạng thái (Status) ● Ngày tạo (Date of Create) ● CVs and Degrees buttons. ● Thao tác (Action: Accept or Reject)

Alternative Sequences/Flows	A1: Approve Application: ○ The moderator accesses the “Quản lý đơn ứng tuyển của giảng viên” page. ○ The system successfully display the list of all Lecturer’s applications, consist of following informations: ■ Họ tên (Full name) ■ Email ■ Tiêu đề (Title) ■ Chức danh (Profession) ■ Mô tả (Short Description) ■ Trạng thái (Status) ■ Ngày tạo (Date of Create) ■ CVs and Degrees buttons. ■ Thao tác (Action: Accept or Reject) ○ The moderator clicks the “Duyệt” button on the Action column to accept the application. ○ The system updates the application status to “Đã chấp nhận”, and sends an email to notify the lecturer of approval. A2: Reject Application: ○ The moderator accesses the “Quản lý đơn ứng tuyển của giảng viên” page. ○ The system successfully display the list of all Lecturer’s applications, consist of following informations: ■ Họ tên (Full name) ■ Email ■ Tiêu đề (Title) ■ Chức danh (Profession) ■ Mô tả (Short Description) ■ Trạng thái (Status) ■ Ngày tạo (Date of Create) ■ CVs and Degrees buttons. ■ Thao tác (Action: Accept or Reject) ○ The moderator clicks the “Từ chối” button on the Action column to reject the application. ○ The system updates the application status to “Đã từ chối”, and sends an email to notify the lecturer of disapproval.
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Exception	E1_No Application Found ● Trigger: Occurs at Normal Flow Step 2 (System retrieves latest application record). ● Sequence: 1. Lecturer navigates to "My Application" section in Dashboard. 2. System attempts to retrieve application record for logged-in user. 3. System detects that no application exists (user never submitted application). 4. System displays empty state screen: a. Empty state illustration (e.g., document icon) b. Message: "Bạn chưa nộp đơn ứng tuyển nào." (MSG-NO-APPLICATION) c. Subtitle: "Hãy nộp đơn ứng tuyển để trở thành giảng viên chính thức." d. Call-to-action button: "Nộp đơn ứng tuyển" (Submit Application) 5. Application detail view not displayed. Postcondition: Empty state displayed; User informed no application exists; Clear guidance to submit new application; Good onboarding experience. E2_Unauthorized_Access ● Trigger: Moderator not logged in or token expired ● Sequence: 1. User attempts to access application management page 2. System validates token/userId 3. System detects userId == null 4. System returns Code 401 - "Unauthorized" Postcondition: Access denied, redirect to login E3_Non_Moderator_Role ● Trigger: Logged-in user is not System Moderator (RoleId ≠ 2) ● Sequence: 1. User attempts to view or update applications 2. System validates _currentUserService.RoleId 3. System detects RoleId ≠ 2 4. System returns Code 400 - "Không thể cập nhật trạng thái ứng tuyển! Kiểm tra quyền truy cập."
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Postcondition: Access denied, moderator privileges required

E4_Application_Not_Found_For_Update

- Trigger: Moderator tries to approve/reject non-existent application
- Sequence:
 1. Moderator clicks "Duyệt" or "Từ chối" button
 2. System queries GetByIdAsync(applicationId)
 3. System returns null
 4. System returns Code 400 - "Không thể cập nhật trạng thái ứng tuyển! Kiểm tra quyền truy cập."

Postcondition: Status not updated, application not found

E5_Status_Update_Failed

- Trigger: Database fails to update application status
- Sequence:
 1. Moderator submits approve/reject action
 2. System calls UpdateStatusAsync()
 3. System encounters database error or SaveChanges() returns false
 4. System returns Code 400 - "Không thể cập nhật trạng thái ứng tuyển! Kiểm tra quyền truy cập."

Postcondition: Status unchanged, error displayed

E6_Email_Notification_Failed

- Trigger: Email service fails after successful status update
- Sequence:
 1. Application status updated successfully
 2. System attempts SendLecturerApplicationStatusEmailAsync()
 3. Email service throws exception
 4. System throws Exception: "Lỗi cập nhật trạng thái đơn ứng tuyển giảng viên !"
 5. System returns Code 500

Postcondition: Status updated in DB but email not sent, error logged

E7_Server_Error

- Trigger: Unexpected system error during retrieval or update
- Sequence:
 1. Moderator performs any action
 2. System encounters unexpected exception

	3. System catches Exception 4. System returns Code 500 + exception message Postcondition: Operation failed, user prompted to retry
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2.6 Shopping cart & payment feature

2.6.1 Buy course

Primary Actors	Student	Secondary Actors	none
Description	The "Buy Course" use case allows a logged-in Student to purchase courses. There are two entry points: Direct Purchase (Mua ngay): Buying a single course directly from the detail page. Cart Checkout (Thanh toán giỏ hàng): Buying multiple courses saved in the cart. Upon success, the system enrolls the student in the course(s) and records the transaction.		
Preconditions	• The Student is logged in. • Case A (Buy Now): Student is on Course Detail page, course is not enrolled.		

	● Case B (Cart): Student is on Cart page, cart is not empty.
Postconditions	● Transaction status is "Success". Student is enrolled in the purchased course(s). If coming from Cart, the purchased items are removed from the Cart. System displays the Payment Success screen.
Normal Sequence/Flow	1. Entry Points: ● Case A: Student clicks "Mua ngay" on Course Detail page. ● Case B: Student clicks "Tiến hành thanh toán" on Cart page. 2. The system displays the Checkout Screen (image_c361d7.jpg) showing: ● List of course(s) to be purchased. ● Total amount (Tổng thanh toán). ● Payment method selection (PayOS/QR). 3. The Student selects the Payment Method (e.g., PayOS). 4. The Student clicks the "Thanh toán" (Pay) button. 5. The system redirects User to the Payment Gateway (or displays QR Code). 6. The Student completes the payment via Banking App. 7. The Payment Gateway returns the transaction status to the system. 8. The system validates the payment and updates the database (Enroll user, create invoice). 9. The system displays the Payment Success Screen. 10. The Student clicks "Vào học ngay" or "Về trang chủ".
Alternative Sequences/Flows	Payment Failed / Cancelled: At step 7, if the payment fails or user cancels: 1. The system displays a Payment Failed Screen. 2. Message: "Thanh toán thất bại hoặc đã bị hủy." 3. Action: "Thử lại" (Retry) or "Quay lại giỏ hàng". Voucher Application (Optional): At step 2, the User may enter a Voucher code (if not applied in Cart). 1. System validates and updates the "Total Amount".
Exception	E1: Authentication Error ● Trigger: In Normal flow, if the user is not logged into the system with an account. ● Sequence: 1. The user is not logged into the system 2. The user adds items to the cart. 3. The user proceeds to checkout. ● System: Displays a toast message informing the user to login. ● Result: The user is redirected to the login page. E2: Validation Error

	● Trigger: In Normal flow, if the user already owns one or more courses in the cart. ● Sequence: 1. The user is logged into the system 2. The user adds items including owned courses to the cart. 3. The user proceeds to checkout. ● System: Displays a toast message “Bạn đã sở hữu khoá học này rồi”. ● Result: The user is notified of an owned course in the cart. E3: Payment Gateway Error ● Trigger: In Normal flow, if the payment gateway encounters an unexpected error. ● Sequence: 1. The user is logged into the system. 2. The user adds items to the cart. 3. The user proceeds to checkout. 4. The 3rd party payment gateway fails unexpectedly. ● System: Displays a toast message “Có lỗi xảy ra. Vui lòng thử lại sau”. ● Result: The user is notified of a payment gateway error.
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2.7 Profile Management Feature

2.7.1 View Profile

Primary Actors	Lecturer	Secondary Actors	none
Description	The "View Lecturer Profile" use case allows a logged-in Lecturer to view their account details. This includes Public Information (visible to students like Title, Profession, Bio) and Private Information (Financial details for payouts like Bank Account).		
Preconditions	● The User is logged in as a Lecturer. ● The User has navigated to the "Profile" or "Account Settings" page		
Postconditions	● The Lecturer's full profile information is displayed. Sensitive financial data is displayed (possibly masked for security).		

Normal Sequence/Flow	1. The Lecturer clicks on their Avatar or "Tài khoản" in the navigation menu. 2. The system retrieves the user's basic info (from User table) and extended info (from Lecturer table). 3. The system displays the Profile Screen divided into sections: ● Basic Info: Avatar, Name, Email. ● Professional Info: Title (Học hàm/học vị), Profession (Nghề nghiệp), Bio. ● Financial Info: Bank Name, Bank Number, Receiver Name. 4. The Lecturer reviews the information. 5. Optional: The Lecturer clicks "Chỉnh sửa" (Edit) to update information.
Alternative Sequences/Flows	Missing Financial Info: At step 3, if the Lecturer hasn't set up payment details yet: 1. The system displays a warning or prompt in the Financial section: "Vui lòng cập nhật thông tin ngân hàng để nhận thanh toán."
Exception	E1_Session Expired ○ Trigger: Occurs at Normal Flow Step 1 (User navigates to Profile page). ○ Sequence: ■ Lecturer's session expires (e.g., JWT token expired). ■ The lecturer attempts to access the Profile page. ■ System validates session token. ■ System detects expired authentication tokens. ■ System prevents profile page access. ■ System displays toast message: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-SESSION-EXPIRED). ■ System redirects Lecturer to login page. ■ After re-authentication, system redirects back to Profile page. Postcondition: Profile not displayed; Lecturer must re-authenticate; Redirected to login; Returns to profile after login.

2.7.2 Update Profile

Primary Actors	Lecturer	Secondary Actors	none
Description	The "Update Lecturer Profile" use case allows a logged-in Lecturer to modify their personal, professional, and financial details. Note that profile picture updates are not supported in this function; users can only update text-based information.		
Preconditions	• The User is logged in as a Lecturer. • The User is viewing the Profile screen. • The User clicks the "Chỉnh sửa" (Edit) button.		
Postconditions	• The Lecturer's text information is updated in the database. The User interface reflects the new changes immediately.		
Normal Sequence/Flow	1. The Lecturer clicks "Chỉnh sửa hồ sơ" on the Profile page. 2. The system displays the Edit Profile Form with current data pre-filled (Avatar is read-only or hidden). 3. The Lecturer modifies the allowed fields: • Updates Contact Info (Full Name, Phone). • Updates Professional Info (Title, Profession, Bio). • Updates Financial Info (Bank Name, Number, Receiver). 4. The Lecturer clicks the "Lưu thay đổi" (Save Changes) button. 5. The system validates the input (e.g., checks required fields, validates phone/bank number format). 6. The system updates the record in the database. 7. The system displays a success notification: "Cập nhật hồ sơ thành công." 8. The system redirects back to the View Profile mode.		
Alternative Sequences/Flows	• Validation Error: At step 5, if the Lecturer enters an invalid phone number (e.g., contains letters): ○ The system highlights the "Phone" field. ○ The system displays error: "Số điện thoại không hợp lệ." ○ The update is blocked until corrected.		
Exception	E1_Validation Error (Invalid Phone Number Format) ○ Trigger: Occurs at Normal Flow Step 5 (System validates input after Lecturer clicks "Lưu thay đổi"). ○ Sequence: ■ System validates phone number field. ■ System detects that phone number contains: ○ Non-numeric characters (letters, special characters)		

	○ Invalid format (not matching Vietnamese phone pattern) ○ Incorrect length (not 10 digits) ○ System prevents profile update. ○ System highlights "Phone" field in red. ○ System displays inline error message: "Số điện thoại không hợp lệ. Vui lòng nhập 10 chữ số." (MSG-VAL-INVALID-PHONE). Postcondition: Profile not updated; Lecturer remains on edit form to correct phone number; No database changes made. E2_Validation Error (Missing Required Field - Full Name) ○ Trigger: Occurs at Normal Flow Step 5 (System validates input after Lecturer clicks "Lưu thay đổi"). ○ Sequence: ○ System validates required fields. ○ System detects that "Full Name" field is empty or contains only whitespace. ○ System prevents profile update. ○ System highlights "Full Name" field in red. ○ System displays inline error message: "Họ và tên là bắt buộc." (MSG-VAL-NAME-REQUIRED). Postcondition: Profile not updated; Lecturer remains on edit form to enter full name; No database changes made.
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2.8 Question Feature

2.8.1 Manage Question

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturer to view a list of existing questions in a Quiz, make edits to question content, or delete questions from the Quiz.		
Preconditions	● The Lecturer must be logged in to the system. ● Quiz has been created. ● The instructor is on the Quiz detail.		
Postconditions	● List of questions updated (edited or deleted).		
Normal Sequence/Flow	1. The Lecturer clicks on a Quiz in the section list. 2. The Lecturer clicks on a specific Quiz item.. 3. The System displays the Quiz interface, including the list of questions (Question Text, Type). 4. The Lecturer reviews the list of questions.		
Alternative Sequences/Flows	A1. Edit Question :		

	- At Step 4, the Lecturer clicks the "Edit" icon on a specific question. - The System displays the "Chỉnh sửa câu hỏi" form with current data pre-filled. - The Lecturer modifies the details (e.g., Text, Answers, Correct Option) and clicks "Cập nhật". - The System validates the new data. - The System updates the question and refreshes the list. A2. Delete Question from Quiz: - At Step 4, the Lecturer clicks the "Delete" icon on a specific question. - The System prompts for confirmation: "Bạn có muốn xóa câu hỏi này không?" - The Lecturer confirms. - The System removes question from quiz. - The System displays a success message.
Exception	E1. Validation Error (Edit): - In A1, if the Lecturer clears the "Câu hỏi".The system display "Vui lòng nhập câu hỏi". - If the Lecturer does not select any correct answer. The system will report an error "Vui lòng ít nhất có 1 đáp án đúng".

2.8.2 Manage Question Bank

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturers to manage a shared question bank for each Section of the course. Functions include: viewing the list, creating new questions , editing, deleting, and importing in bulk from Excel files.		
Preconditions	- The Lecturer must be logged in to the system. - The Lecturer has at least one Course and Section created.		
Postconditions	Question bank data is updated.		
Normal Sequence/Flow	- Lecturers navigate the "Ngân hàng đề" page. - The system displays a list of the lecturer's courses in the left column. - Lecturer selects a Course. - The system loads and displays a list of Sections belonging to that course. - The lecturer selects a specific Chapter. - The system displays a list of questions in that chapter (including columns:Câu hỏi, Ngày tạo, Ngày cập nhật, Hành động).		
Alternative Sequences/Flows	A1: Create New Question : - At the question list interface (Step 6), the Lecturer clicks the "Add New" button. - The system displays the Modal "Create New Question".		

	○ The Lecturer enters the Question Content (using Rich Text Editor). ○ The Lecturer selects the Question Type (Single Choice / Multiple Choice). ○ The Lecturer enters the Answers and selects the correct answer.(Optional) The Lecturer can upload a photo for the question or each answer. ○ The Lecturer clicks the "Create Question" button. ○ The system checks the data (Validation). ○ The system saves the new question to the bank and updates the displayed list. ● A2: Edit Question : ○ At the question list, the Lecturer clicks the "Edit" button (pencil icon) on the line of the question he wants to edit. ○ The system displays the Modal "Chỉnh sửa câu hỏi" with the old data already filled in. ○ The lecturer changes the information. ○ The lecturer clicks the "Lưu" button. ○ The system updates the question and notifies success. ● A3: Delete Question ○ In the question list, the lecturer clicks the "Delete" button (trash can icon). ○ The system displays a confirmation dialog box (Popconfirm). ○ The lecturer confirms the deletion. ○ The system deletes the question and removes it from the display list. ● A4: Import from Excel ○ In the list interface, the lecturer clicks the "Nhập từ Excel" button. ○ The system displays the Modal "Nhập câu hỏi từ excel". ○ The lecturer clicks the "Tải file mẫu" button to download the standard template file (quiz_questions_template.xlsx) provided by the system. ○ The lecturer fills in the question information into the Excel file according to the correct template format. ○ The lecturer returns to the modal and uploads the Excel file with the data filled in. ○ The system reads the file, checks the format and adds valid questions to the bank. ○ The system notifies the input result and refreshes the question list.
Exception	E1: Validation Error ● Trigger: In A1, A2, A4 flow, if the Lecturer submitted invalid Question Bank field values(too many characters, blank fields, invalid number of correct questions). ● Sequence: 1. The Lecturer opens the Create/Update form or Import window.

	2. The Lecturer enters/imports invalid field values(too many characters, blank fields, invalid number of correct questions). 3. The Lecturer clicks “Lưu”. ● System: Displays a toast message informing the Lecturer of the error. ● Result: The Question Bank managing operations is blocked. E2: Authentication Error ● Trigger: In Normal, A1, A2, A3, A4 flow, if the user is not logged into the system with an account. ● Sequence: 1. An user accesses the system without logging in. 2. The user tries to access the managing page of a course with links. ● System: Redirect the user to the login page. ● Result: The Question Bank view page is blocked, the user is redirected to the login page. E3: Authorization Error ● Trigger: In Normal, A1, A2, A3, A4 flow, if the logged in user role is not Lecturer. ● Sequence: 1. An user logged into the system 2. The user tries to access the managing page of an unowned course with links. ● System: Display the notice “Truy cập bị từ chối”. ● Result: The Question Bank view page is blocked. E4: Question Not Found ● Trigger: In A3 flow, if the deleting question is not found. ● Sequence: 1. The Lecturer logged into the system. 2. The Lecturer accesses the Question Bank manage page. 3. The Lecturer clicks Delete on an already deleted question. ● System: Display a toast message “Không tìm thấy câu hỏi”. ● Result: The Question Bank view page is refreshed, with the deleted Question Banks disappeared from the list. E5: Invalid File Format Type ● Trigger: In A1, A2, A4 flow, if the uploaded file format is not valid (A1, A2: JPEG, PNG; A4: Excel). ● Sequence: 1. The Lecturer logged into the system. 2. The Lecturer accesses the Question Bank manage page. 3. The Lecturer opens the create/update or import Question Bank window. 4. The Lecturer uploads invalid files (A1, A2: JPEG, PNG; A4: Excel).
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	• System: Display a toast message “Chỉ được phép nhập file hình ảnh” or “Chỉ được phép nhập file Excel”. • Result: The Question Bank creation is blocked.
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2.9 AI Feature

2.9.1 AI Lesson Q&A

Primary Actors	Student	Secondary Actors	AI(Gemini), Qdrant
Description	Student is logged in and viewing a lesson. Student uses the AI Q&A feature to ask questions about the lesson content and receives answers based on the video subtitles of that lesson.		
Preconditions	• Student has logged into the system • Student has enrolled in the course containing this lesson • Course has AI support enabled • Lesson exists and is active • AI(Gemini) and Qdrant Service are operational		
Postconditions	• Student receives an AI answer based on the lesson content		
Normal Sequence/Flow	AI Lesson Q&A 1. Student is viewing the lesson (Lesson detail page) 2. Student enters a question in the input field "Trợ lý AI cho bài học này" 3. Student clicks the "Gửi câu hỏi" button 4. System validates request: checks that the question is not empty 5. System checks authentication: verifies Student is logged in 6. System checks authorization: verifies user is a Student 7. System checks that lesson exists and is active 8. System checks that course has AI support enabled 9. System checks that Student has enrolled in this course 10. System sends question to AI (gemini) to create embedding vector 11. System searches in Qdrant Vector Database with lessonId filter and score threshold 12. System retrieves top-k results (default topK = 5) with score >= threshold 13. System sorts chunks by chunkIndex to maintain order 14. System combines context from chunks into a text string 15. System sends question and context to AI (gemini) to generate answer 16. System receives answer from AI (gemini) 17. System formats response including answer 18. System displays answer to Student		

Alternative Sequences/Flows	Question empty 1. System detects request.Question is null or empty 2. System returns error 400: "Please enter a question." 3. System displays error message to Student 4. Student must re-enter the question Course AI Support Not Enabled 1. System detects course.IsAiSupport != true 2. System returns error 400: "Course has not enabled AI support." 3. System displays error message to Student 4. Student cannot use this feature for this course
Exception	E1_AI Service Error ○ Trigger: Normal Flow Step 10 (Create embedding vector) or Step 15 (Generate answer) ○ Sequence: ■ System sends request to AI Service (Gemini) for embedding or chat completion ■ System encounters an exception: ● Network error (connection timeout, no response) ● API error (invalid API key, quota exceeded, rate limit) ● Service unavailable (Gemini API down) ● Timeout (request exceeds timeout threshold) ■ System catches the exception and logs error details (error message, timestamp, lessonId, userId) ■ System returns error 400: "An error occurred: [error message]" ■ System displays error message to Student: "Sorry, the AI service is temporarily unavailable. Please try again later." ■ System does not create embedding or generate answer Postcondition: Student remains on the lesson detail page, question is not processed, no answer is displayed, student can retry with the same or a different question after the service recovers

3. Functional Requirements

Below are the main functional requirements; for more detail, please refer to the following content:

3. Functional Requirements in Report 3

3.1 Course Feature

3.1.1 View Detail Course

3.1.1.1 Screen: Course Detail Page

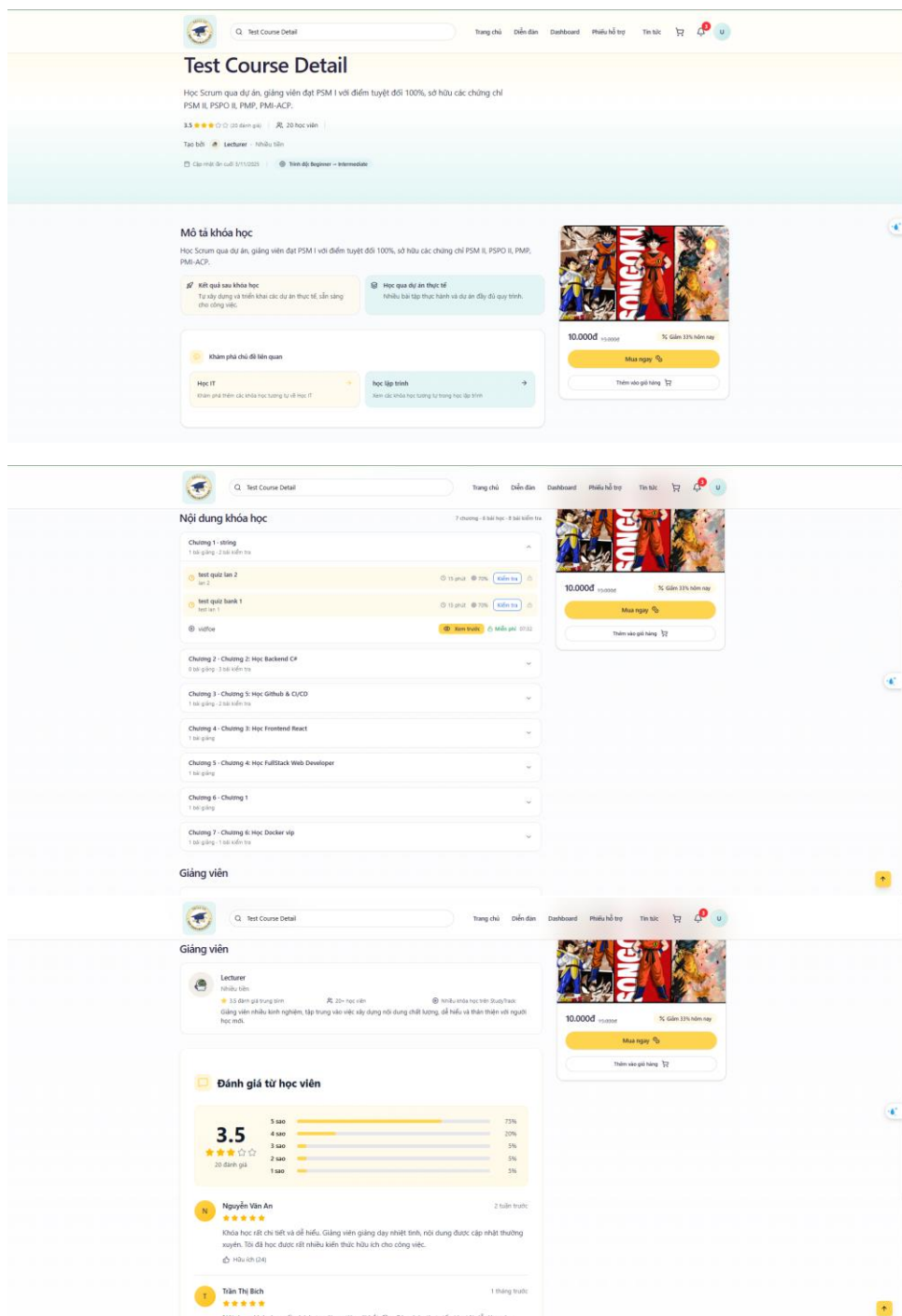


Figure 3.3.1.1 - course details page screen interface

This screen allows the Student to:

- **View Course Overview:** view comprehensive details of the course including title, description, learning outcomes, and average ratings.
- **Explore Syllabus:** browse the curriculum structure (chapters and lessons), check durations, and watch preview videos (if available).
- **View Instructor Details:** view the instructor's profile information, bio, number of students, and their reputation.
- **View Reviews:** view ratings and feedback from other students who have taken the course.
- **Enroll or Purchase:** add the course to the shopping cart or purchase it immediately to gain access to the full content.

Mapped Use Cases: UC-ViewCourseDetail, UC-AddToCart, UC-EnrollCourse (Buy Now), UC-ViewReviews.

Field Name	Description
"Created By" Link	Link. Action: Clicking the instructor's name redirects the user to the Instructor Profile page.
"Mua ngay" Button	Button. Action: Adds the course to the cart and immediately redirects the user to the Checkout/Payment page.
"Thêm vào giỏ hàng" Button	Button. Action: Adds the course to the cart, updates the cart badge, and displays a success toast notification. User stays on the current page.
Related Tags	Link. Action: Clicking a tag redirects the user to a search result page filtered by that specific keyword/topic.
Chapter Accordion	Button/Control. Action: Toggles the chapter section to expand or collapse the list of lessons inside.
"Xem trước" Button	Button. Action: Opens a video player popup to watch the trial lesson content (only visible for "Trial" lessons).
"Hữu ích" Button	Button. Action: Allows the user to "Like" or mark a review as helpful.

3.1.2 View Course List

3.1.2.1 Screen: Course Catalog Page

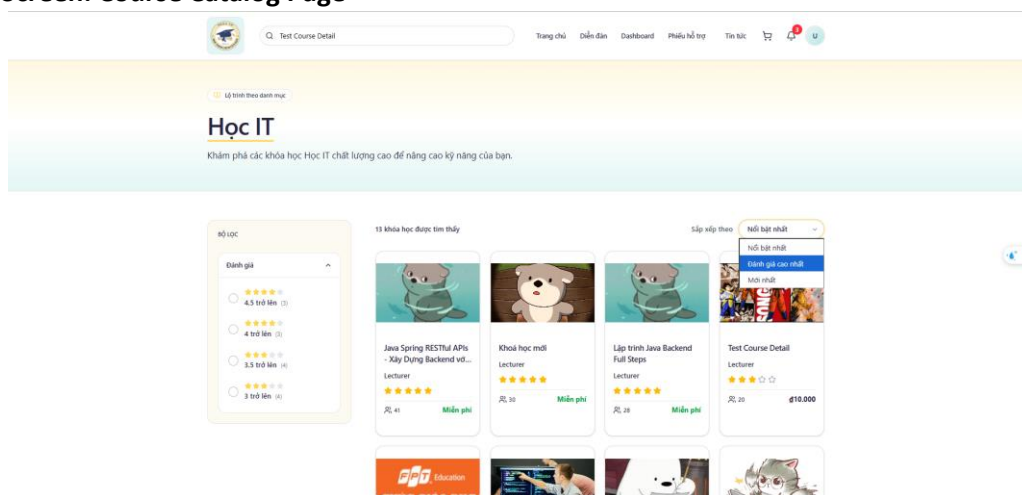


Figure 3.3.2.1 - course catalog page screen interface

This screen allows the Student to:

- **View Category Information:** see the title and description of the selected category (e.g., IT, Business) to understand the learning path.
- **Filter Courses:** narrow down the course list using sidebar filters (e.g., by Rating) to find relevant content.
- **Sort Courses:** organize the displayed courses by criteria such as "Most Featured", "Highest Rated", or "Newest".
- **Browse Course Cards:** view a grid of course summaries including thumbnail, title, instructor, rating, student count, and price.
- **Navigate to Detail:** click on a specific course card to access the Course Detail Page.

Field Name	Description
Sidebar Filter	Left panel for filtering results
Tiêu đề Bộ lọc	Label. Section header "FILTER" (BỘ LỌC).
Bộ lọc Xếp hạng	Radio Button Group. Allows filtering by star rating (e.g., "4.5 stars & up"). Displays the count of courses next to each option.
Phần Thanh công cụ	Tools above the grid
Số lượng Kết quả	Label. Shows total number of courses found (e.g., "13 courses found").
Menu thả xuống Sắp xếp	Dropdown List. Options include: 1. Featured (Nổi bật nhất) 2. Highest Rated (Đánh giá cao nhất) 3. Newest (Mới nhất).
Lưới Khóa học	List of course items
Thẻ Khóa học	Container. Clickable area representing a single course.
Hình thu nhỏ	Image. Cover image of the course.
Tiêu đề Khóa học	Link. Title of the course. Max 2 lines.
Tên Giảng viên	Label. Name of the lecturer.
Số sao xếp hạng	Icon. Visual representation of the average rating (Yellow stars).
Số lượng Học viên	Icon & Label. Icon representing users and the number of enrolled students (e.g., 41).

3.1.3 View My Course

3.1.3.1 Screen: My Learning Dashboard

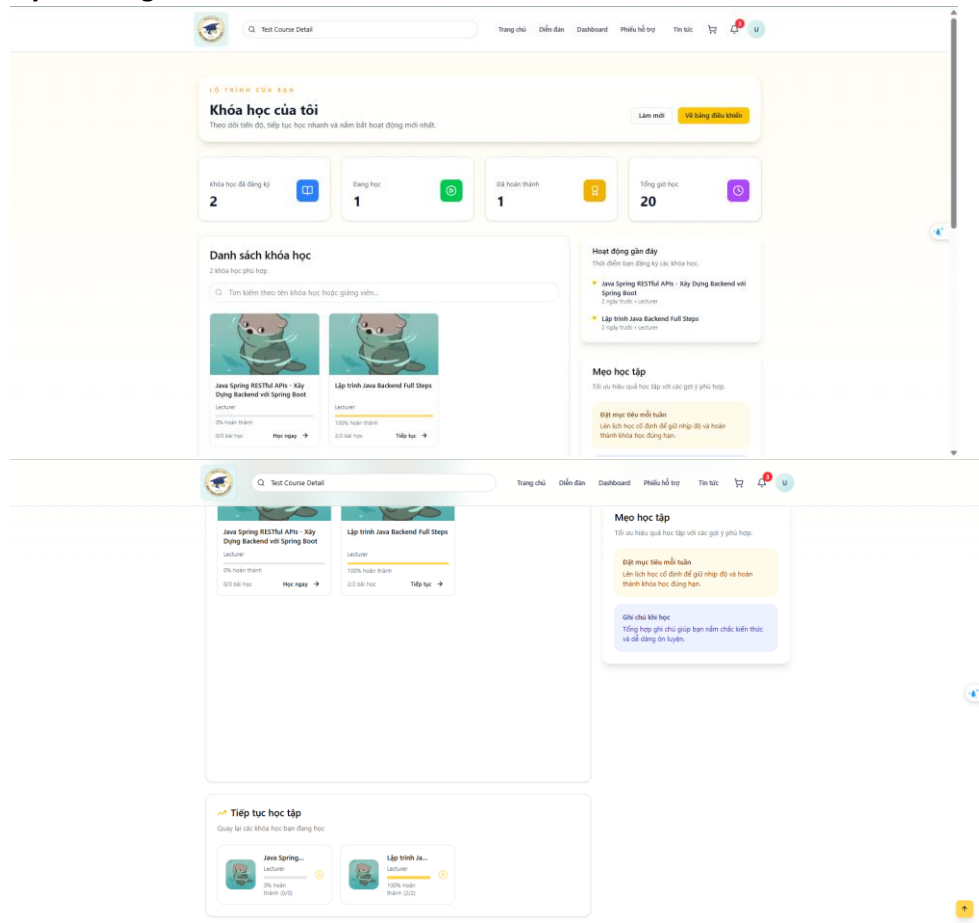


Figure 3.3.3.1 - my course dashboard screen interface

This screen allows the Student to:

- **Track Learning Overview:** view statistics on enrolled courses, courses in progress, completed courses, and total learning hours.
- **Search Enrolled Courses:** find specific courses within their personal library using keywords (course name or instructor).
- **Resume Learning:** view the progress of current courses and click "Continue" or "Start Now" to enter the learning player immediately.
- **View Recent Activities:** monitor recent interactions or updates related to their courses.
- **Get Study Tips:** view helpful advice (e.g., setting goals, taking notes) displayed on the sidebar to improve learning efficiency.

Field Name	Description
"Làm mới" Button	Button. Action: Reloads the latest learning progress and data from the server.
"Về bảng điều khiển" Button	Button. Action: Navigates back to the main user dashboard.
Ô tìm kiếm	Textbox. Action: Allows the user to type keywords to filter the course list by name or instructor.

Field Name	Description
Tiêu đề/Hình thu nhỏ khóa học	Link/Image. Action: Clicking either the title or the image redirects the user to the Course Player to view details or study.
Học ngay	Button. Label: "Start Now" or "Continue". Action: Immediately launches the Lesson Player. - "Start Now": For courses with 0% progress. - "Continue": For courses with >0% progress.
Trình chiếu khóa học	List/Swipe. Action: Allows horizontal scrolling/swiping to view more courses in the "Continue Learning" section.

3.2 Quiz Feature

3.2.1 External Interfaces

3.2.1.1 User Interfaces

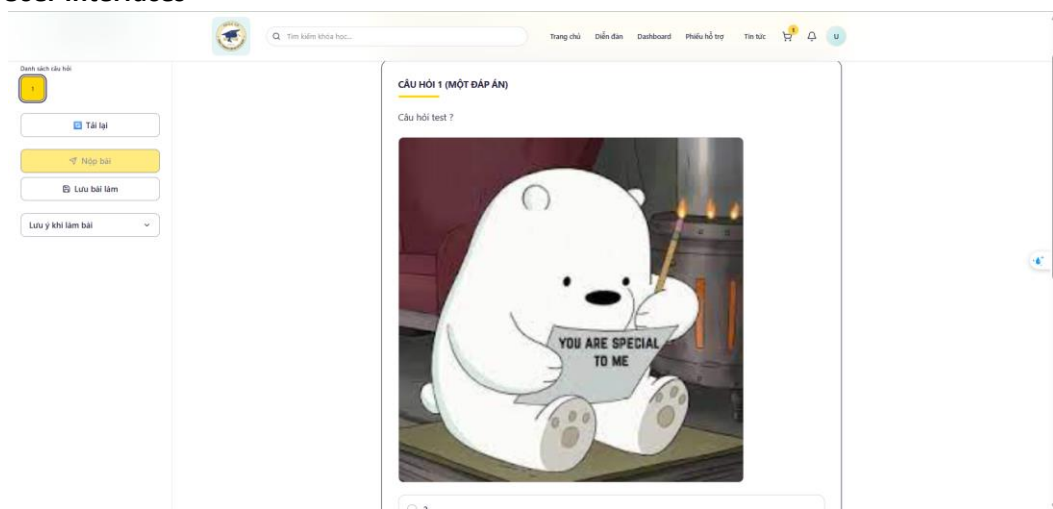


Figure 3.6.1.1 - user screen interface

This screen is the active environment for students to take a quiz. It allows the Student to:

- **View Question:** read the question content, including text and media (images), and identify the question type (e.g., Single Choice, Multiple Choice).
- **Navigate Questions:** use the sidebar question list to jump between questions or track which questions have been answered.
- **Select Answers:** choose options corresponding to the question.
- **Save Progress:** manually save their work to prevent data loss before submission.
- **Submit Quiz:** finalize the attempt and submit answers for grading.

Field Name	Description
Question List Grid	Button/Grid. Action: Clicking a number allows the user to jump directly to that specific question.
"Tải lại" Button	Button. Label: "Tải lại". Action: Refreshes the quiz interface to reload content or images if errors occur.
"Nộp bài" Button	Button. Label: "Nộp bài". Action: Triggers the submission process. Displays a confirmation popup before finalizing the grade.

"Lưu bài làm" Button	Button. Label: "Lưu bài làm". Action: Saves the currently selected answers to the server without submitting the quiz.
"Lưu ý khi làm bài" Dropdown	Button/Accordion. Label: "Lưu ý khi làm bài". Action: Expands or collapses the section to show/hide quiz rules and instructions.
Tùy chọn trả lời	Radio Button / Checkbox. Action: Users click to select one (Radio) or multiple (Checkbox) answers for the current question.

3.2.2 View Detail Result

3.2.2.1 Quiz Result Page

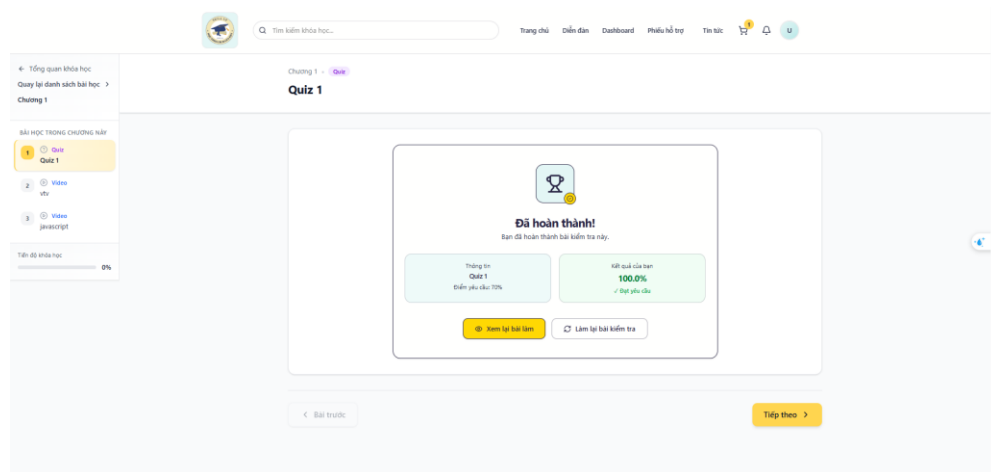


Figure 3.6.2.1 -quiz result screen interface

This screen displays the outcome after a student submits a quiz. It allows the Student to:

- View Performance: instantly see the completion status, achieved score (percentage), and pass/fail result.
- Review Attempt: click "Review" to see detailed feedback on which questions were answered correctly or incorrectly.
- Retry Quiz: check if retakes are allowed and click "Retry" to start a new attempt to improve the score.
- Navigate Forward: proceed to the next lesson in the course syllabus.

Field Name	Description
"Xem lại bài làm" Button	Button. Label: "Xem lại bài làm". Action: Redirects the user to the review mode to see correct/incorrect answers and explanations.
"Làm lại bài kiểm tra" Button	Button. Label: "Làm lại bài kiểm tra". Action: Resets the current answers and starts a new quiz session (if retakes are allowed).
"Bài tiếp theo" Button	Button. Label: "Bài tiếp theo" (Next Lesson). Action: Navigates immediately to the next lesson or video in the course syllabus.

3.3 Cart Feature

3.3.1. Add Cart Item

3.3.1.1 Course Detail Page

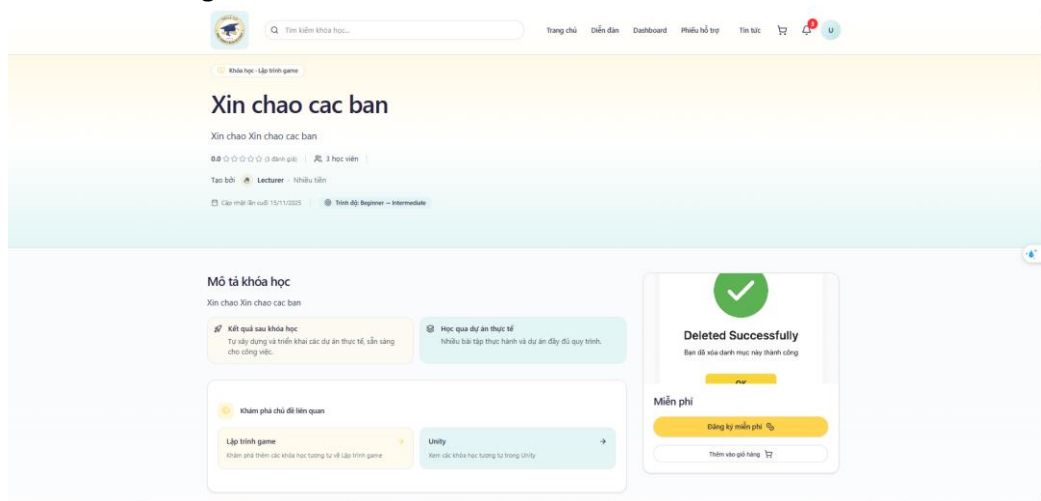


Figure 3.7.1.1 -course detail page screen interface

This function allows the Student to add a course to their shopping cart without immediately checking out.

- **Action:** Click the "Add to Cart" (Thêm vào giỏ hàng) button.
- **System Behavior:** The system validates the request, adds the course to the cart session, updates the cart icon badge number, and displays a success notification (Toast message). The user remains on the current page to continue browsing.

Field Name	Description
"Thêm vào giỏ hàng" Button	Button. Action: Adds item to cart. Visible if the user has not purchased the course yet.
Cart Icon Badge	Icon/Label. Header component. Updates immediately (e.g., from 0 to 1) upon successful addition.

3.4 Shopping cart & payment Feature

3.4.1 Buy course

3.4.1.1 Buy Now

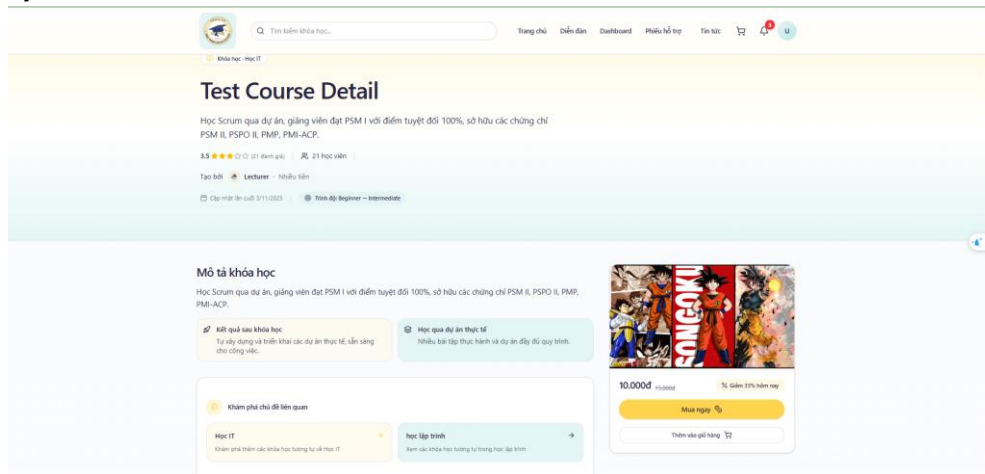


Figure 3.8.1.1 -buy course screen interface

This function allows the Student to purchase a specific course immediately without adding it to the cart.

- Trigger: Click the "Mua ngay" (Buy Now) button on the sticky purchase card.
- System Behavior:
 1. Authentication Check: Checks if the user is logged in. If not, redirects to Login.
 2. Order Creation: Creates a temporary order containing only this single course.
 3. Redirection: Redirects the user directly to the Payment Gateway interface to complete the transaction.

Field Name	Description
"Mua ngay" Button	Button. Label: "Mua ngay". Action: Triggers the immediate purchase flow for the single item.
Price Display	Label. The specific price used for the order calculation.

3.4.2 Checkout from Cart

3.4.2.1 Shopping Cart Page

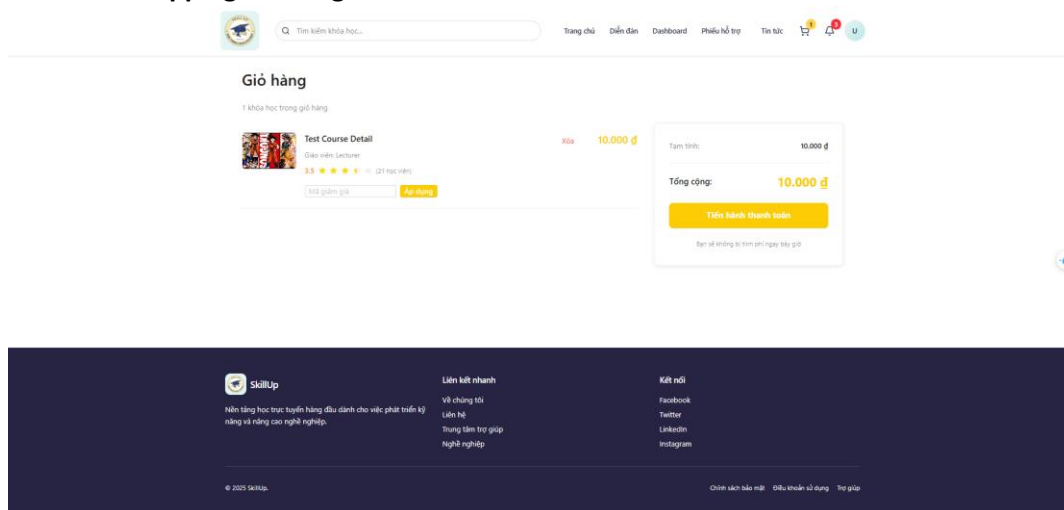


Figure 3.8.2.1 -cart page screen interface

This function allows the Student to finalize the purchase for all items currently in the shopping cart.

- Trigger: Click the "Tiến hành thanh toán" (Proceed to Checkout) button.
- System Behavior:
 1. Validation: Checks if the cart is not empty and validates item availability.
 2. Calculation: Finalizes the total amount after applying any vouchers.
 3. Redirection: Redirects the user to the Payment Gateway interface to pay for the entire cart bundle.

Field Name	Description
"Tiến hành thanh toán" Button	Button. Label: "Tiến hành thanh toán". Action: Initiates the checkout process for the entire cart.
Tổng số tiền	Label. The final sum calculated from all items minus vouchers. This is the amount passed to the payment gateway.

3.5 Lecturer Application Feature

3.5.1 Create Lecturer Application

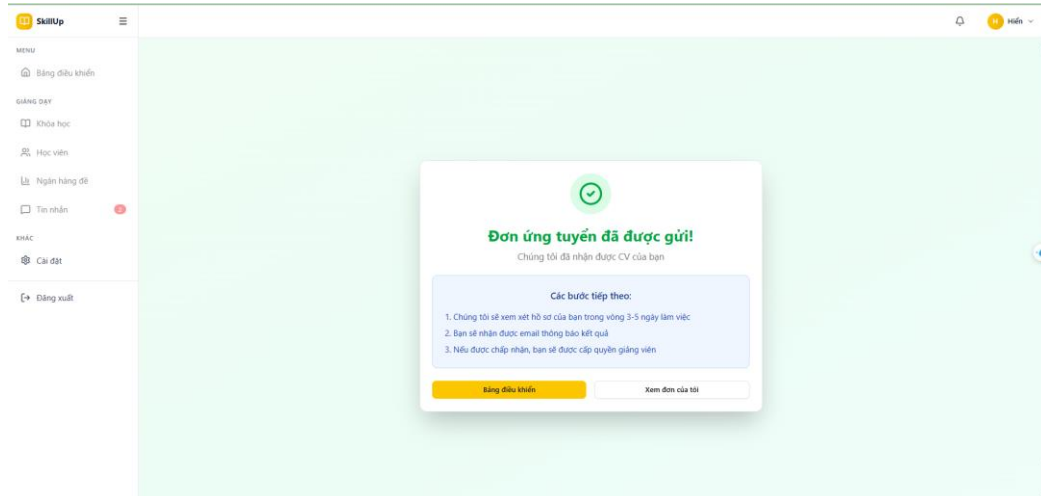


Figure 3.10.1 - create lecturer application screen interface

This modal appears automatically after a user successfully submits their lecturer application form. It serves to:

- **Confirm Submission:** notify the user that their application and CV have been received by the system.
- **Inform Next Steps:** outline the review process (timeline of 3-5 days, email notification, permission granting).
- **Navigate:** provide options to return to the main dashboard or review the specific application details just submitted.

Field Name	Description
"Bảng điều khiển" Button	Button. Label: "Bảng điều khiển" (Dashboard). Action: Closes the modal and redirects the user to the main User Dashboard.
"Xem đơn của tôi" Button	Button. Label: "Xem đơn của tôi" (View My Application). Action: Redirects the user to the Application Detail/Status page to track progress.

3.5.2 View detail Lecturer Application

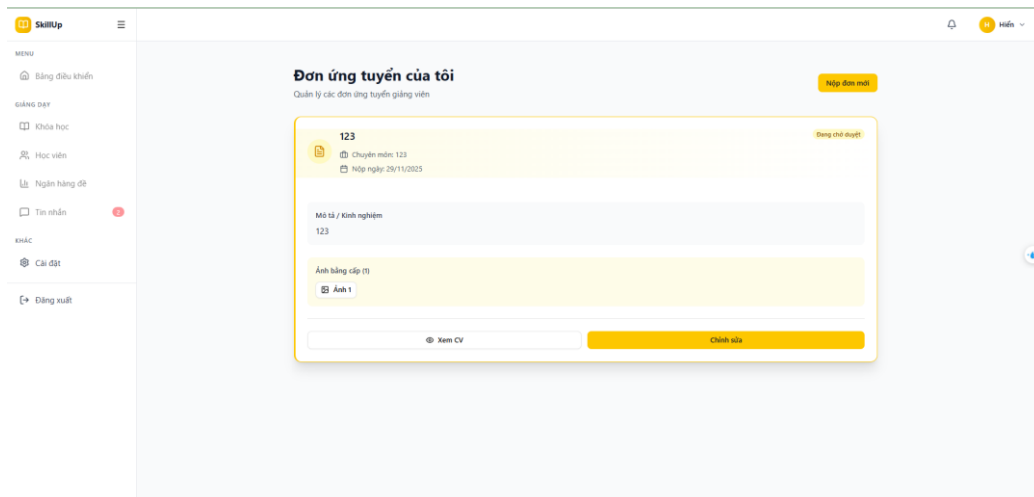


Figure 3.10.2 -Detail lecturer application screen interface

This screen allows the User to track the progress of their lecturer application. It allows the User to:

- **Check Status:** view the current status of the application (e.g., Pending, Approved, Rejected).
- **Review Information:** review the submitted details, including personal info and the attached CV/Portfolio.
- **View Feedback:** read any notes or reasons for rejection provided by the Admin (if applicable).
- **Cancel Application:** retract the application if it is still in the "Pending" state (if the system permits).
- **Navigate Back:** return to the dashboard or home page.

Field Description

Field Name	Description
"Back" Button	Button. Action: Navigates back to the User Dashboard or previous page.
CV/Portfolio Link	Link/Button. Action: Downloads or opens the uploaded CV file (PDF/Doc) for review.
"Cancel Application" Button	Button. Label: "Hủy đơn" (Cancel). Action: Withdraws the application. Typically enabled only when status is "Pending". Triggers a confirmation popup.
"Re-apply" Button	Button. Label: "Gửi lại đơn" (Re-apply). Action: Navigates to the Registration Form to create a new application (visible only if the previous application was Rejected).

3.5.3 View Lecturer Application List

Quản lý đơn ứng tuyển giảng viên

Xoá bộ lọc Tải lại

Tiêu đề	Mô tả	Chức danh	Email	Trạng thái	Ngày tạo	CV & Bằng cấp	Thao tác
Giảng viên	Giảng viên	Web Developer	hocpttk@gmail.com111	Đã chấp nhận	09:24 02/12/2025		Duyệt Từ chối
123	123	123	collinsbest30026@gmail.com	Chờ xử lý	10:40 29/11/2025		Duyệt Từ chối
Pro top 1	Pro top 1	Pro top 1	quatbthe176382@fpt.edu.vn11...	Từ chối	23:46 28/11/2025		Duyệt Từ chối
hi	hi	hi	holastaynow@gmail.com	Đã chấp nhận	23:13 17/11/2025		Duyệt Từ chối
Giaos vien	thủ việc	Web dev	hhhhhhh@gmail.com	Đã chấp nhận	14:52 11/11/2025		Duyệt Từ chối
Senior	hhhhhhafaf	Web dev	hhhy@gmail.com	Đã chấp nhận	14:38 11/11/2025		Duyệt Từ chối
Học sinh	Học sinh	Học sinh	butienquat2k3@gmail.com	Từ chối	20:57 09/11/2025		Duyệt Từ chối
Top 1	Top 1	Top 1	hhhh@gmail.com	Đã chấp nhận	00:32 07/11/2025		Duyệt Từ chối
senior	sss	Web	testgiaovien@gmail.com	Đã chấp nhận	21:15 05/11/2025		Duyệt Từ chối
Fresher	Fresher	Web	holastaynow11@gmail.com	Đã chấp nhận	22:31 30/10/2025		Duyệt Từ chối

11 bản ghi < 1 2 >

Figure 3.10.3 -Lecturer application list screen interface

This screen allows the System Moderator to view the list of all lecturer applications. It allows the System Moderator to:

- Check Status: view the current status of the application (e.g., Pending, Approved, Rejected).
- Review Information: review the submitted details, including personal info and the attached CV/Portfolio.
- Handle Application: accept or reject the application if it is still in the "Pending" state.

Field Description

Field Name	Description
Tìm kiếm	Type: string . Action: filter and display all applications with matching information.
Xoá bộ lọc	Type: Button. Action: Reset all applying filters.
Tải lại	Type: Button. Action: Refresh the table data.
File icon	Type: Button. Action: Open the application's CVs popup window.
Trophy icon	Type: Button. Action: Open the application's Degrees popup window.
Duyệt	Type: Button. Action: Update the application status to Accepted.
Từ chối	Type: Button. Action: Update the application status to Rejected.

3.6 Question & Answer Feature

3.6.1 Manage Question & Answer

3.6.1.1 Create question bank

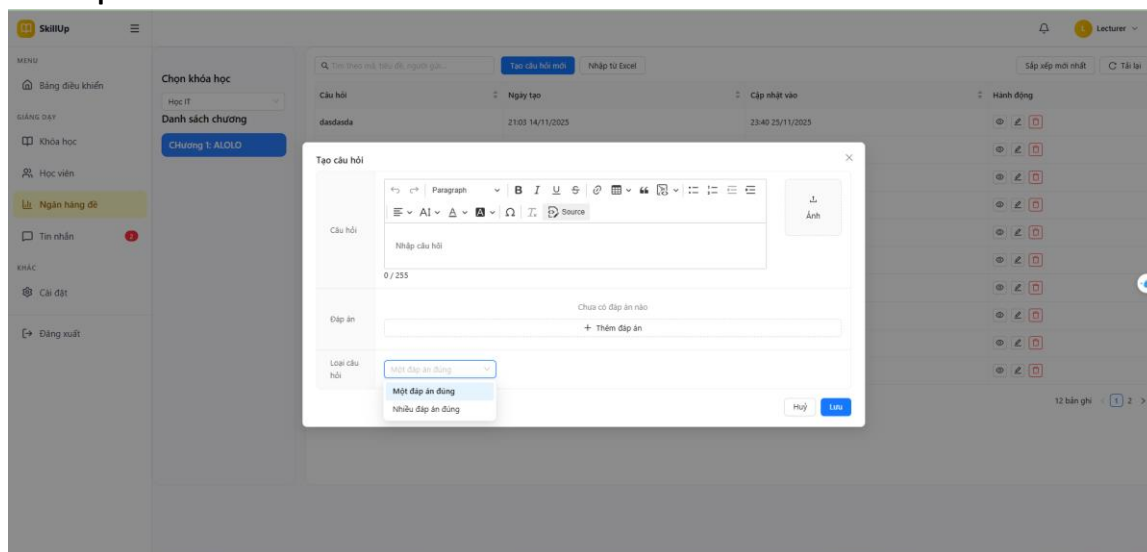


Figure 3.11.1.1 -Create question bank screen interface

This screen allows the Instructor to create a new question and manage its answer options simultaneously.

- **Question Content:** Enter the question text and media.
- **Manage Answers:** Create answer options, delete unnecessary options, and define the correct result.
- **Save:** Store the question and its answers to the database.

Field Description

Field Name	Description
Quiz Title Input	Textbox. Action: Allows the instructor to type the name of the quiz (e.g., "Final Exam").
"Add Question" Button	Button. Action: Adds a new empty question block to the list.
Question Type Dropdown	Dropdown. Action: Selects the format of the question (e.g., Single Choice, Multiple Choice).
Question Text Area	Text Area. Action: Input field for the main content of the question.
"Upload Image" Button	Button. Action: Uploads an image to attach to the specific question (optional).
Answer Option Input	Textbox. Action: Input fields to type the possible answers (Option A, B, C...).
Correct Answer Checkbox/Radio	Checkbox/Radio. Action: Marks which option is the correct answer. Essential for auto-grading.
"Delete Question" Icon	Button. Action: Removes the specific question from the quiz.

"Save Quiz" Button	Button. Action: Validates all fields (must have title, questions, and correct answers) and saves the quiz data.
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3.6.1.2 Update question bank

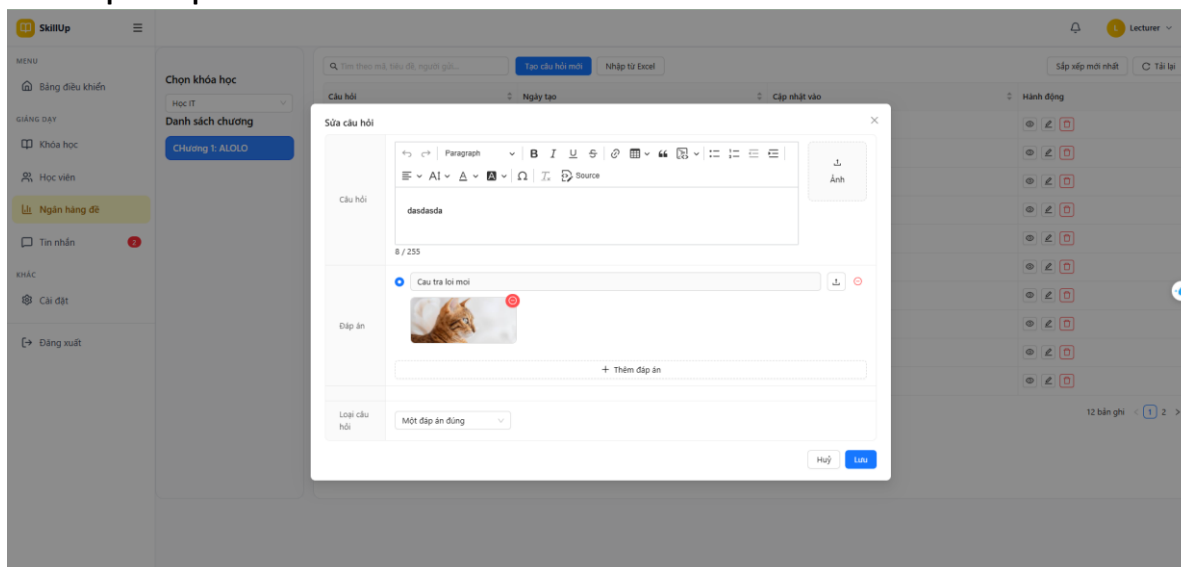


Figure 3.11.1.2 -Update question bank screen interface

This screen allows the Instructor to modify an existing question and its answers.

- **Modify Question:** Edit the text or media.
- **Modify Answers:** Rename answer options (Update), add new options (Create), or remove options (Delete).
- **Update:** Save changes to the database.

Field Name	Description
"Update" Button	Button. Action: Validates the modified data and overrides the existing quiz record in the database. Displays a "Updated Successfully" toast.
"Delete Answer" Icon	Button/Icon. Action: Removes a specific answer option (e.g., Option D) from a question.
"Cancel" Button	Button. Action: Discards any unsaved changes and returns to the Course Curriculum page.

3.6.1.3 View question bank

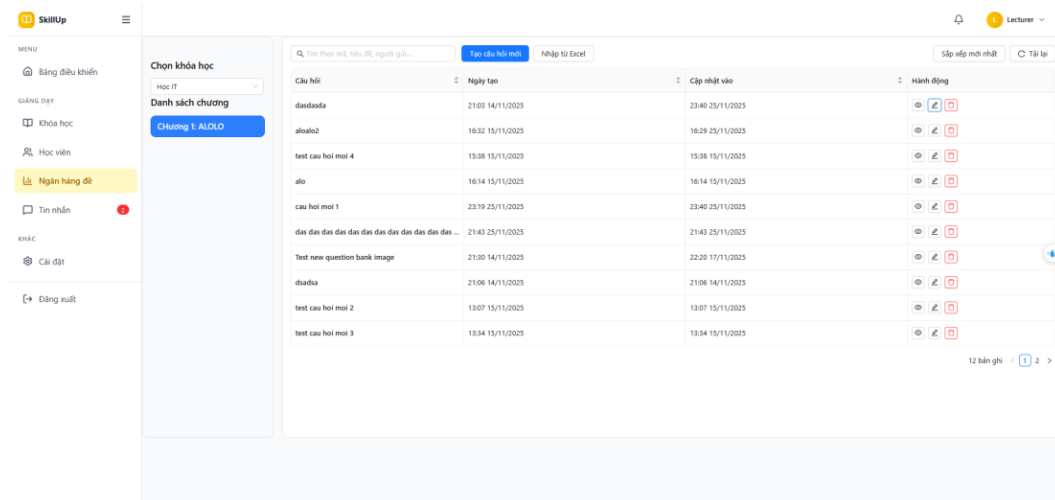


Figure 3.11.1.3 - question bank screen interface

This screen allows the Instructor to preview how a specific question appears to students.

- Review: Check question text, image, and options.
- Actions: Quick access to Edit or Delete this question.

Field Name	Description
"Edit" Button	Button. Action: Navigates to the Quiz Editor screen to modify settings or questions.
"Preview" Button	Button. Action: Opens a simulation of the quiz (Student View) so the instructor can test the flow.
"Delete" Button	Button. Action: Permanently removes the quiz from the course. Triggers a confirmation popup.
"Back" Button	Button. Action: Returns to the Course Curriculum/Section list.
Question Accordion/Toggle	Button/Icon. Action: Expands a specific question row to reveal its answer options and the correct answer key.

3.6.1.4 Delete question bank

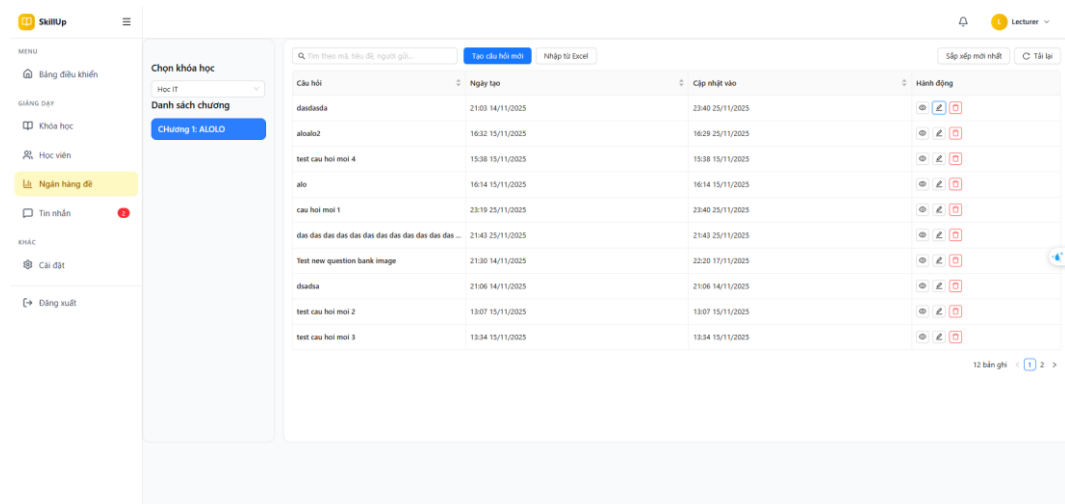


Figure 3.11.1.4 -delete question bank screen interface

This function removes a question permanently.

- Confirm: User must confirm the action in a popup.

Field Name	Description
"Delete" Icon	Button/Icon. Action: Triggers the delete confirmation popup for the selected question.
"Confirm Delete" Button	Button. (Inside Modal). Action: Permanently removes the question from the quiz data and closes the modal.
"Cancel" Button	Button. (Inside Modal). Action: Closes the confirmation popup without deleting the question.

3.6.1.5 Add question from excel

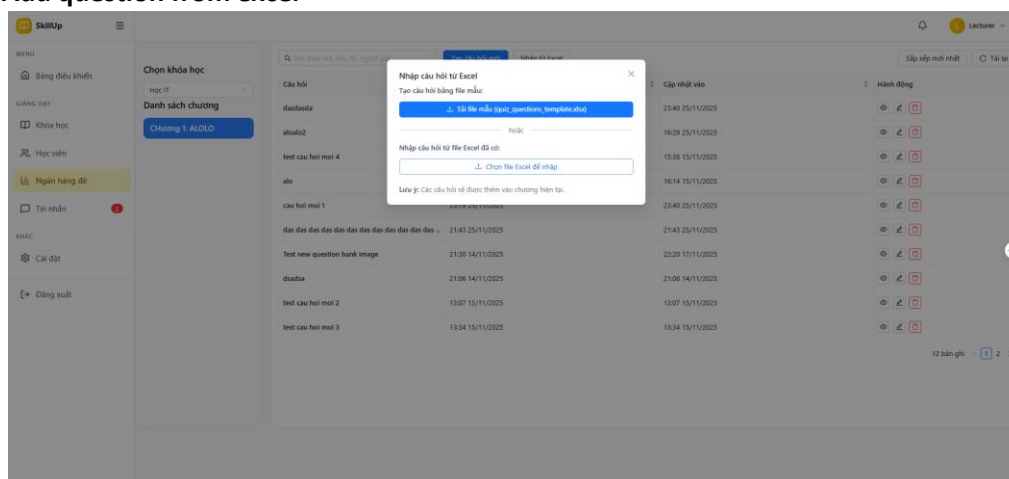


Figure 3.11.1.5 -add question from excel screen interface

This function allows the Instructor to bulk upload questions using a pre-formatted Excel or CSV file.

- **Download Template:** Get the standard file format required by the system.
- **Upload File:** Select and upload the filled data file.
- **Process:** The system validates the file and creates questions automatically.

Field Name	Description
"Download Template" Link	Link/Button. Action: Downloads the sample Excel/CSV file containing the required column headers (Question, Option A, Option B, Correct Answer, etc.).
"Select File" Button	Button/Input. Action: Opens the operating system's file picker to select a local file (usually restricted to .xlsx, .xls, .csv).
"Upload" Button	Button. Action: Submits the selected file to the server for processing and data import.
"Cancel" Button	Button. Action: Closes the import modal without performing any action.

3.7 Quiz Feature

3.7.1 Manage quiz

3.7.1.1 Create quiz

Quiz Cuối cùng

aa

Điểm đạt (%) 80 Thời gian (phút) 30

Tạo quiz Hủy

+ Thêm nội dung

Figure 3.12.1.1 -create quiz screen interface

This function allows the Instructor to initialize a new quiz.

- Input Data: The instructor fills in the quiz title, description, time limit, and passing score into empty fields.
- Action: Clicking "Create" saves the new quiz to the database.

Field Name	Description
Quiz Title Input	Textbox. Action: Input field for the new quiz name.
Description Area	Text Area. Action: Input field for quiz instructions.
Time Limit Input	Number Input. Action: Sets the duration (in minutes).
Passing Score Input	Number Input. Action: Sets the minimum percentage to pass.
"Hủy" Button	Button. Action: Validates that required fields are filled and creates the new quiz record.
"Hủy" Button	Button. Action: Discards input and returns to the quiz list.

3.7.1.2 Update quiz

Quiz Cuối cùng

aa

Điểm đạt (%) 80 Thời gian (phút) 30

Tạo quiz Hủy

+ Thêm nội dung

Figure 3.12.1.2 -update quiz screen interface

This function allows the Instructor to modify an existing quiz's settings.

- Load Data: The system pre-fills the form with the current quiz details.
- Modify: The instructor edits the information (e.g., changing the time limit).
- Action: Clicking "Update" overrides the old settings.

Field Name	Description
Quiz Title Input	Textbox. Action: Edit the existing quiz name.
Description Area	Text Area. Action: Edit existing instructions.
Time Limit Input	Number Input. Action: Modify the duration value.
Passing Score Input	Number Input. Action: Modify the passing percentage.
"Tạo quiz" Button	Button. Action: Saves the changes to the system. Displays a success message.
"Hủy" Button	Button. Action: Discards changes and returns to the quiz list without saving.

3.7.1.3 View quiz

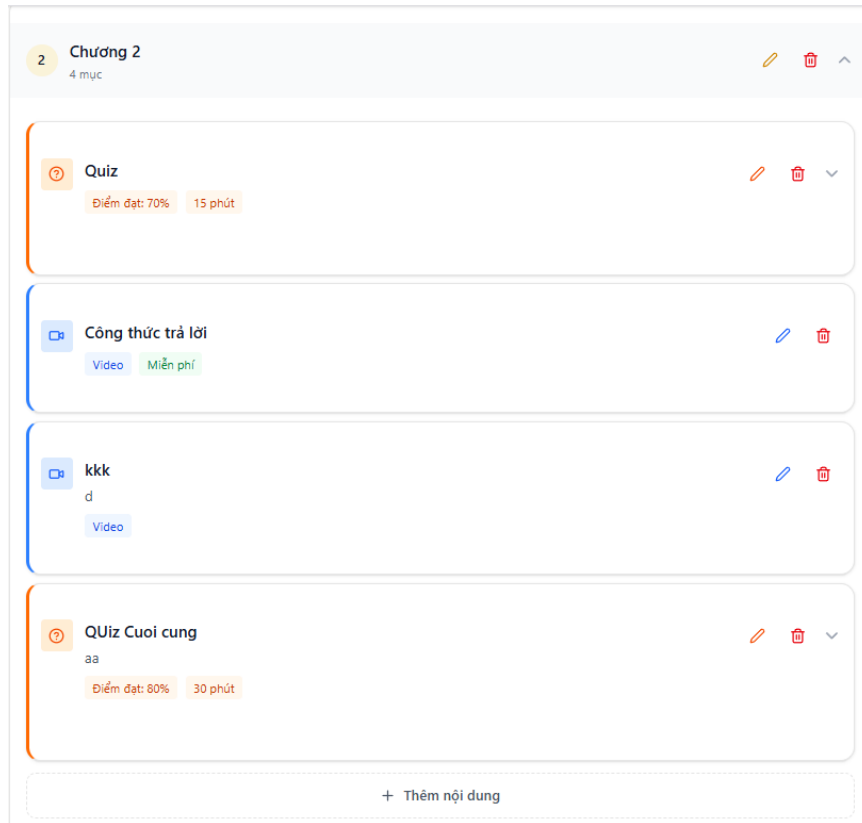


Figure 3.12.1.3 - quiz screen interface

This screen displays all quizzes created within the course. Instructors can track status, search, and access actions like Edit or Delete.

Field Name	Description
"Chỉnh sửa" Icon	Button. Action: Navigates to the Quiz Editor to modify settings (Time, Title, etc.).
"xóa" Icon	Button. Action: Triggers the delete confirmation popup for a specific quiz.
Status Toggle	Switch. Action: Toggles the quiz availability (Draft/Published).

3.7.1.4 Delete quiz

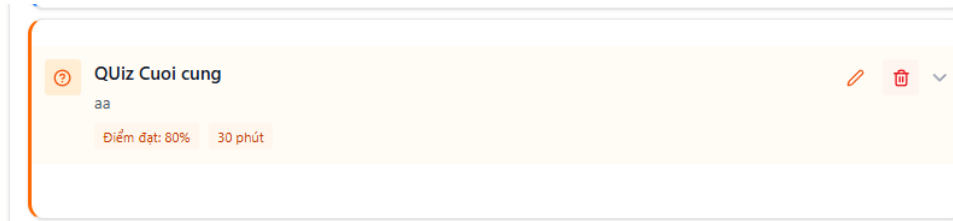


Figure 3.12.1.4 -delete quiz screen interface

This function removes a quiz from the course. Requires strict confirmation to prevent data loss

Field Name	Description
"Confirm Delete" Button	Button. Action: Permanently deletes the quiz and all associated questions/results.
"Cancel" Button	Button. Action: Closes the popup without deleting.

3.7.1.5 Add question to quiz

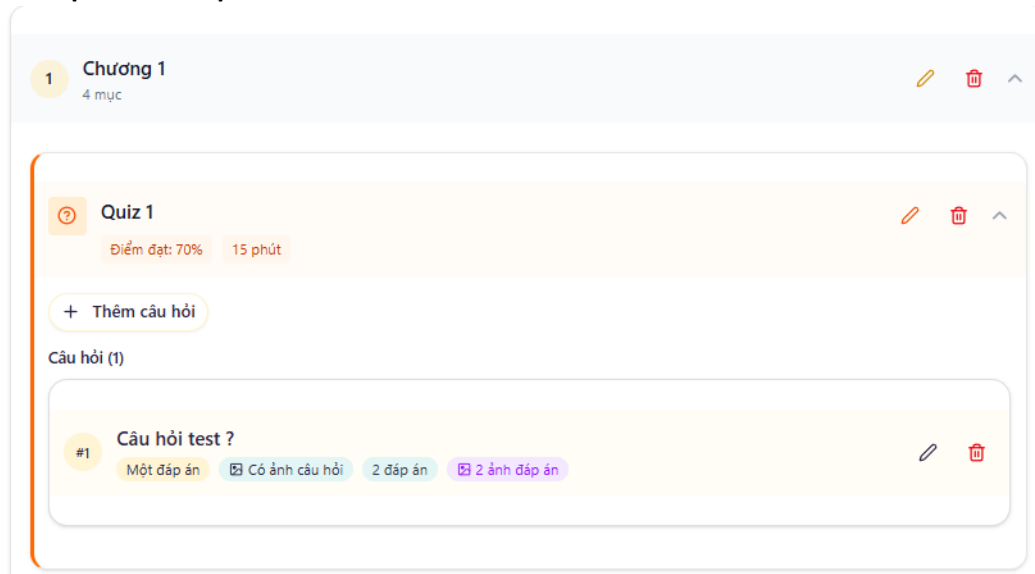


Figure 3.12.1.3 - add question to quiz screen interface

This function allows the Instructor to populate the quiz with questions.

- Methods: Instructors can either create a new question from scratch or select existing questions from the Question Bank.
- Selection: Search and select multiple questions to add at once.
- Action: Confirm to link selected questions to the current quiz.

Field Name	Description
"Thêm câu hỏi mới" Button	Button. Action: Opens the Question Creator form to make a brand new question and add it directly to this quiz.
"Thêm từ ngân hàng câu hỏi" Button	Button. Action: Opens a modal displaying the Question Bank to select existing questions.
Ô tìm kiếm	Textbox. Action: Filters questions in the bank by content or tags.
Question Checkbox	Checkbox. Action: Selects specific questions from the list to add.
"Thêm câu hỏi đã chọn" Button	Button. Action: Adds all checked questions to the quiz list.

3.8 Manage question Feature

3.8.1 Create question

Figure 3.15.1 - create question screen interface

This function allows the Instructor to input content for a new question.

- **Input Data:** Enter the question text, upload optional media (images), and define answer options.
- **Logic:** Select the correct answer(s) to enable auto-grading.
- **Action:** Save the question to the Question Bank or Quiz.

Field Name	Description
Question Content Area	Text Area. Action: Input field for the question description/text. Supports rich text (bold, italic).
"Upload Image" Button	Button. Action: Opens file picker to attach an illustration image to the question.
Answer Option Input	Textbox. Action: Input fields for answer choices (e.g., Option A, Option B).
Correct Answer Selector	Radio/Checkbox. Action: Marks the specific option as the correct answer. - Radio: For Single Choice questions.- Checkbox: For Multiple Choice questions.
"Add Option" Button	Button. Action: Adds a new answer row (e.g., adds Option E) to the list.
"Delete Option" Icon	Button. Action: Removes an answer row (e.g., remove Option D).
"Save" Button	Button. Action: Validates inputs (must have text & correct answer) and saves the question.
"Cancel" Button	Button. Action: Discards inputs and closes the form.

3.8.2 Update question

1 Chương 1
1 mục

Chương 1
1

Điểm đạt: 70% 15 phút

+ Thêm câu hỏi

Câu hỏi (1)

Chỉnh sửa câu hỏi

Câu hỏi *

1

Mô tả / Gợi ý (tùy chọn)

1

Loại câu hỏi *

☒ Một đáp án đúng ☐ Nhiều đáp án đúng

Hình ảnh câu hỏi (tùy chọn)

Tải lên hình ảnh

Đáp án *

☒ 1

1

Figure 3.15.2 - update question screen interface

This function allows the Instructor to modify an existing question.

- **Load Data:** The system automatically pre-fills the form with the current question text, attached image, answer options, and the correct answer key.
- **Modify:** Instructors can edit the text, replace/remove images, change answer choices, or select a different correct answer.
- **Action:** Clicking "Update" saves the changes to the database.

Field Name	Description
"Update" Button	Button. Action: Validates the modified inputs and overrides the existing question record. Displays a success toast.
Question Content Area	Text Area. Action: Edit the existing question description/text.
"Change/Remove Image" Button	Button. Action: Replaces the current image with a new one or removes it entirely.
Answer Option Input	Textbox. Action: Edit the text of existing answer options.
Correct Answer Selector	Radio/Checkbox. Action: Change which option is marked as the correct answer.
"Delete Option" Icon	Button. Action: Removes a specific answer row from the list.
"Add Option" Button	Button. Action: Adds a new answer row if more options are needed.
"Cancel" Button	Button. Action: Discards all unsaved changes and returns to the question list.

3.8.3 Delete question

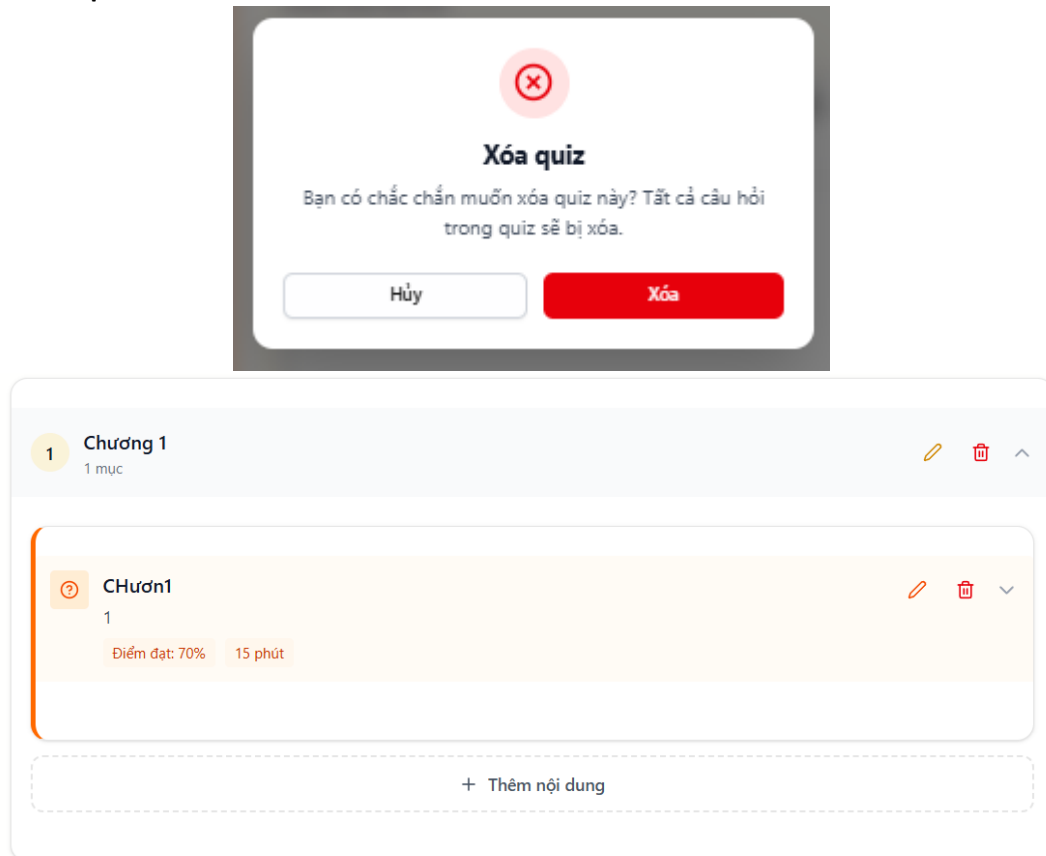


Figure 3.15.3 - delete question screen interface

This function allows the Instructor to permanently remove a question from the bank or quiz.

- **Trigger:** Click the "Delete" (Trash bin) icon on a question row.
- **System Behavior:** Displays a confirmation modal. If confirmed, the question is removed immediately.

Field Name	Description
"Delete" Icon	Button/Icon. Action: Triggers the delete confirmation popup for the specific question.
"Confirm Delete" Button	Button. (Inside Modal). Action: Permanently deletes the question record from the database.
"Cancel" Button	Button. (Inside Modal). Action: Closes the popup without performing the deletion.

3.8.4 View question

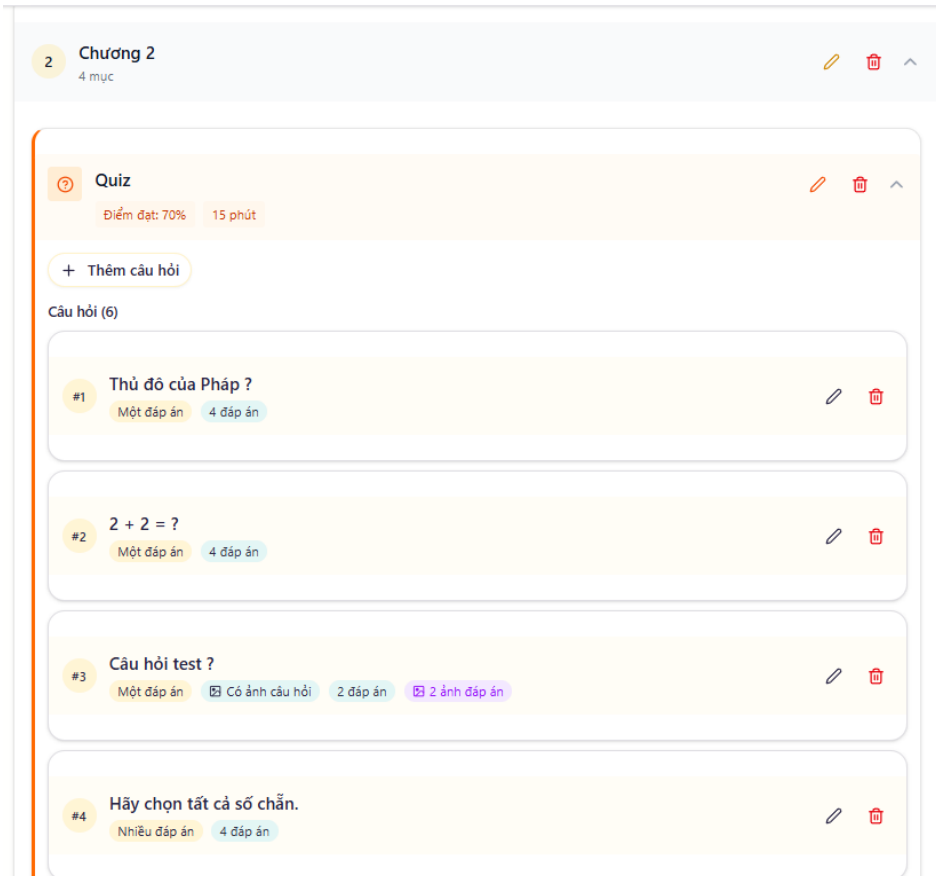


Figure 3.15.4 - list question screen interface

This function allows the Instructor to review the full details of a specific question.

- **View Content:** Displays the question text, attached images, answer options, and highlights the correct answer.
- **Purpose:** Ensures accuracy before adding the question to a quiz.

Field Name	Description
"Edit" Button	Button. Action: Navigates to the Question Editor screen to modify the content.
"Delete" Button/Icon	Button. Action: Triggers the delete confirmation popup to remove this question.
"Close" or "Back" Button	Button. Action: Closes the preview modal or returns to the question bank list.

4. Non-Functional Requirements

4.1 External Interfaces

4.1.1 User Interfaces

UI-1: All screens shall have clear information display and easy to use layout.

UI-2: All designs have been implemented by the front-end team based on available resources.

4.1.2 Software Interfaces

SI-1: OLP-SkillUp Interface sends access requests via OAuth2; OLP-SkillUp Back-end verifies authentication tokens.

4.1.3 Hardware Interfaces

No hardware interfaces have been identified

4.1.4 Communication Interfaces

No communication interfaces have been identified.

4.2 Quality Attributes

4.2.1 Usability

APIs:

- APIs are designed following RESTful standards to facilitate seamless integration with client systems, ensuring clear and easy-to-use CRUD (Create, Read, Update, Delete) operations. The APIs support search, filtering, and reporting features through well-structured and extensible endpoints.

React User Interface:

- The UI is optimized for users, ensuring a friendly, intuitive, and easy-to-navigate experience. It integrates multilingual support (English and Vietnamese), allowing users to switch languages during their session. The interface ensures fast search, filtering, and report display, with modularized and reusable components.

4.2.2 Performance

The React frontend is optimized with lazy loading, memoization, and minimizing unnecessary rendering.

5. Requirement Appendix

5.1 Business Rules

These are sample business rules for the Authentication function. For details of all business rules, please refer to section 5.1, Report 3 - Software Requirement Specification.

ID	Rule Definition
BR-AUTH-01	Inactive or Banned accounts cannot login
BR-AUTH-02	Only roles Student and Lecturer can register via public registration
BR-AUTH-03	OTP must not be used before and must be valid
BR-AUTH-04	Account must be InActive status to verify OTP
BR-AUTH-05	Old password must be correct to change password

5.2 Application Messages List

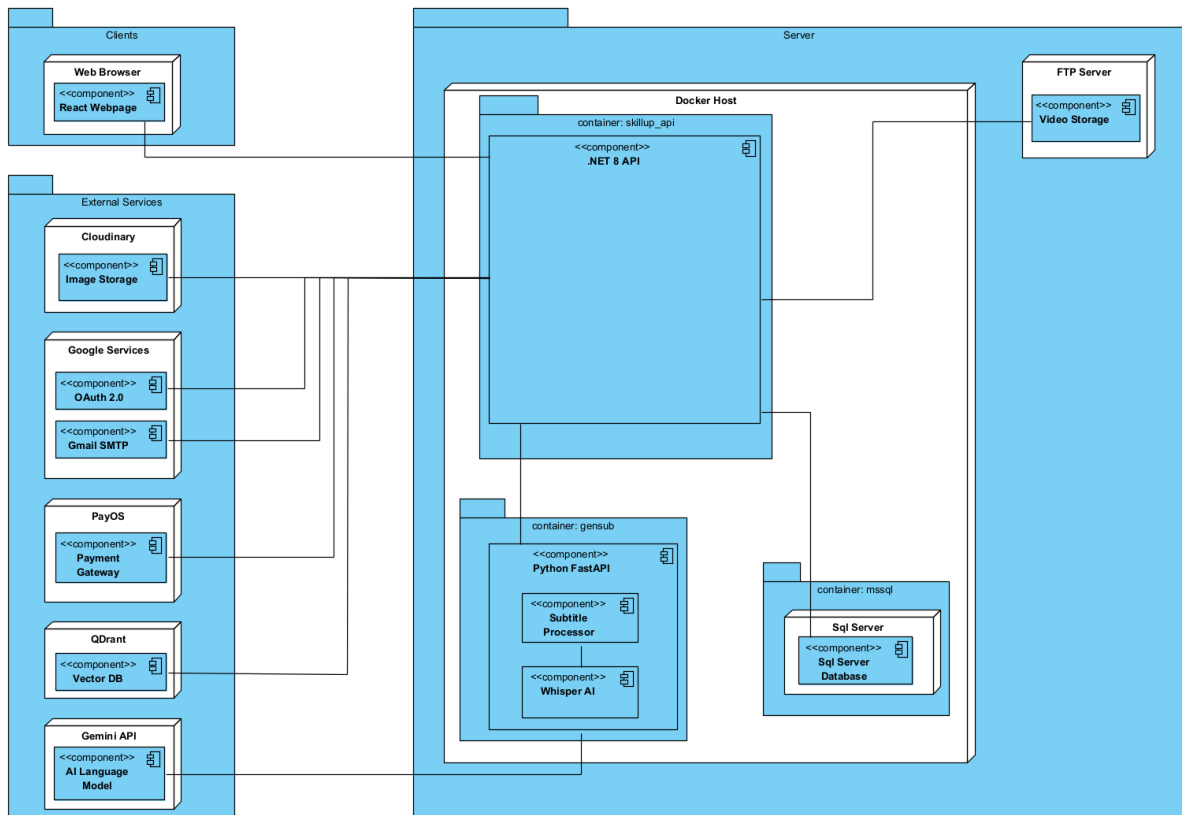
These are sample message codes for the Authentication function. For details of all message codes, please refer to section 5.2, Report 3 - Software Requirement Specification.

#	Message code	Message Type	Context	Content
1	MSG01	Toast (Success, Green)	User successfully logs into the system	<i>Đăng nhập thành công</i>
2	MSG02	Toast (Success, Green)	User logs in using Google OAuth	<i>Đăng nhập Google thành công</i>
3	MSG03	Toast (Error, Red)	User login fails due to invalid credentials	Đăng nhập thất bại. Vui lòng thử lại!
4	MSG04	Toast (Error, Red)	User logs in but email not verified	<i>Tài khoản chưa được kích hoạt</i>
5	MSG05	Toast (Success, Green)	User successfully registers an account	<i>Đăng ký thành công! Vui lòng kiểm tra email để xác thực tài khoản.</i>

IV. Software Design Description

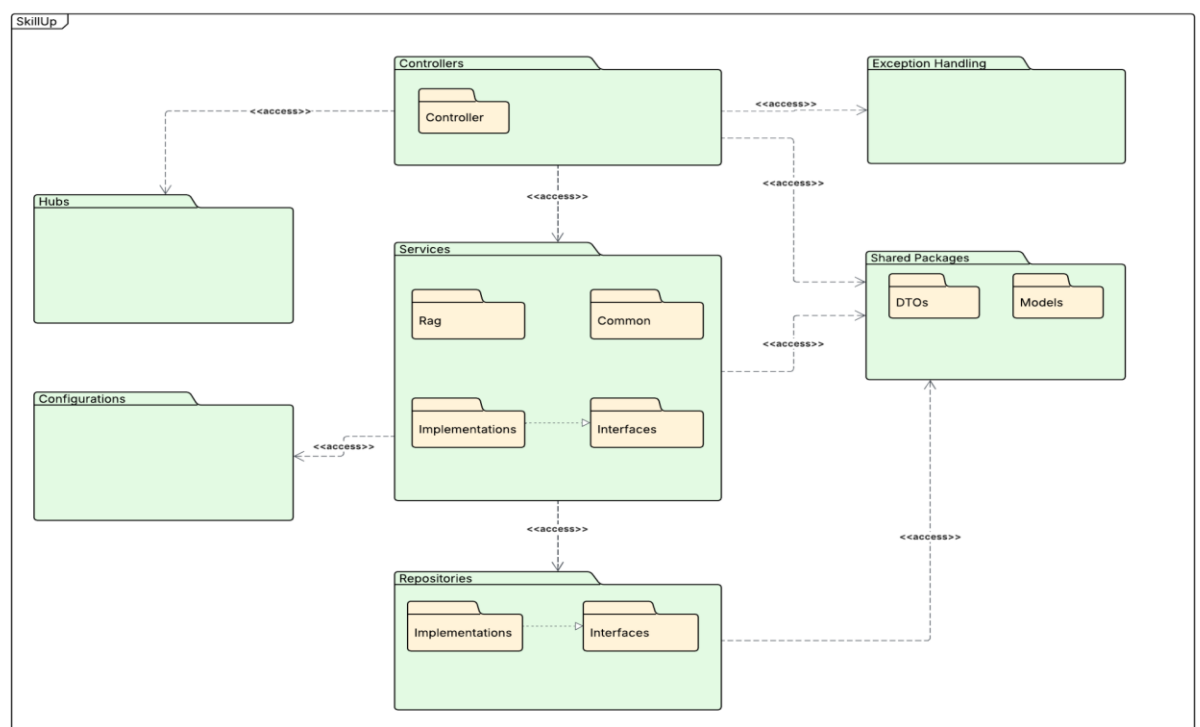
1. System Design

1.1 System Architecture



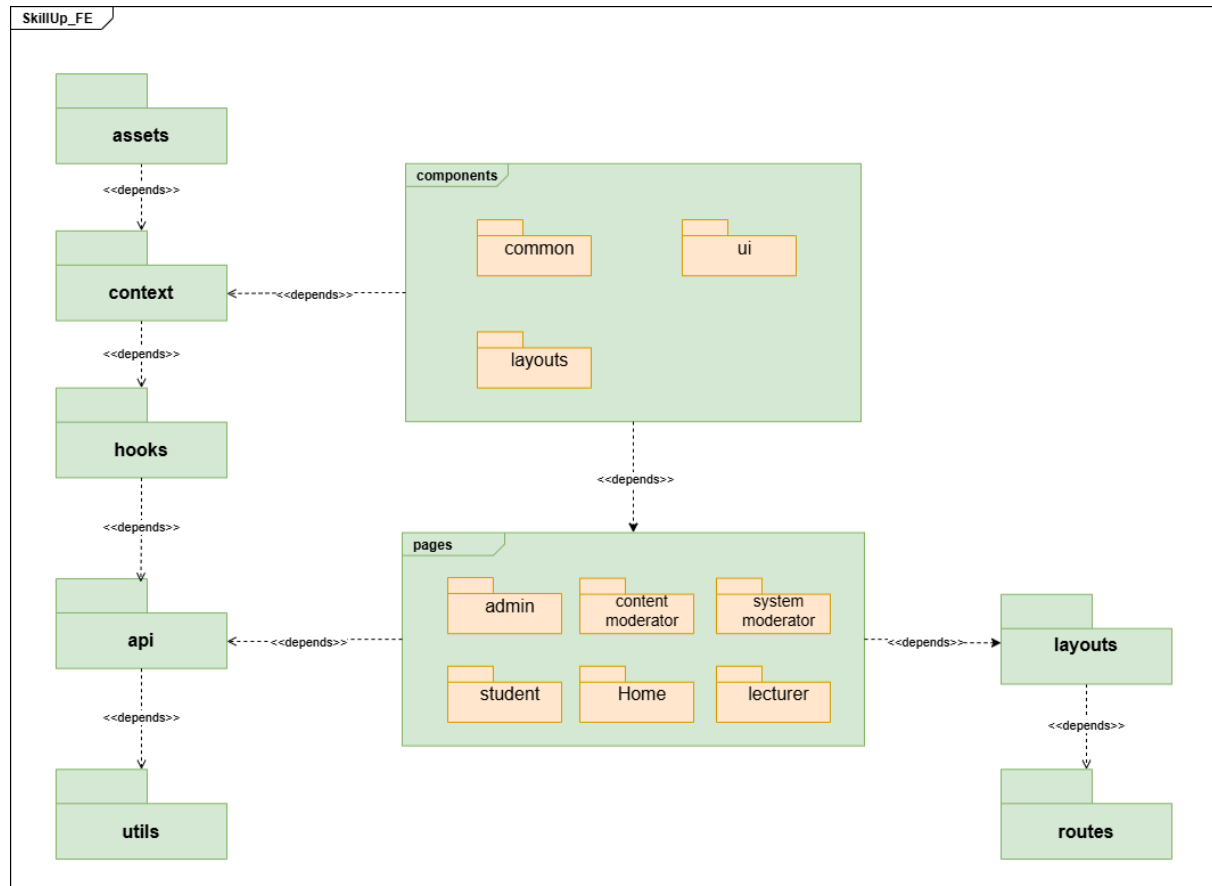
1.2 Package Diagram

1.2.1 Backend Package Diagram



No	Package	Description
01	Controllers	This package serves as the entry point for the application. It handles incoming HTTP requests from the client (UI), validates inputs, and orchestrates the flow by calling the appropriate Services. It is responsible for returning the final HTTP responses to the user.
02	Hubs	This package manages real-time communication using SignalR. It allows the server to push instant updates or notifications to connected clients without waiting for a request.
03	Exception Handling	This is a centralized module for managing errors. It captures runtime exceptions across the application to ensure the system fails gracefully, logs the issues for debugging, and returns standardized error messages to the client.
04	Services	This represents the Business Logic Layer. It contains the core intelligence of the application. Defines contracts and logic to decouple dependencies, provides modules for AI-driven features or advanced search capabilities, shared logic or utilities used within the service layer.
05	Shared Packages	This package contains data structures used across multiple layers of the application to ensure consistency. DTOs (Data Transfer Objects): Objects designed to carry data between processes (e.g., from the API to the client) without exposing the internal database structure. Models: The core domain entities representing the business data.
06	Configurations	This package manages the setup and environment settings for the application. It handles dependency injection, database connection strings, and other global settings required for the services to run.
07	Repositories	This is the Data Access Layer. It abstracts the direct interaction with the database. Interfaces: Defines the methods for data access (e.g., Get, Save, Delete). Implementations: The actual code that executes queries against the database. This layer ensures that the business logic (Services) does not need to know the specific details of how data is stored or retrieved.

1.2.2 Frontend Package Diagram



No	Package	Description
1	assets	Static resources package containing images, icons, fonts, and media files used throughout the application. Serves as the foundation for visual elements. No dependencies on other packages.
2	context	Global state management package containing React Context providers (CartContext for shopping cart state management). Provides shared state across the application, syncs with localStorage for guest users. Depends on assets for potential icons/images.
3	hooks	Custom React hooks package providing reusable logic for components. Contains hooks for data fetching, state management, and side effects. Depends on the store to consume context hooks like useCart().
4	api	API service layer package that encapsulates all backend HTTP requests. Contains service modules: cartAPI (cart operations),

		postAPI (forum posts), newsAPI (news CRUD), voucherAPI (voucher validation), lecturerAPI (lecturer profiles), and others. All services use configured axiosInstance with JWT token handling and automatic refresh. Depends on hooks for custom logic.
5	utils	Utility functions package providing helper methods used across the application. Contains guestCart.js (localStorage-based cart operations for non-authenticated users), formatting helpers, validation functions, and other reusable utilities. Depends on api for backend interactions when needed.
6	components	Reusable UI components package organized into subpackages: header (navigation and top bar components), sidebar (sidebar/drawer menu components), footer (footer component with links and info). Contains shared UI elements from shadcn/ui library (Button, Card, Input, Dialog) and feature-specific components (CourseCard, PostCard, CommentSection). Depends on store (accesses CartContext) and assets (uses images/icons).
7	pages	Page-level components package organized by user roles: admin (admin dashboard, category management, user management), moderator (content moderation, post reports, lecturer applications), student (student dashboard, my courses, learning interface, quiz taking), lecturer (lecturer dashboard, course creation, curriculum management, pricing). Each subpackage contains complete page views with business logic. Depends on components (imports reusable UI), api (backend calls), store (global state), and hooks (custom logic).

2. Database Design

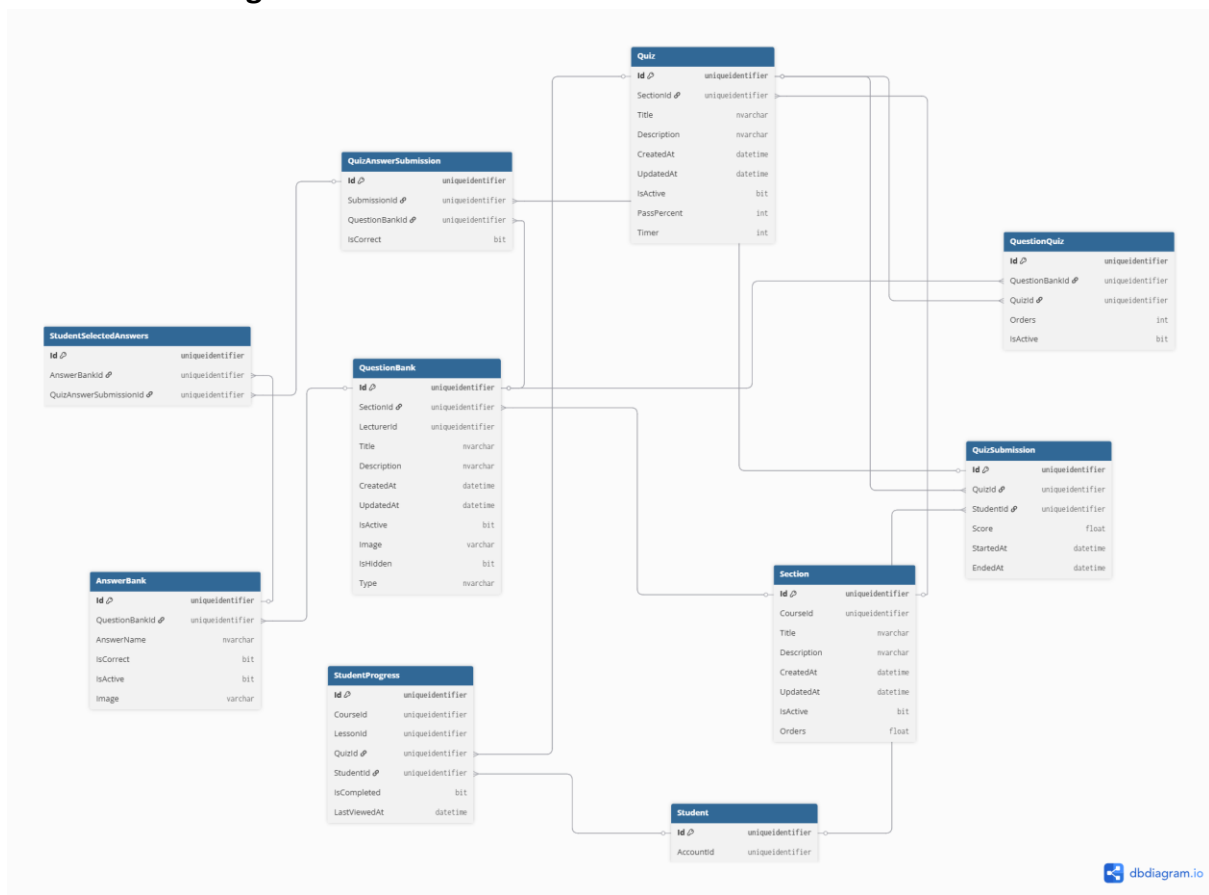


Figure 2.1. - data tables related to quiz

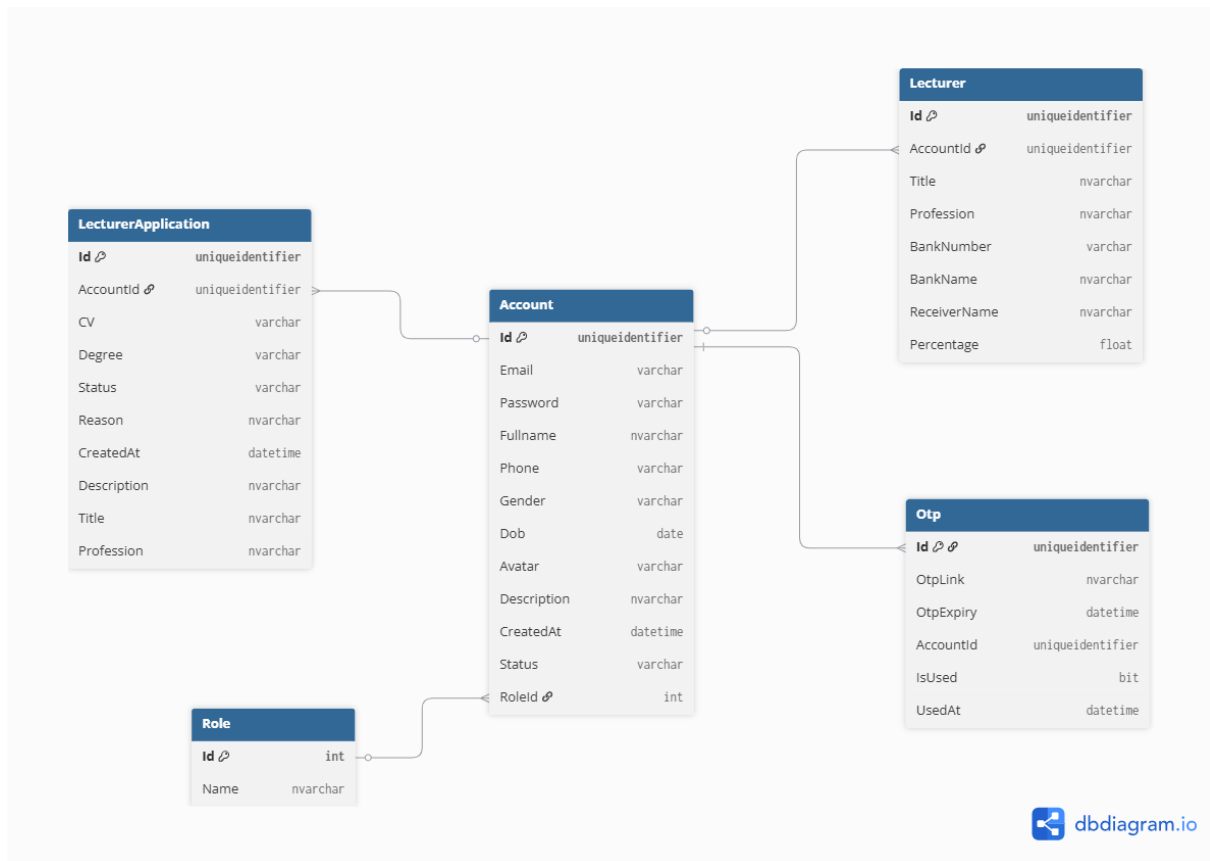


Figure 2.2. - data tables related to lecturer application

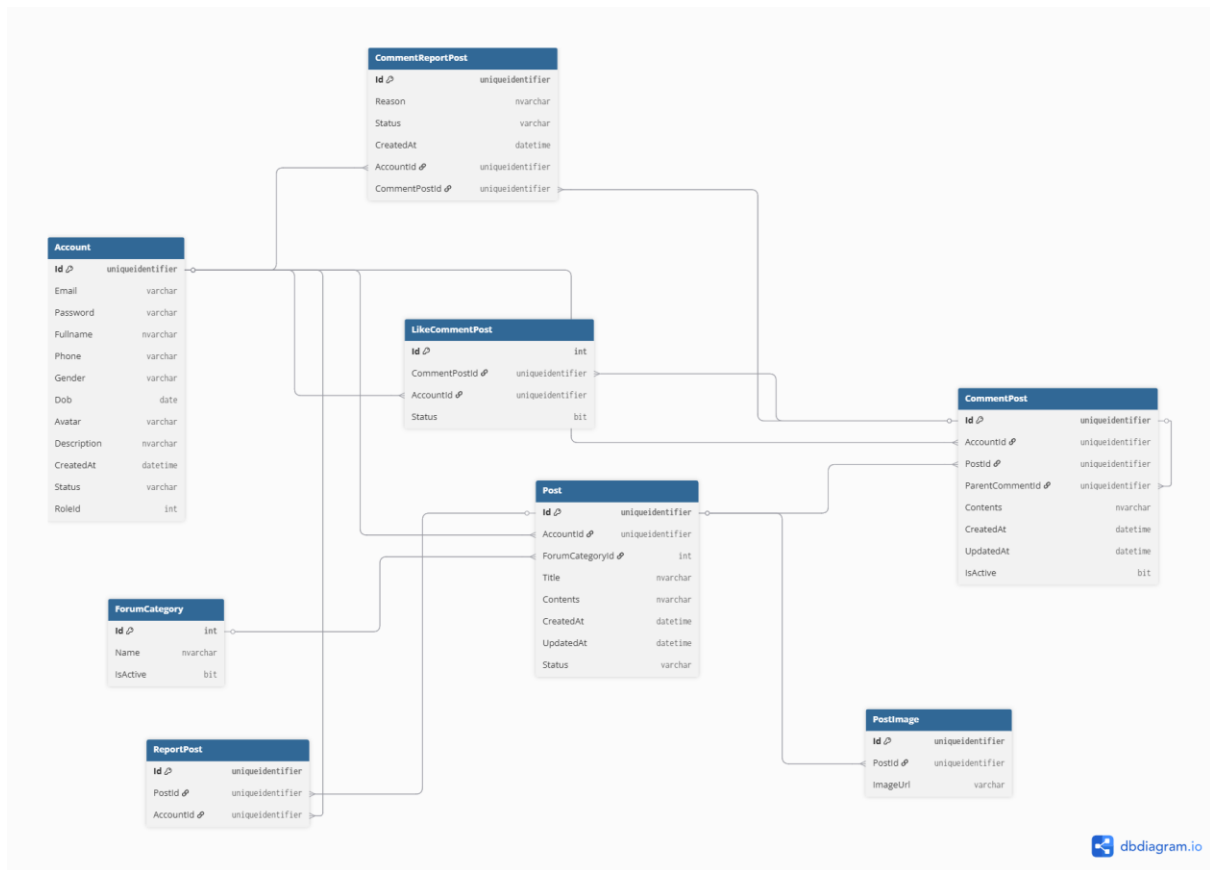


Figure 2.3. - data tables related to forum

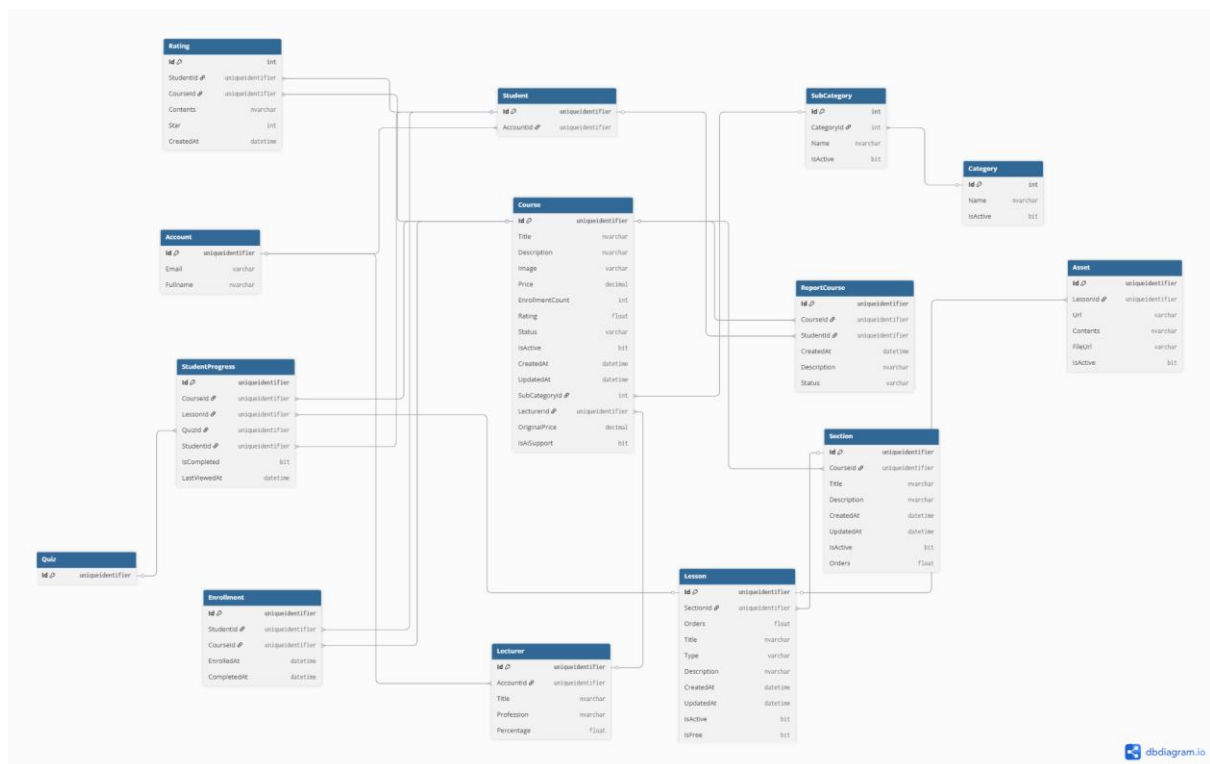


Figure 2.4. - data tables related to course

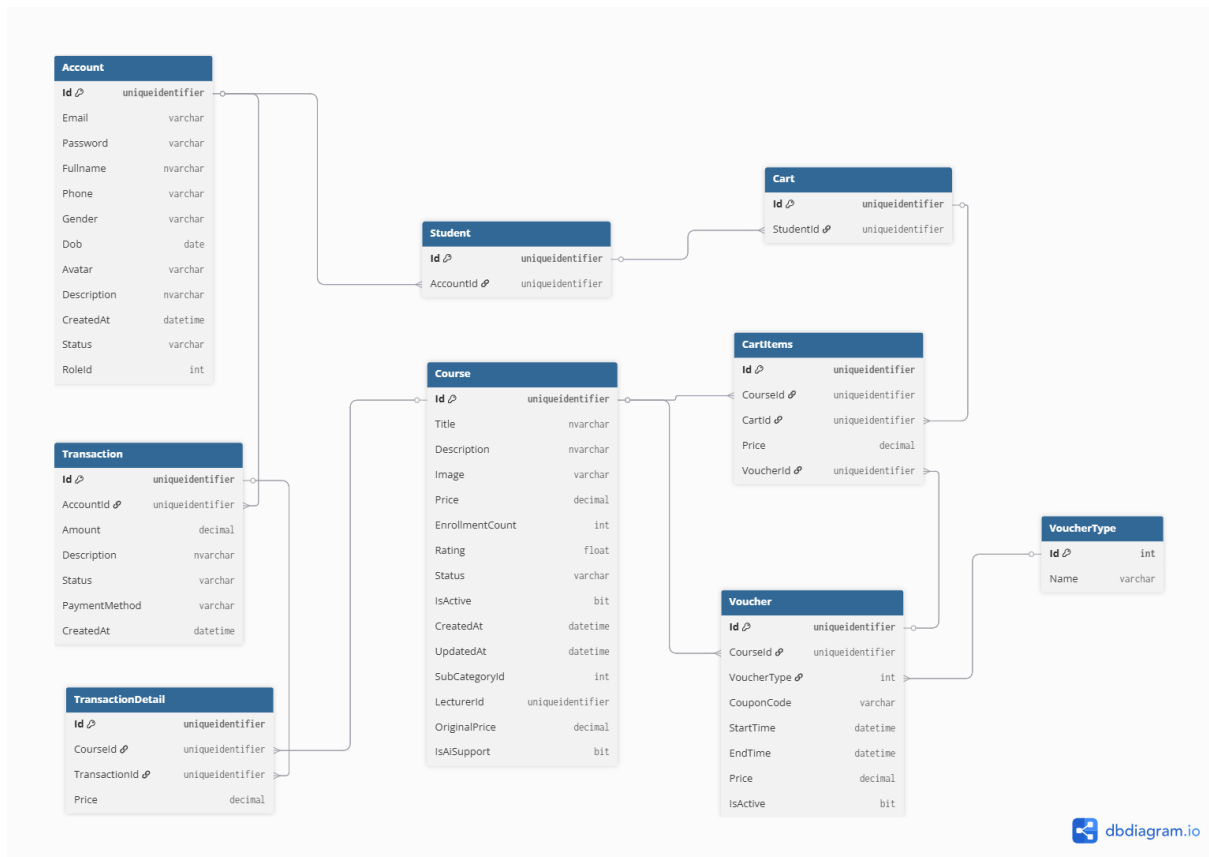


Figure 2.5. - data tables related to shopping cart & payment

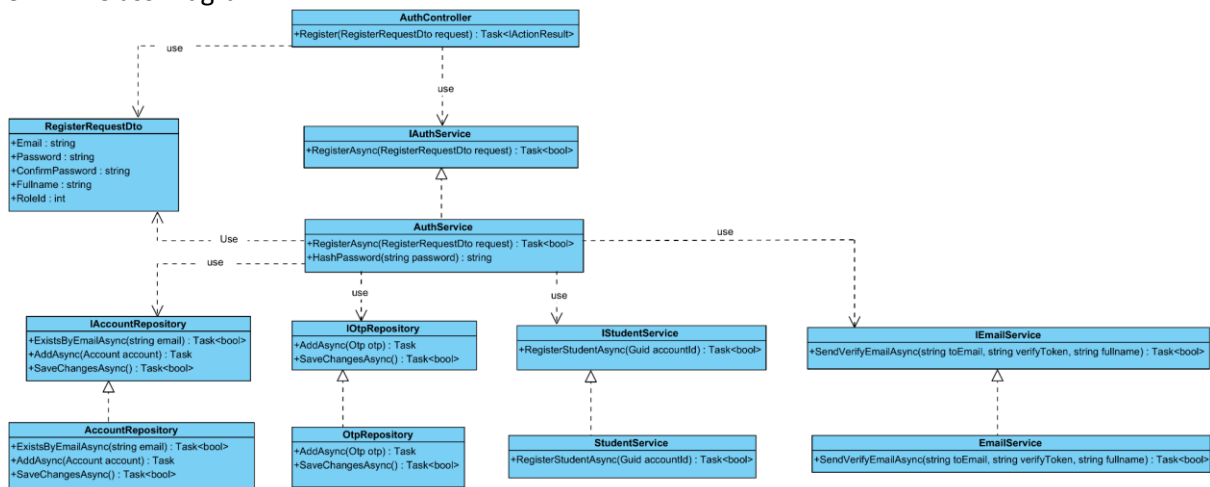
A detailed description of the tables in the database is found in section II.2 Database Design (**Report 4-Software Design Specification**).

3. Detailed Design

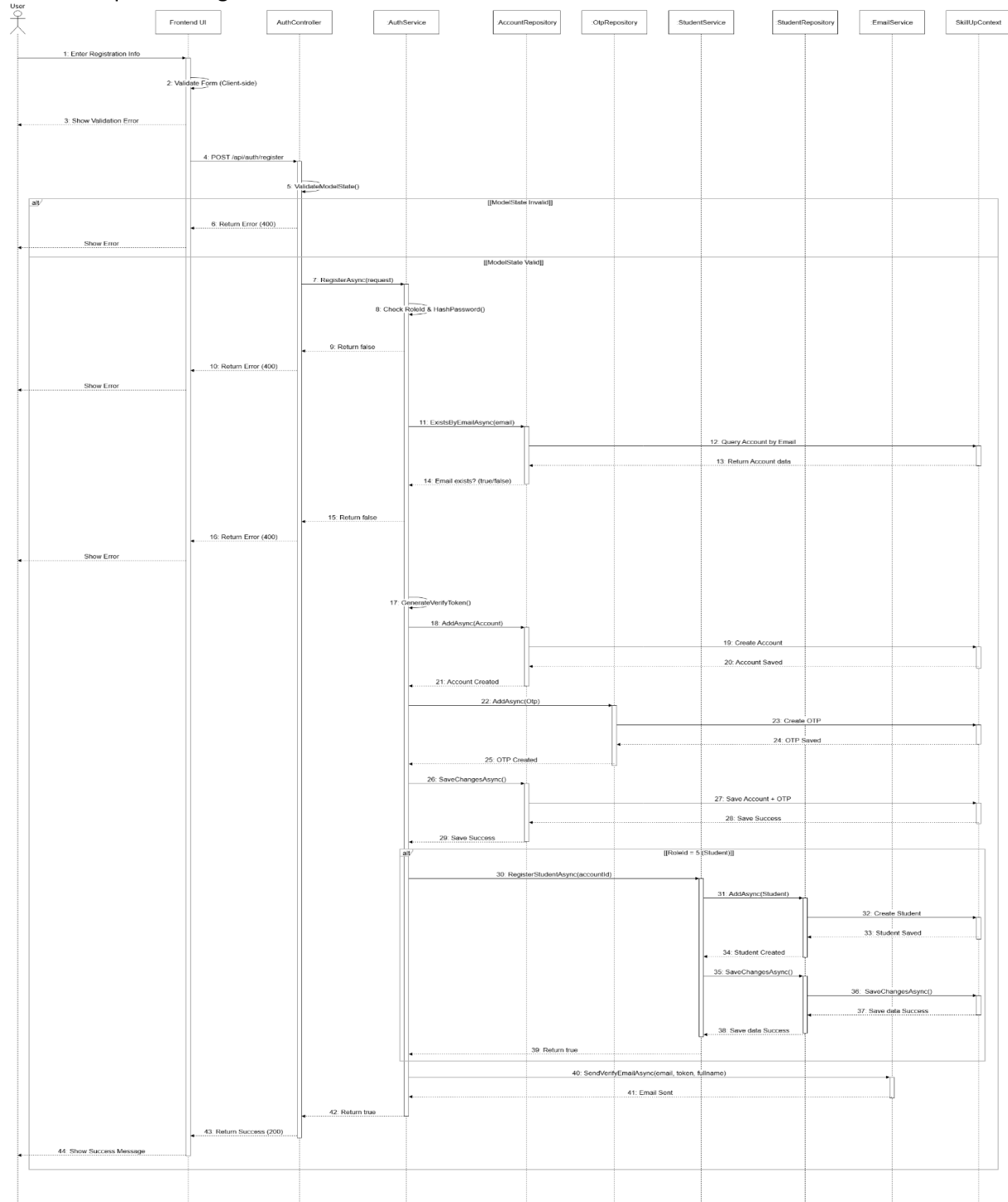
3.1 Common Feature

3.1.1 Register

3.1.1.1 Class Diagram



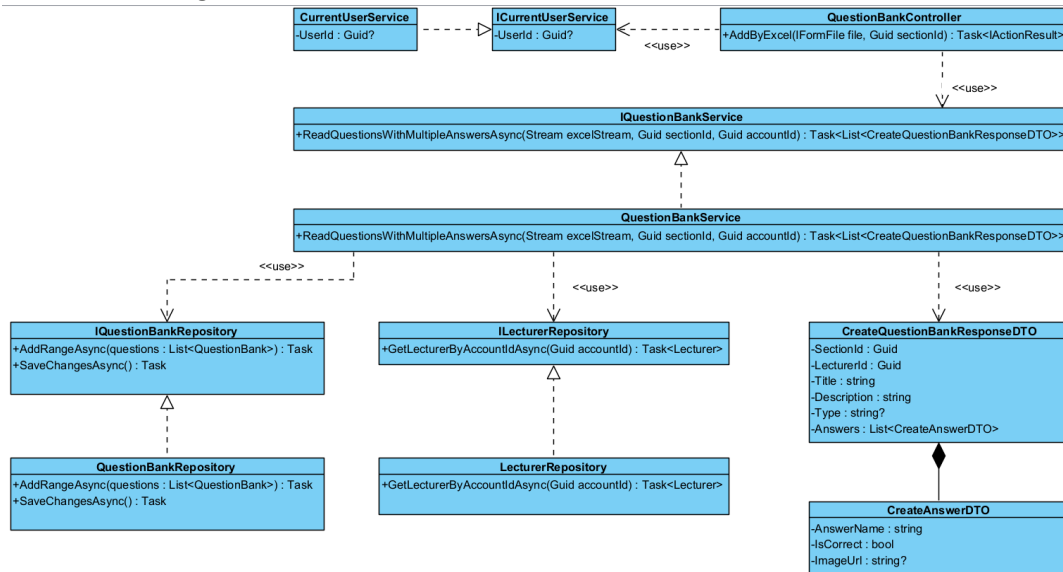
3.1.1.2 Sequence Diagram



3.2 Question Bank Feature

3.2.1 Import Question Bank by Excel

3.2.1.1 Class Diagram:



```
sequenceDiagram
    actor Lecturer
    participant Frontend UI
    participant QuestionBank Controller
    participant QuestionBank Service
    participant LecturerRepository as :Lecturer Repository
    participant QuestionBankRepository as :QuestionBank Repository
    participant SkillUpContext

    Note over Lecturer: 1. Click "Nhập từ Excel", Select File
    Frontend UI->>QuestionBank Controller: 2. POST /api/QuestionBank/add-by-excel
    Note over QuestionBank Controller: [File is Null or not .xlsx]
    QuestionBank Controller->>Frontend UI: 3. Return 400 Bad Request
    Frontend UI->>Lecturer: 4. Toast Error
    Note over QuestionBank Controller: [Valid File]
    Note over QuestionBank Controller: [User Not Authenticated]
    QuestionBank Controller->>Frontend UI: 5. Return 401 Unauthorized
    Frontend UI->>Lecturer: 6. Toast Error
    Note over QuestionBank Controller: [User Authenticated]
    QuestionBank Controller->>QuestionBank Service: 7. Call ReadQuestionsWith MultipleAnswersAsync()
    QuestionBank Service->>LecturerRepository: 8. GetLecturerByAccountId Async(accountId)
    LecturerRepository-->>QuestionBank Service: 9. Return Lecturer Entity
    Note over QuestionBank Service: [Lecturer Not Found]
    QuestionBank Service->>QuestionBank Controller: 10. Throw Exception
    QuestionBank Controller->>Frontend UI: 11. Return 500 Error
    Frontend UI->>Lecturer: 12. Toast Error
    Note over QuestionBank Service: [Lecturer Valid]
    Note over QuestionBank Service: [For Each Row (Starting Row 3)]
    loop
        QuestionBank Service->>QuestionBank Service: 13. Read Question Text & Answers (A-G)
        QuestionBank Service->>QuestionBank Service: 14. Parse Correct Options
        Note over QuestionBank Service: [Invalid Options (e.g. "H")]
        QuestionBank Service->>QuestionBank Controller: 15. Throw Exception ("Row X: Đáp án không hợp lệ")
        QuestionBank Controller->>Frontend UI: 16. Return 500 Error
    end
    QuestionBank Service->>QuestionBank Service: 17. Determine Type (Single/Multi)
    QuestionBank Service->>QuestionBank Service: 18. Instantiate QuestionBank & AnswerBanks
    QuestionBank Service->>QuestionBank Service: 19. Add to List<QuestionBank>
    QuestionBank Service->>QuestionBankRepository: 20. Call AddRangeAsync(questionBanks)
    QuestionBankRepository->>SkillUpContext: 21. Create Question Banks
    QuestionBank Service->>QuestionBankRepository: 22. Call SaveChangesAsync()
    QuestionBankRepository->>SkillUpContext: 23. Save Changes
    SkillUpContext-->>QuestionBankRepository: 24. Return Success
    QuestionBank Service->>QuestionBank Controller: 25. Return (void)
    QuestionBank Controller->>Frontend UI: 26. Return List<ResponseDTO>
    Frontend UI->>Lecturer: 27. Return 200 OK (Success Message)
    Frontend UI->>Lecturer: 28. Toast Success
    Frontend UI->>Frontend UI: 29. Close Modal & Refresh List
```

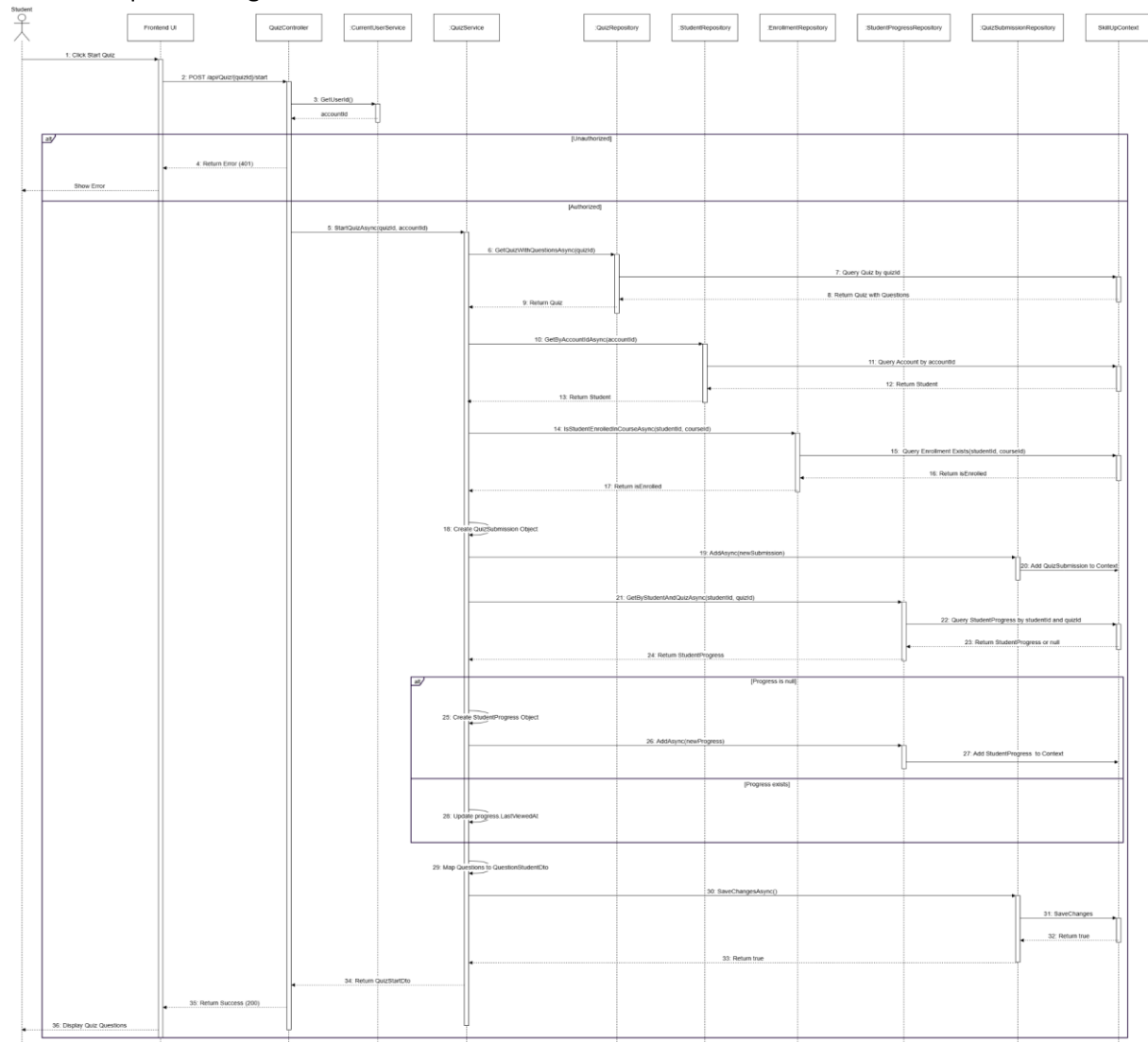
3.3.1 Do Quiz

The diagram illustrates the relationships between various components in a quiz system. Key classes include:

- QuizController**: Manages quiz operations, interacting with **QuizService** and **QuizRepository**.
- QuizService**: Provides business logic for quizzes, interacting with **QuizRepository** and **QuizAttemptRepository**.
- QuizRepository**: Manages the database for quizzes, interacting with **QuizController** and **QuizService**.
- QuizAttemptRepository**: Manages the database for quiz attempts, interacting with **QuizService** and **QuizAttemptService**.
- QuizAttemptService**: Provides business logic for quiz attempts, interacting with **QuizAttemptRepository** and **QuizAttemptController**.
- QuizAttemptController**: Manages quiz attempt operations, interacting with **QuizAttemptService** and **QuizAttemptRepository**.
- QuizQuestionRepository**: Manages the database for quiz questions, interacting with **QuizRepository** and **QuizQuestionService**.
- QuizQuestionService**: Provides business logic for quiz questions, interacting with **QuizQuestionRepository** and **QuizQuestionController**.
- QuizQuestionController**: Manages quiz question operations, interacting with **QuizQuestionService** and **QuizQuestionRepository**.
- QuizSubmissionRepository**: Manages the database for quiz submissions, interacting with **QuizSubmissionService** and **QuizSubmissionController**.
- QuizSubmissionService**: Provides business logic for quiz submissions, interacting with **QuizSubmissionRepository** and **QuizSubmissionController**.
- QuizSubmissionController**: Manages quiz submission operations, interacting with **QuizSubmissionService** and **QuizSubmissionRepository**.
- QuizProgressRepository**: Manages the database for quiz progress, interacting with **QuizProgressService** and **QuizProgressController**.
- QuizProgressService**: Provides business logic for quiz progress, interacting with **QuizProgressRepository** and **QuizProgressController**.
- QuizProgressController**: Manages quiz progress operations, interacting with **QuizProgressService** and **QuizProgressRepository**.

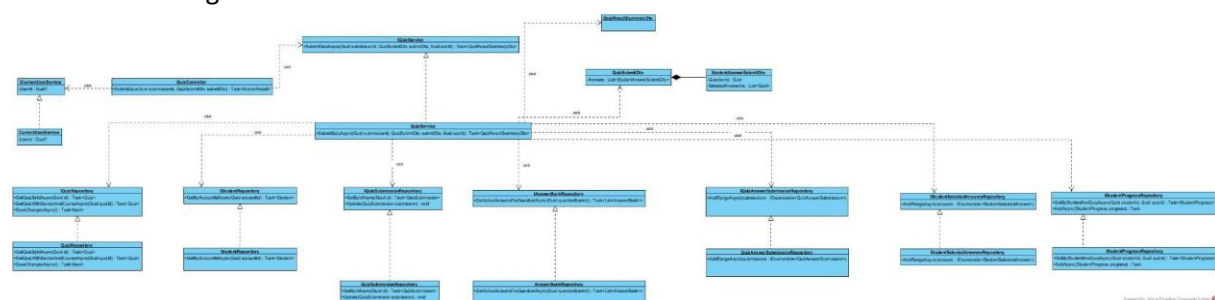
The diagram uses solid lines for associations and dashed lines for dependencies. It also includes a legend for the different types of relationships.

3.3.1.2 Sequence Diagram

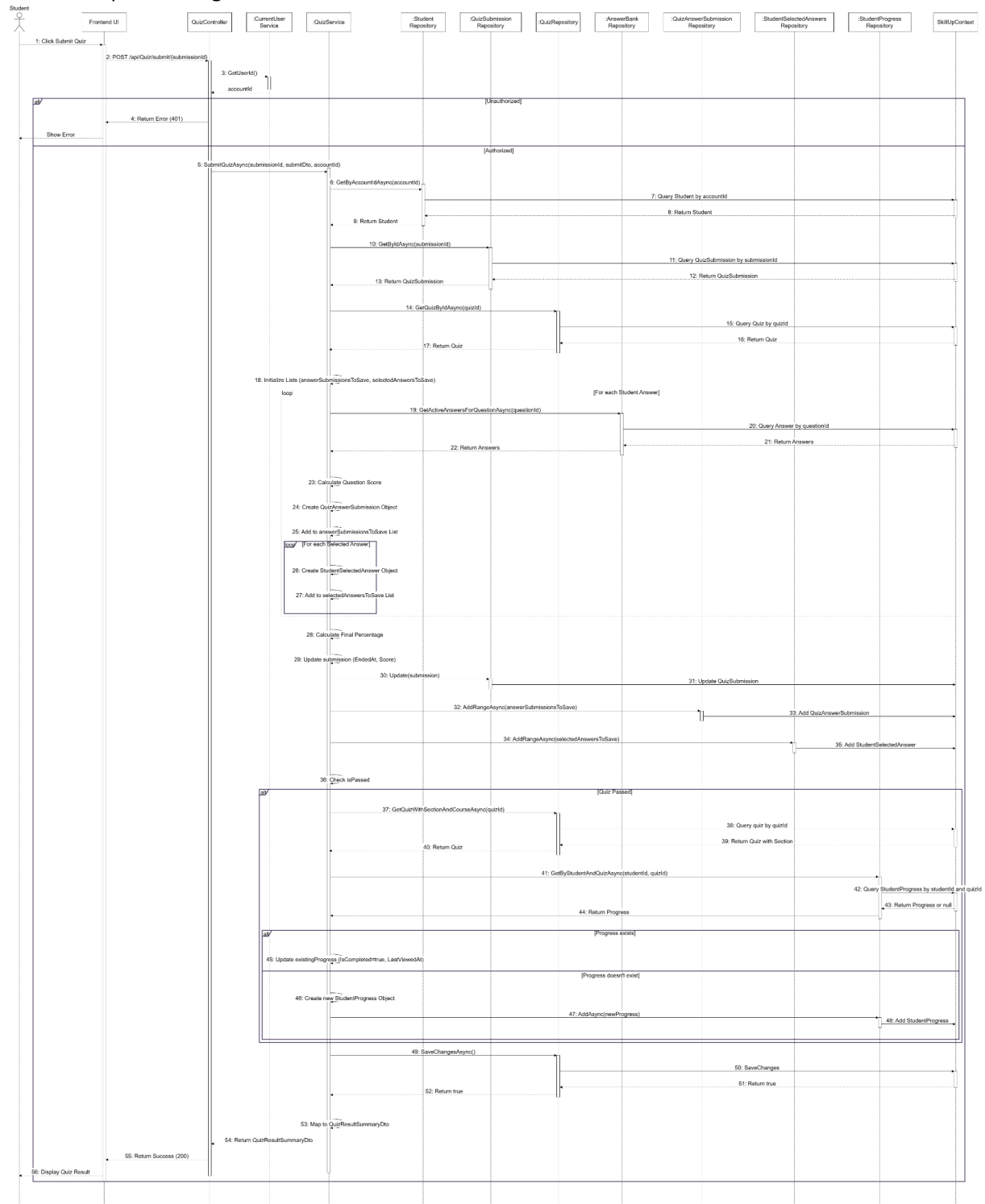


3.3.2 Submit Quiz

3.3.2.1 Class Diagram

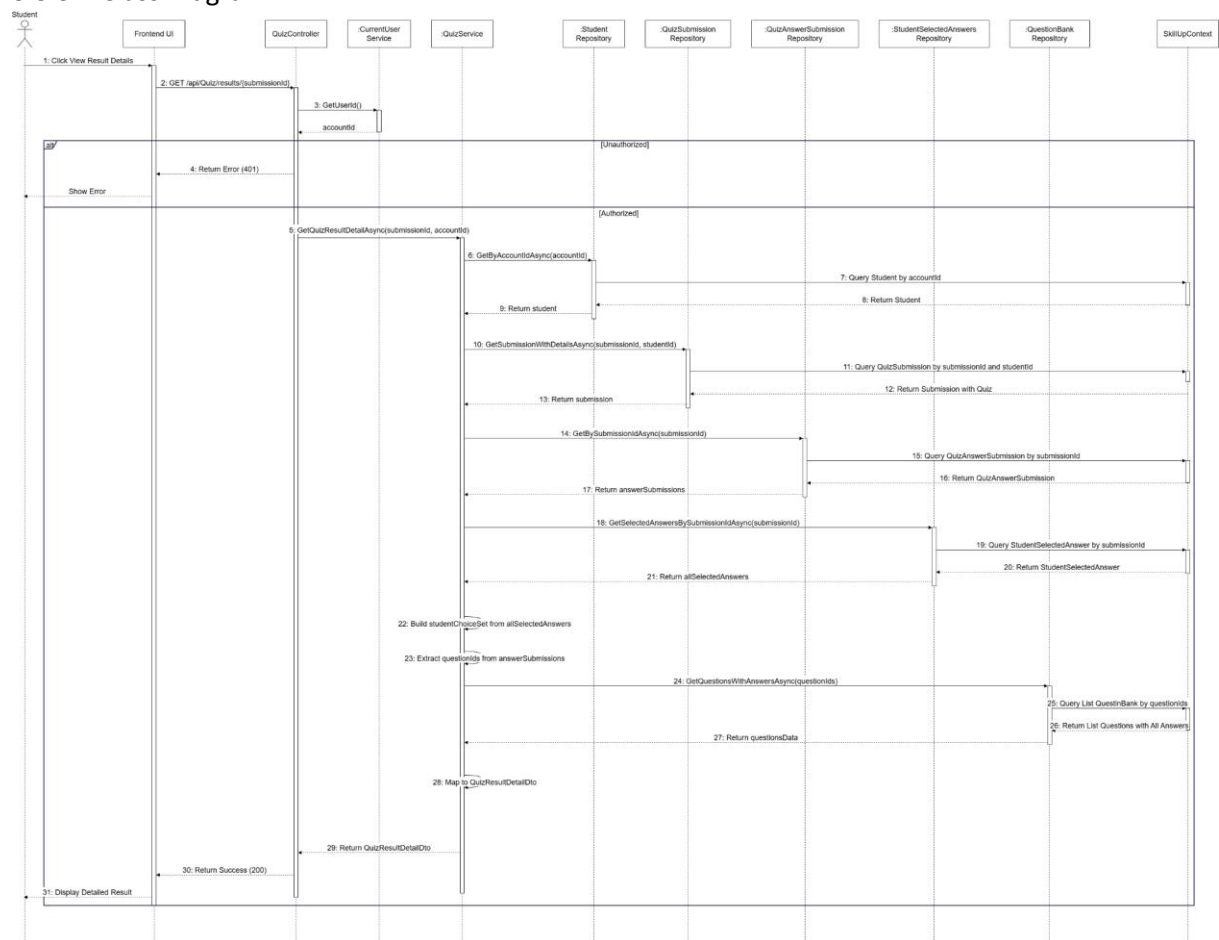


3.3.2.2 Sequence Diagram

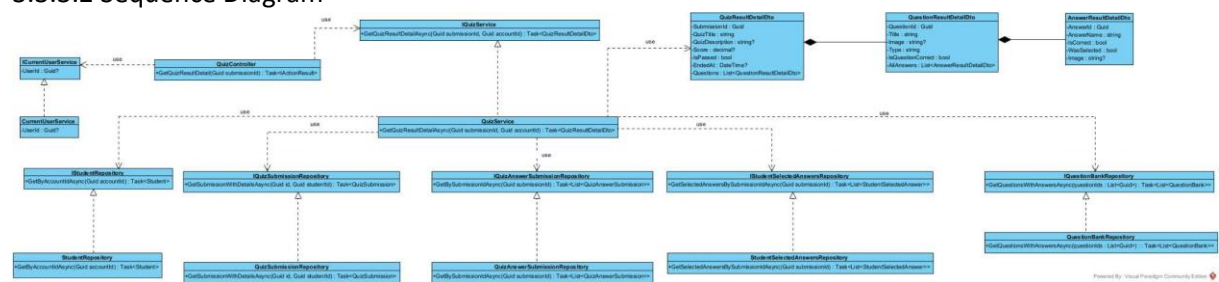


3.3.3 Get detail quiz result

3.3.3.1 Class Diagram



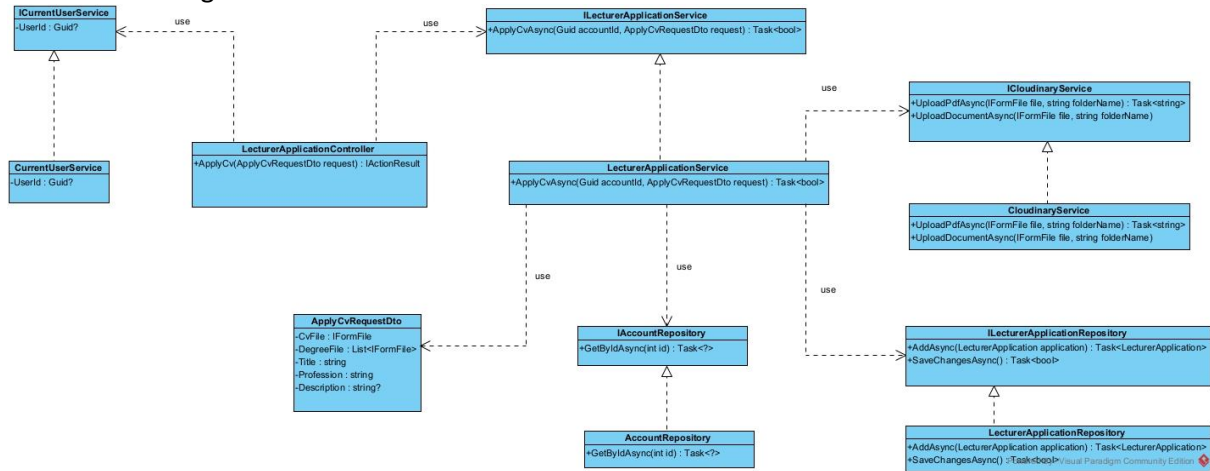
3.3.3.2 Sequence Diagram



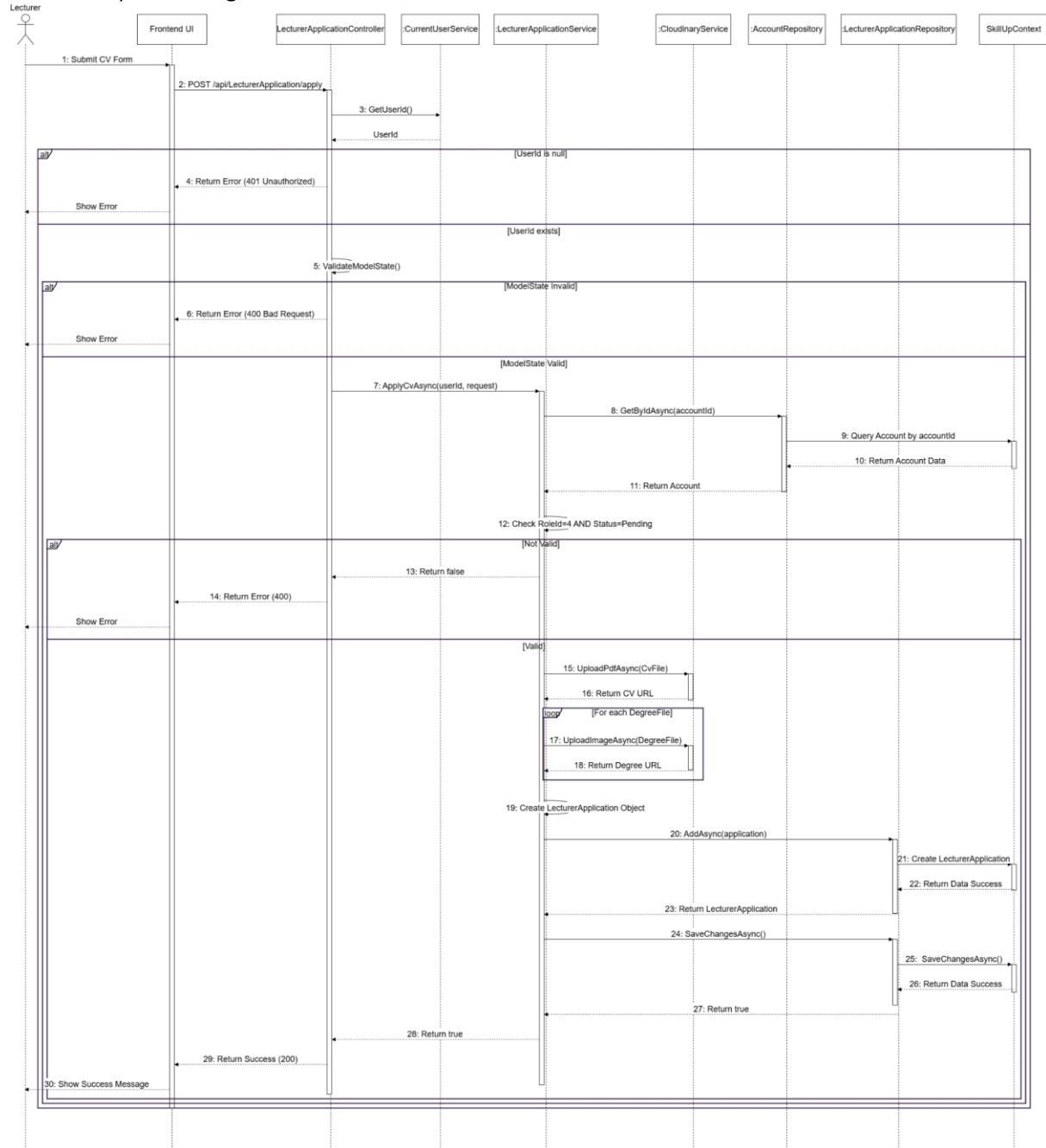
3.4 Lecturer Application Feature

3.4.1 Apply Lecturer Application

3.4.1.1 Class Diagram

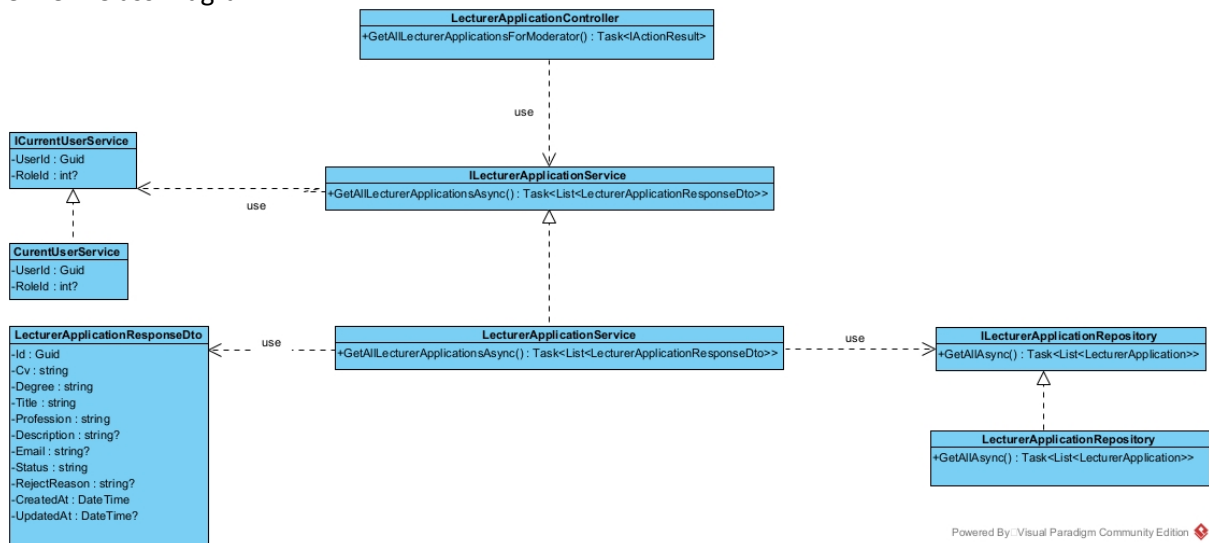


3.4.1.2 Sequence Diagram

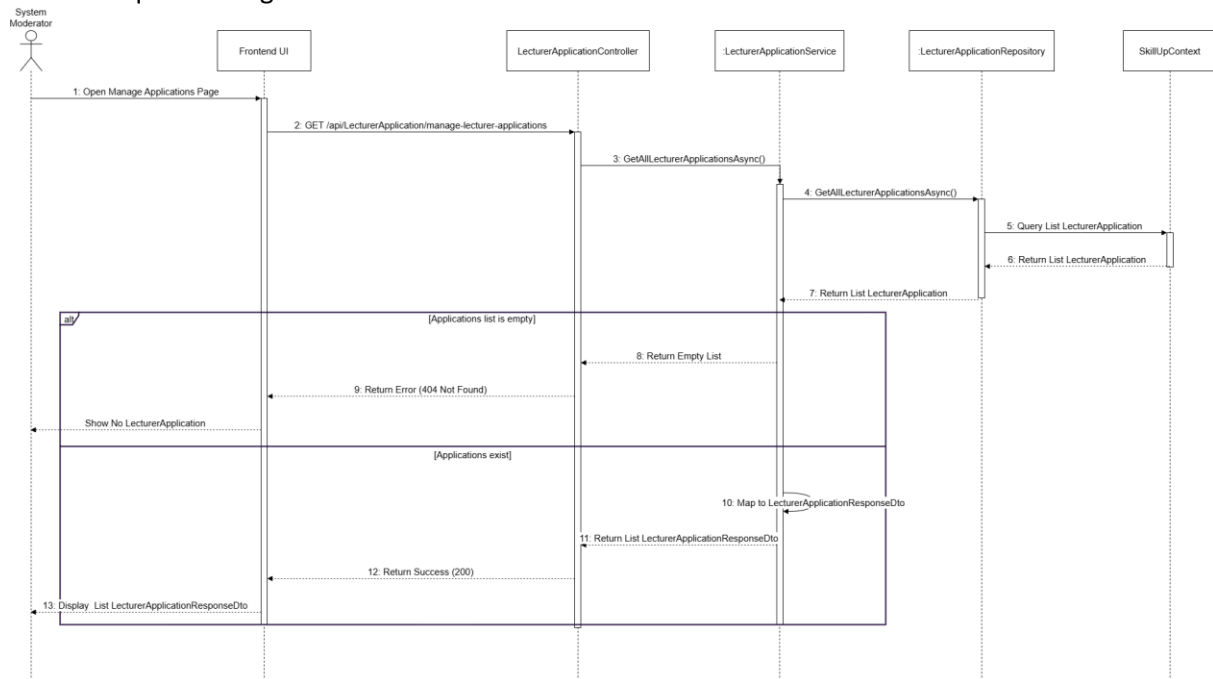


3.4.3 Get All Lecturer Applications

3.4.3.1 Class Diagram

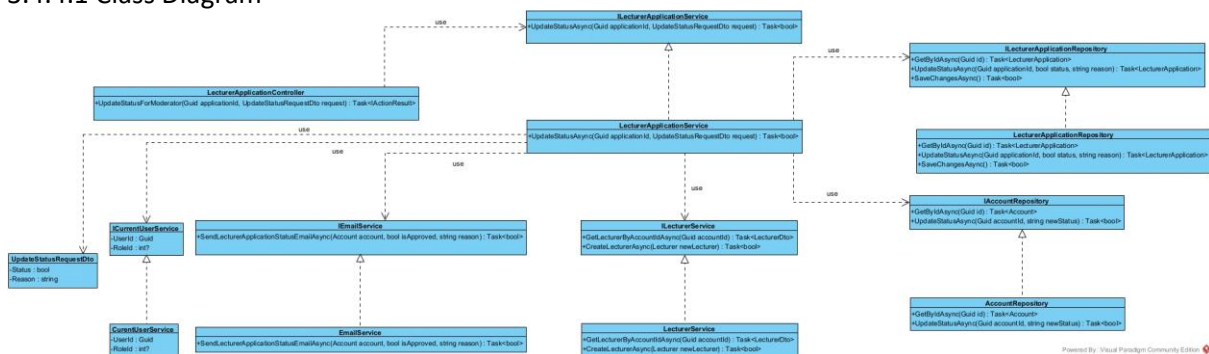


3.4.3.1 Sequence Diagram



3.4.4 Accept Lecturer Application

3.4.4.1 Class Diagram



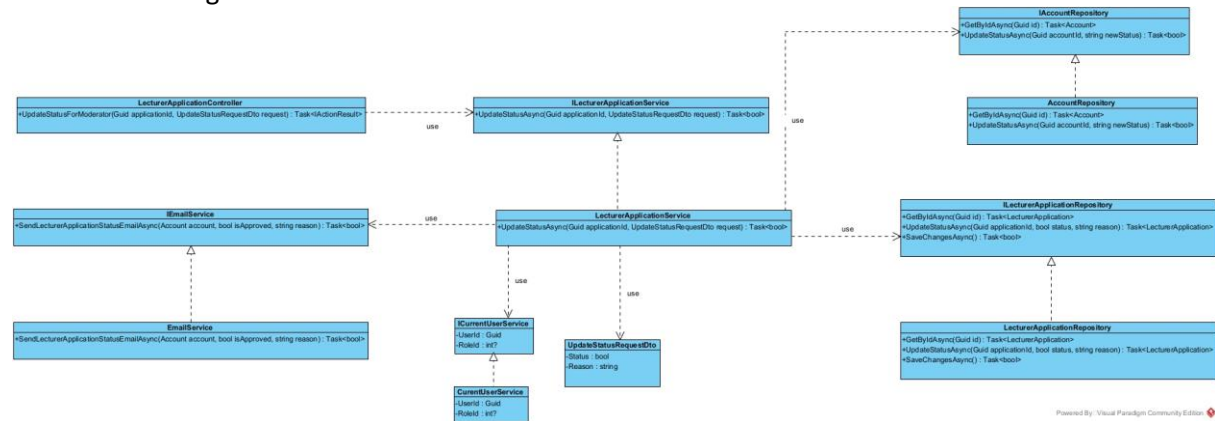
```
sequenceDiagram
    actor User as User Auditor
    participant Frontend UI
    participant LectureApplicationController
    participant LectureApplicationService
    participant LectureService
    participant EmailService
    participant AccountRepository
    participant LectureApplicationRepository
    participant StatusContext

    Note over Frontend UI: 1 Click accept
    Frontend UI->>LectureApplicationController: 2 PUT /update-status/{id} (Status=true)
    Note over Frontend UI: 3 Return Error
    LectureApplicationController-->>Frontend UI: Show Error

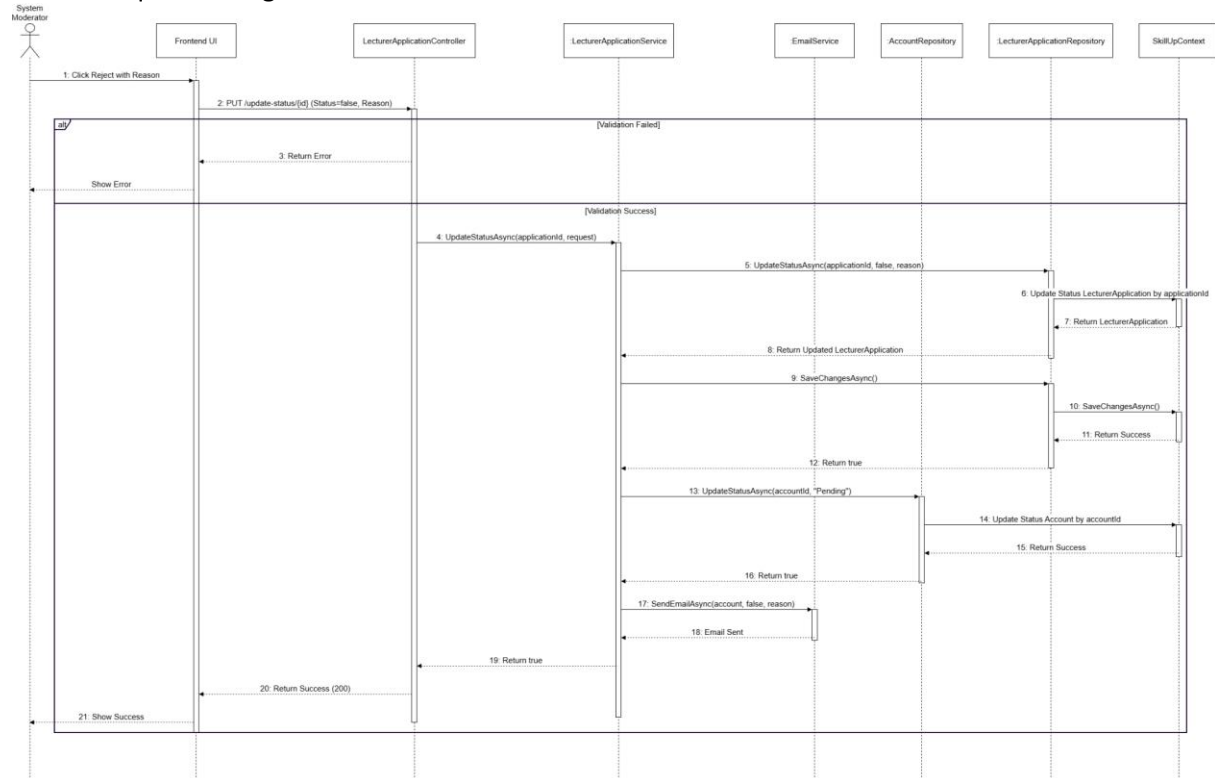
    Note over Frontend UI: 4 UpdateStatus(applicationId, request)
    LectureApplicationController->>LectureApplicationService: 4 UpdateStatus(applicationId, request)
    Note over Frontend UI: (Validation Failed)
    LectureApplicationService-->>Frontend UI: Show Error

    Note over Frontend UI: (Validation Success)
    LectureApplicationService->>LectureApplicationRepository: 5 UpdateStatus(applicationId, true, reason)
    LectureApplicationRepository->>StatusContext: Update Status Lecture/application by applicationId
    StatusContext-->>LectureApplicationRepository: 7 Return Lecture/application
    LectureApplicationRepository-->>LectureApplicationService: 8 Return Updated Lecture/application
    LectureApplicationService->>AccountRepository: 9 SaveChanges(applicationId)
    AccountRepository->>StatusContext: 10 SaveChanges(applicationId)
    StatusContext-->>AccountRepository: 11 Return Success
    AccountRepository-->>LectureApplicationService: 12 Return true
    LectureApplicationService->>LectureService: 13 CreateLecture(applicationId, newLecture)
    LectureService->>StatusContext: 14 Create Lecture
    StatusContext-->>LectureService: 15 Return Success
    LectureService-->>LectureApplicationService: 16 Return true
    LectureApplicationService->>EmailService: 17 UpdateStatus(applicationId, accountId, "Active")
    EmailService->>StatusContext: 18 Update Status Account by accountId
    StatusContext-->>EmailService: 19 Return Success
    EmailService-->>LectureApplicationService: 20 Return true
    LectureApplicationService->>EmailService: 21 SendEmail(applicationId, true, reason)
    EmailService-->>LectureApplicationService: 22 Email Sent
    LectureApplicationService-->>LectureApplicationController: 23 Return true
    LectureApplicationController-->>Frontend UI: 24 Return Success (200)
    Frontend UI-->>User: 25 Show Success
```

3.4.5.1 Class Diagram



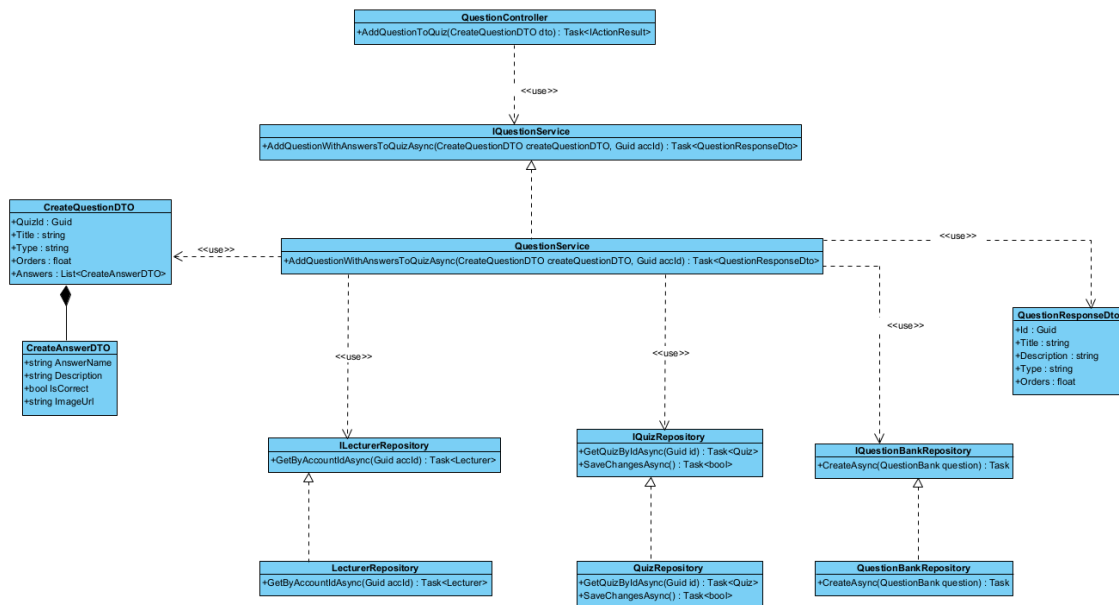
3.4.5.2 Sequence Diagram



3.5 Question Feature

3.5.1 Add question to quiz

3.5.1.1 Class Diagram



```
sequenceDiagram
    actor User
    participant Frontend UI
    participant QuizController
    participant QuizService
    participant LectureRepository
    participant SectionRepository
    participant QuizRepository
    participant SMCQuizContent

    User->>Frontend UI: 1. Enter Quiz info (Title, Time, Priority) & Click "Create"
    Frontend UI->>QuizController: 2. Check Quiz Validation
    QuizController->>QuizService: 3. POST /api/Quiz/CreateQuiz
    QuizService->>QuizController: 4. Check Topic & Module Date
    QuizService->>QuizService: [Valid Request]
    QuizService->>QuizService: [Invalid Request]
    QuizService->>LectureRepository: 6. GetQuizContent(topic, module)
    LectureRepository->>SectionRepository: 7.1. Query Lecture by Topic/Module
    SectionRepository-->>LectureRepository: 7.2. Return Entry
    LectureRepository-->>QuizService: 7.3. Lecture Data
    QuizService->>SectionRepository: 8. GetSectionEntryByTopicAndModule
    SectionRepository->>QuizRepository: 8.1. Query Section with Course Details
    QuizRepository-->>SectionRepository: 8.2. Return Entry
    SectionRepository-->>QuizService: 8.3. Section Data
    QuizService->>QuizController: 9. Create Quiz (topic, module, section)
    QuizController->>QuizService: 10. Save Quiz (topic, module, section)
    QuizService->>QuizRepository: 11. Save Quiz (topic, module, section)
    QuizRepository-->>QuizService: 12. Return Quiz ID
    QuizService->>QuizController: 13. Return Quiz ID
    QuizController->>Frontend UI: 14. Redirect to Quiz Detail / Show Success
    QuizService->>QuizController: 15. Return Quiz ID
```

3.5.2.1 Class Diagram

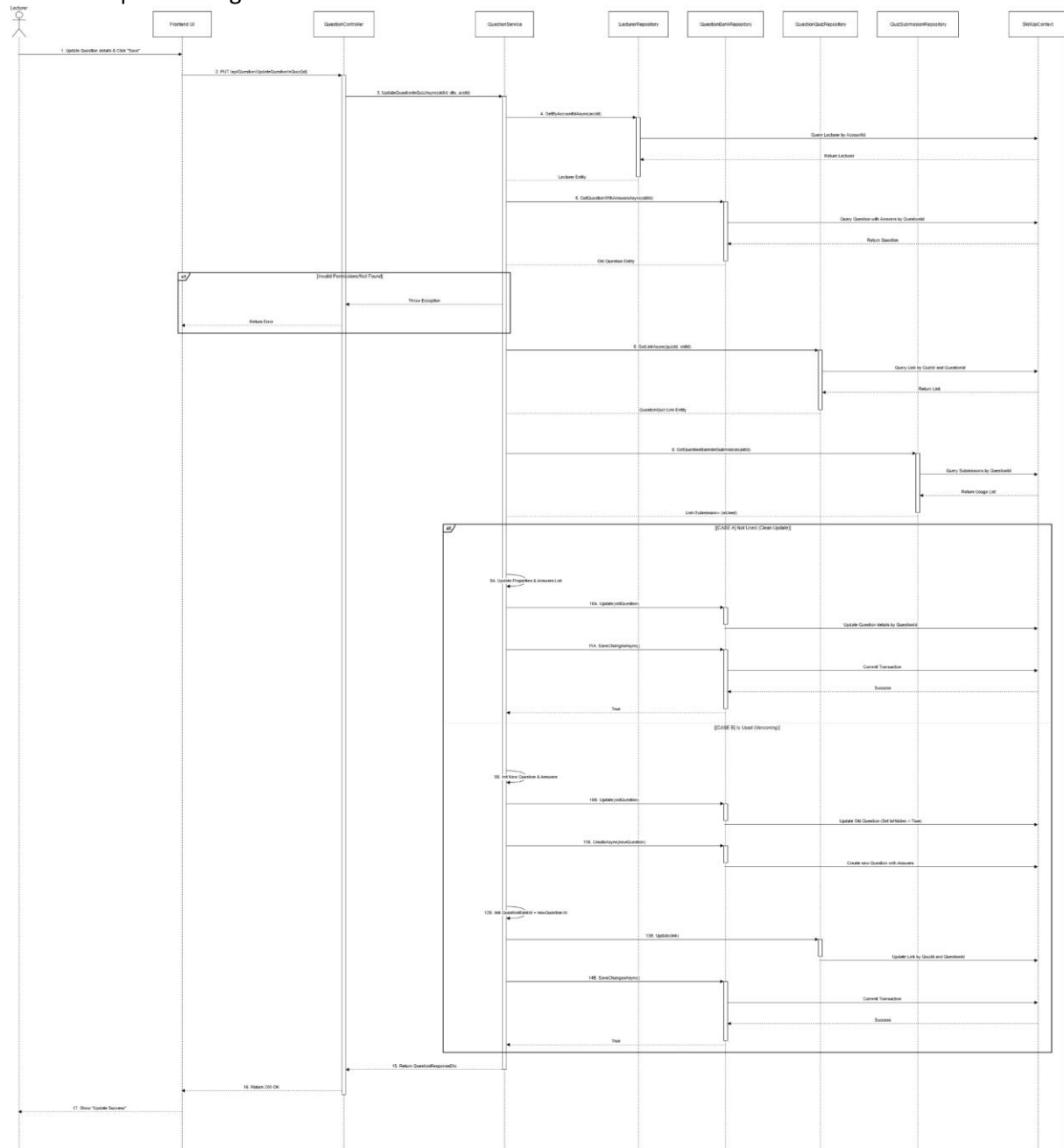
```

classDiagram
    class QuestionController {
        +UpdateQuestionInQuiz(Guid questionId, UpdateQuestionDTO dto) : Task
    }
    class QuestionService {
        +UpdateQuestionInQuizAsync(Guid oldQuestionId, UpdateQuestionDTO dto, Guid accid) : Task
    }
    class QuestionService {
        +UpdateQuestionInQuizAsync(Guid oldQuestionId, UpdateQuestionDTO dto, Guid accid) : Task
    }
    class UpdateQuestionDTO {
        +QuizId : Guid
        +Title : string
        +Type : string
        +Orders : float
        +Answers : List<UpdateAnswerDTO>
    }
    class UpdateAnswerDTO {
        +string AnswerName
        +bool IsCorrect
        +string ImageUrl
    }
    class IQuestionBankRepository {
        +GetByIdAsync(Guid id) : Task
        +CreateAsync(QuestionBank question) : Task
    }
    class QuestionBankRepository {
        +GetByIdAsync(Guid id) : Task
        +CreateAsync(QuestionBank question) : Task
    }
    class ILecturerRepository {
        +GetQuizByIdAsync(Guid id) : Task
        +SaveChangesAsync() : Task
    }
    class LecturerRepository {
        +GetQuizByIdAsync(Guid id) : Task
        +SaveChangesAsync() : Task
    }
    class IQuizRepository {
        +GetLinkAsync(Guid quizId, Guid qBankId) : Task
        +Update(QuestionQuiz questionQuiz) : void
    }
    class QuizRepository {
        +GetLinkAsync(Guid quizId, Guid qBankId) : Task
        +Update(QuestionQuiz questionQuiz) : void
    }
    class QuestionResponseDto {
        +Id : Guid
        +Title : string
        +Description : string
        +Orders : double
        +Image : string
        +Type : string
        +Answers : List<AnswerResponseDto>
    }
    class AnswerResponseDto {
        +Id : Guid
        +Content : string
        +IsCorrect : bool
    }

    QuestionController --> QuestionService : <<use>>
    QuestionService <|-- QuestionService
    QuestionService --> UpdateQuestionDTO : <<use>>
    QuestionService --> QuestionResponseDto : <<use>>
    UpdateQuestionDTO --> UpdateAnswerDTO : <<use>>
    UpdateQuestionDTO --> IQuestionBankRepository : <<use>>
    UpdateQuestionDTO --> ILecturerRepository : <<use>>
    UpdateQuestionDTO --> IQuizRepository : <<use>>
    UpdateQuestionDTO --> AnswerResponseDto : <<use>>
    QuestionBankRepository --|> IQuestionBankRepository
    LecturerRepository --|> ILecturerRepository
    QuizRepository --|> IQuizRepository
    AnswerResponseDto --|> QuestionResponseDto

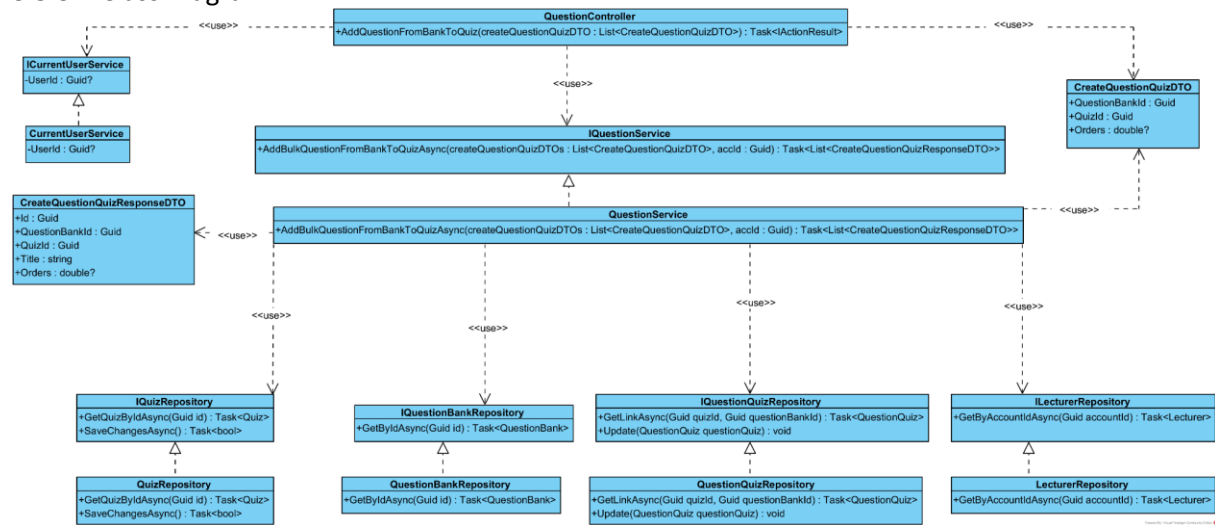
```

3.5.2.2 Sequence Diagram

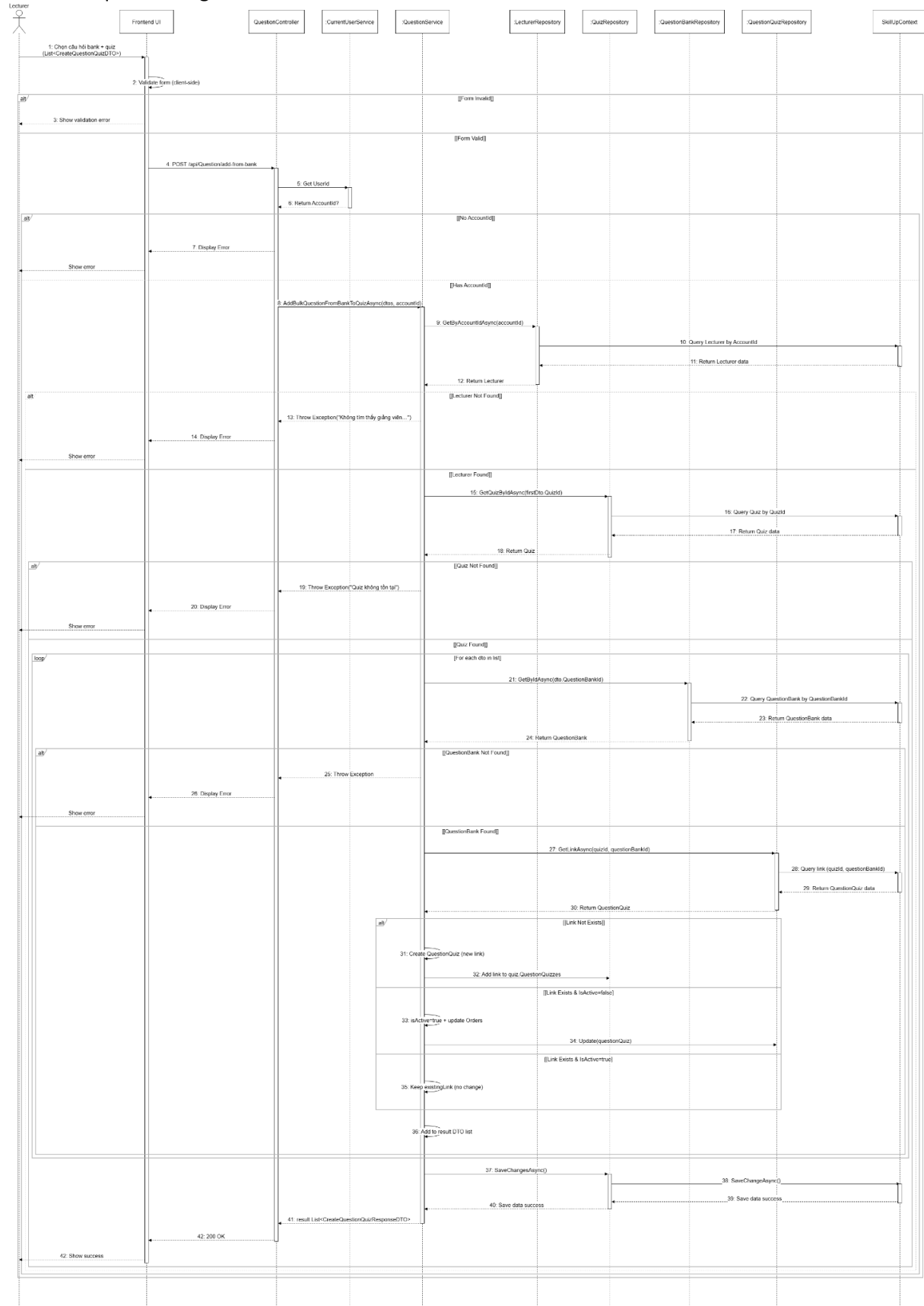


3.5.3 Add From bank

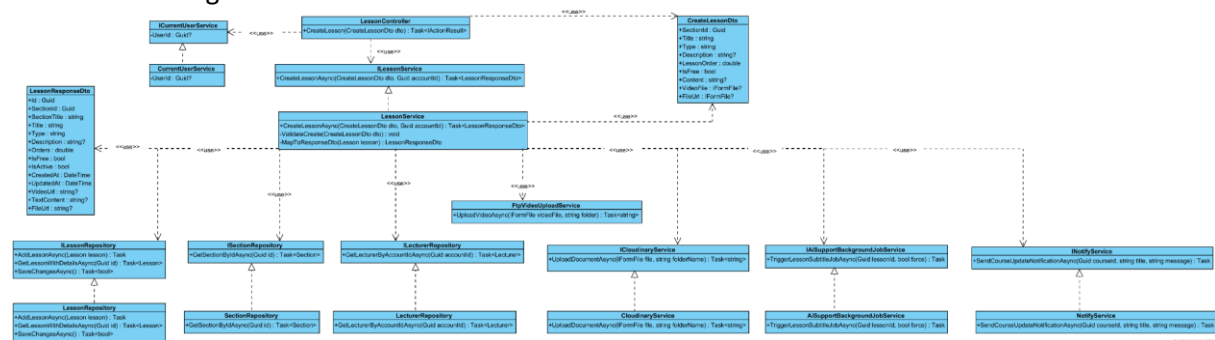
3.5.3.1 Class Diagram



3.5.3.2 Sequence Diagram



3.6.1.1 Class Diagram

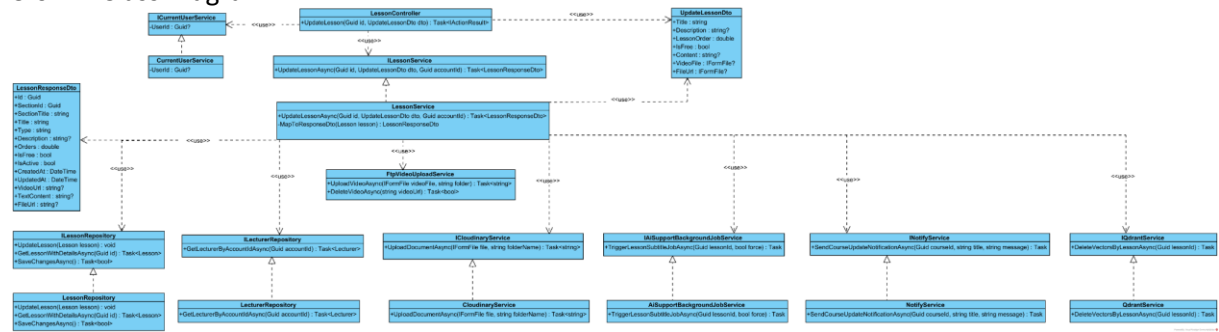


OLP-SkillUp – Report 7



3.6.2 Update Lesson

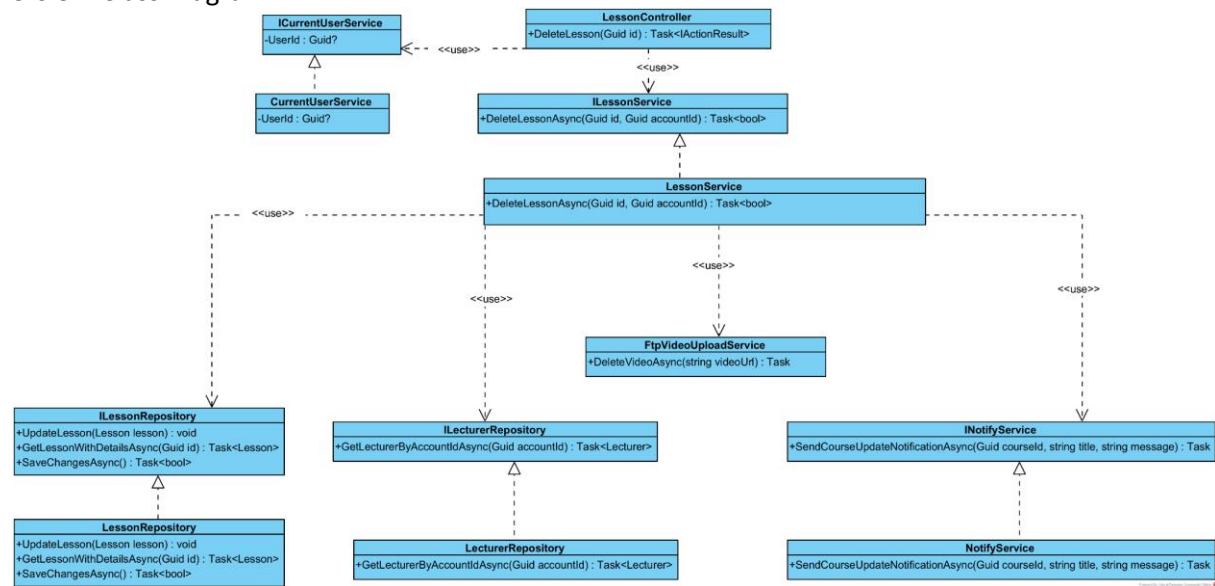
3.6.2.1 Class Diagram



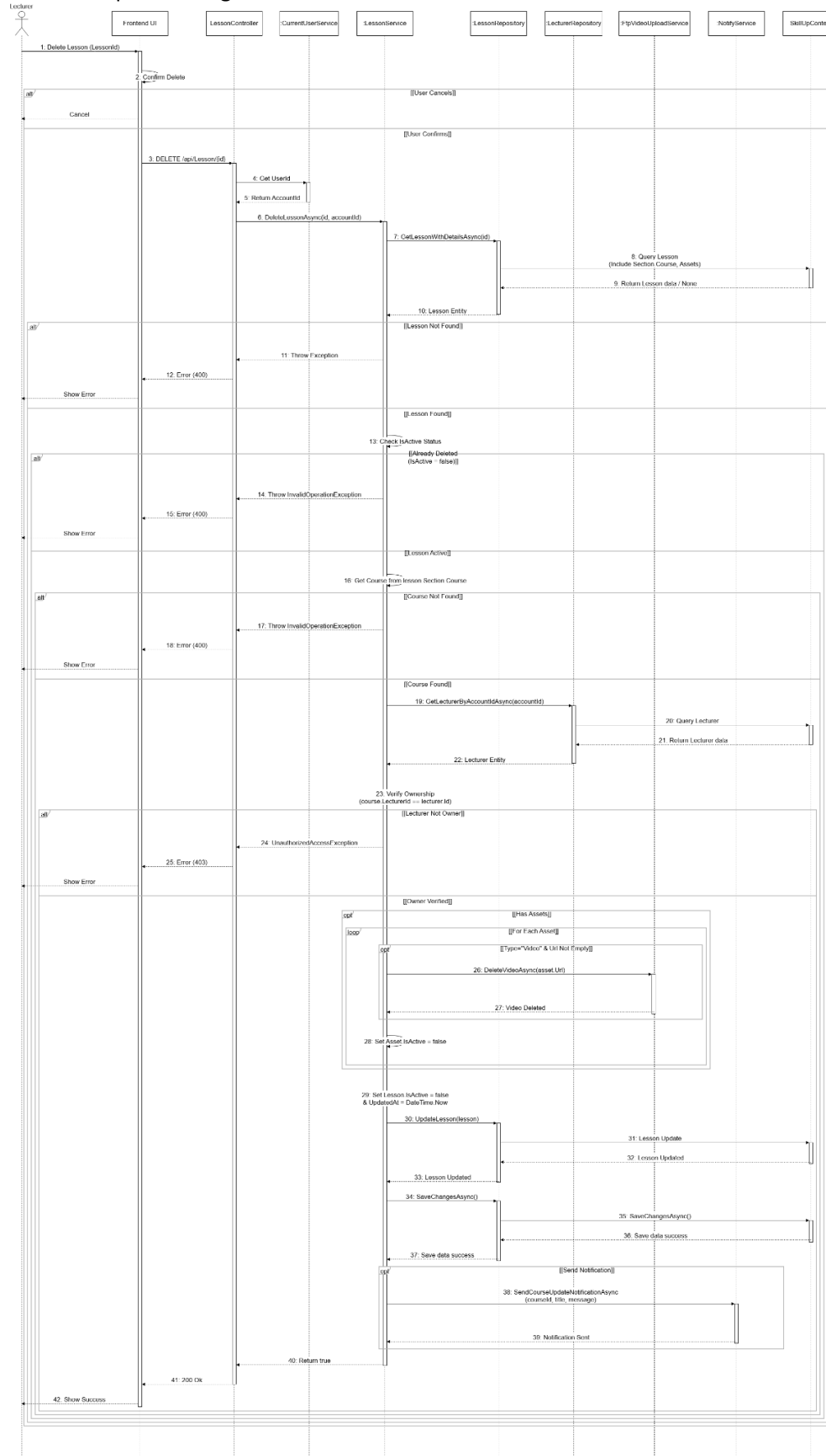
[illegible]

3.6.3.1 Class Diagram

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----



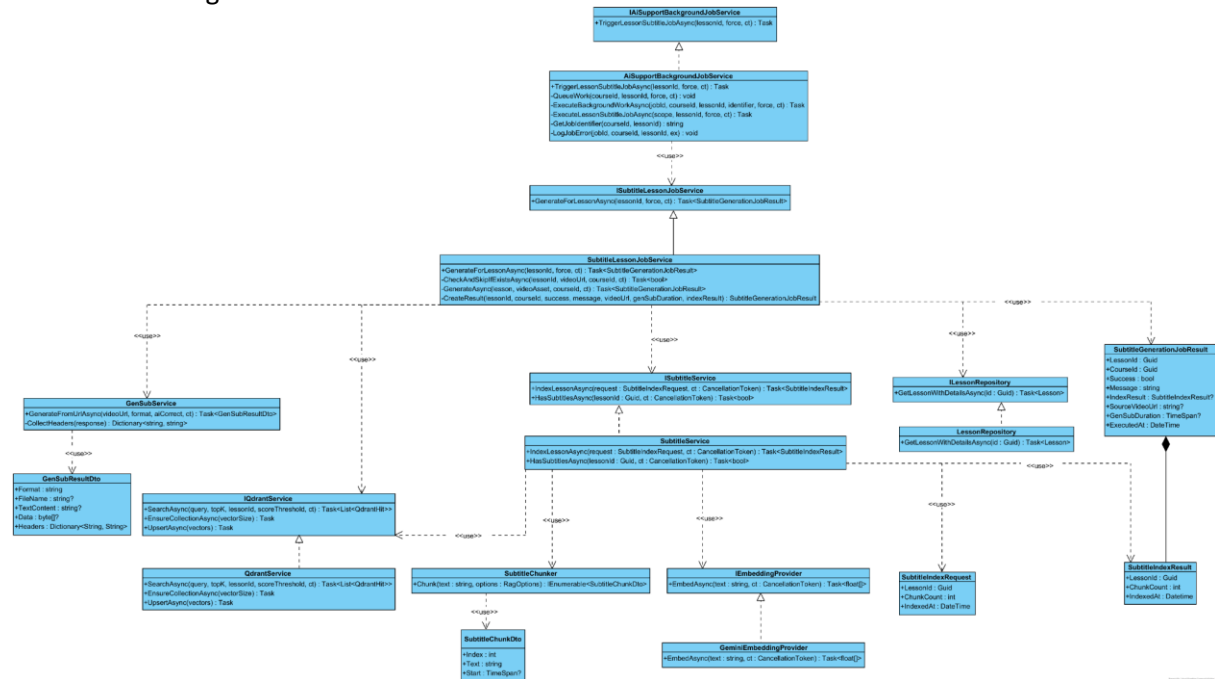
3.6.3.2 Sequence Diagram



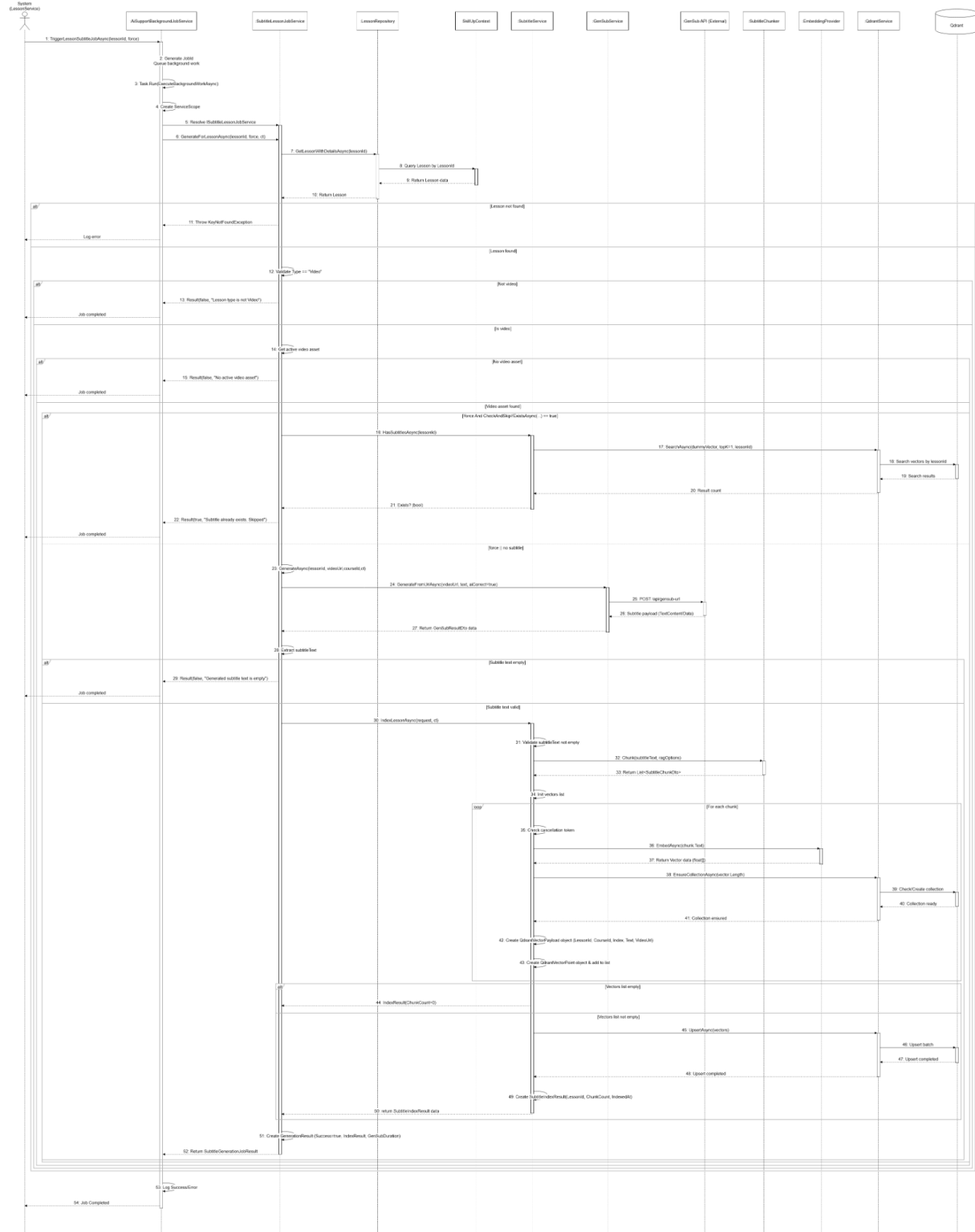
3.7 AI Feature

3.7.1 Subtitle Generation Background Job

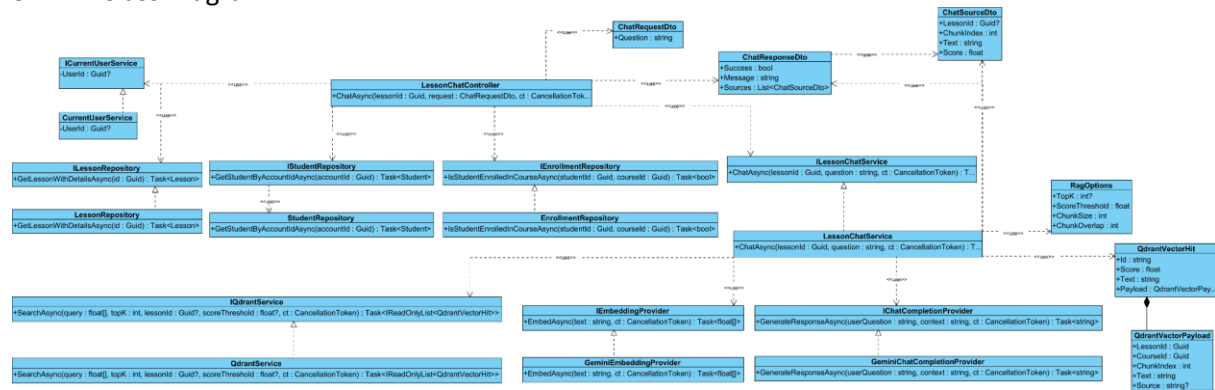
3.7.1.1 Class Diagram



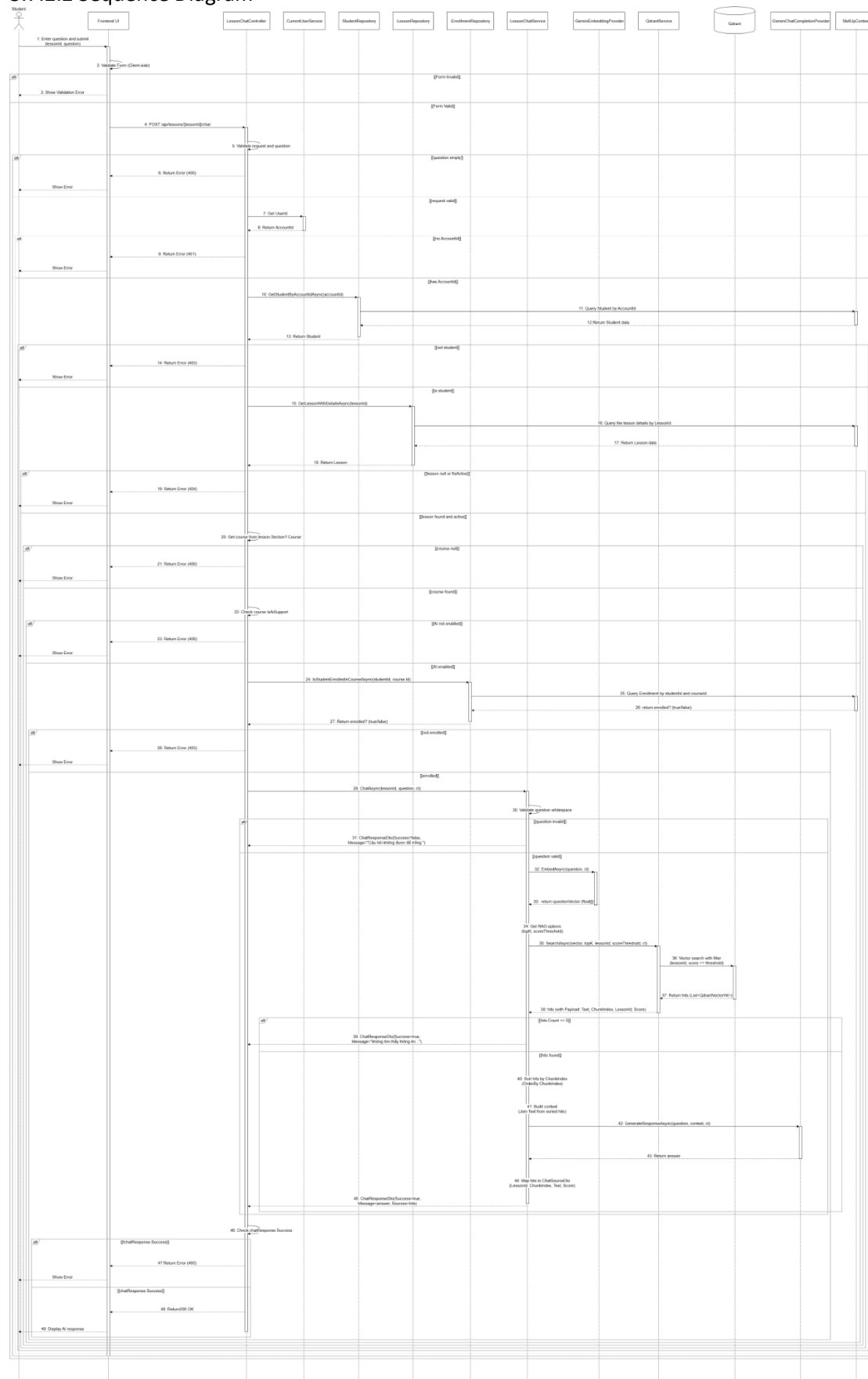
3.7.1.2 Sequence Diagram



3.7.2.1 Class Diagram



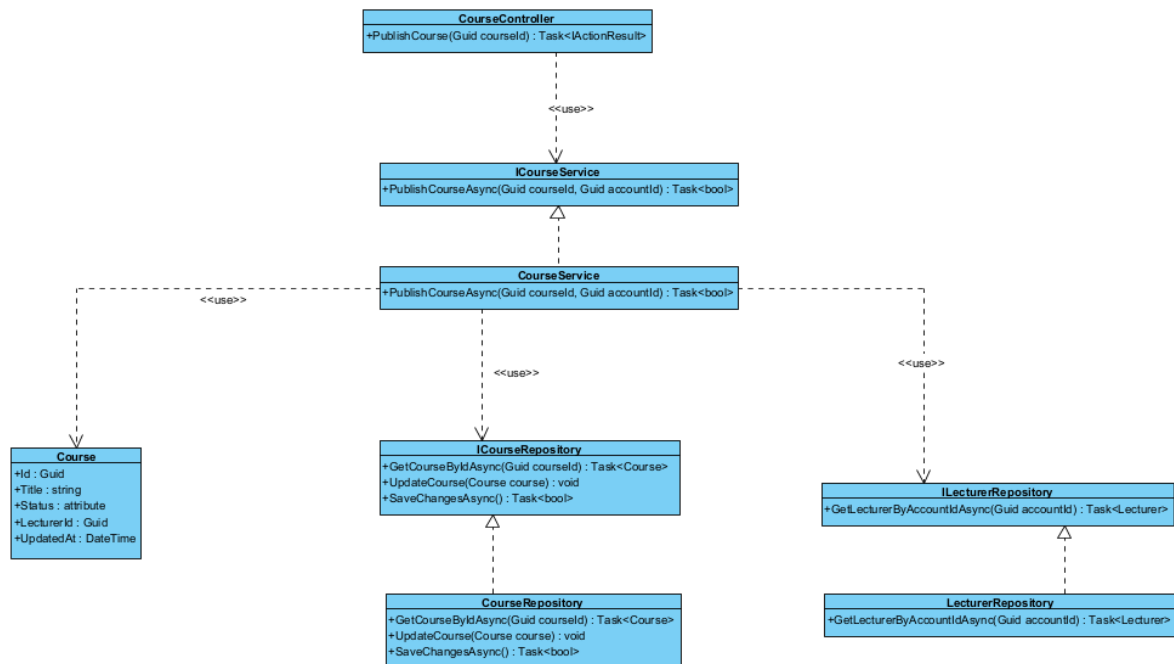
3.7.2.2 Sequence Diagram



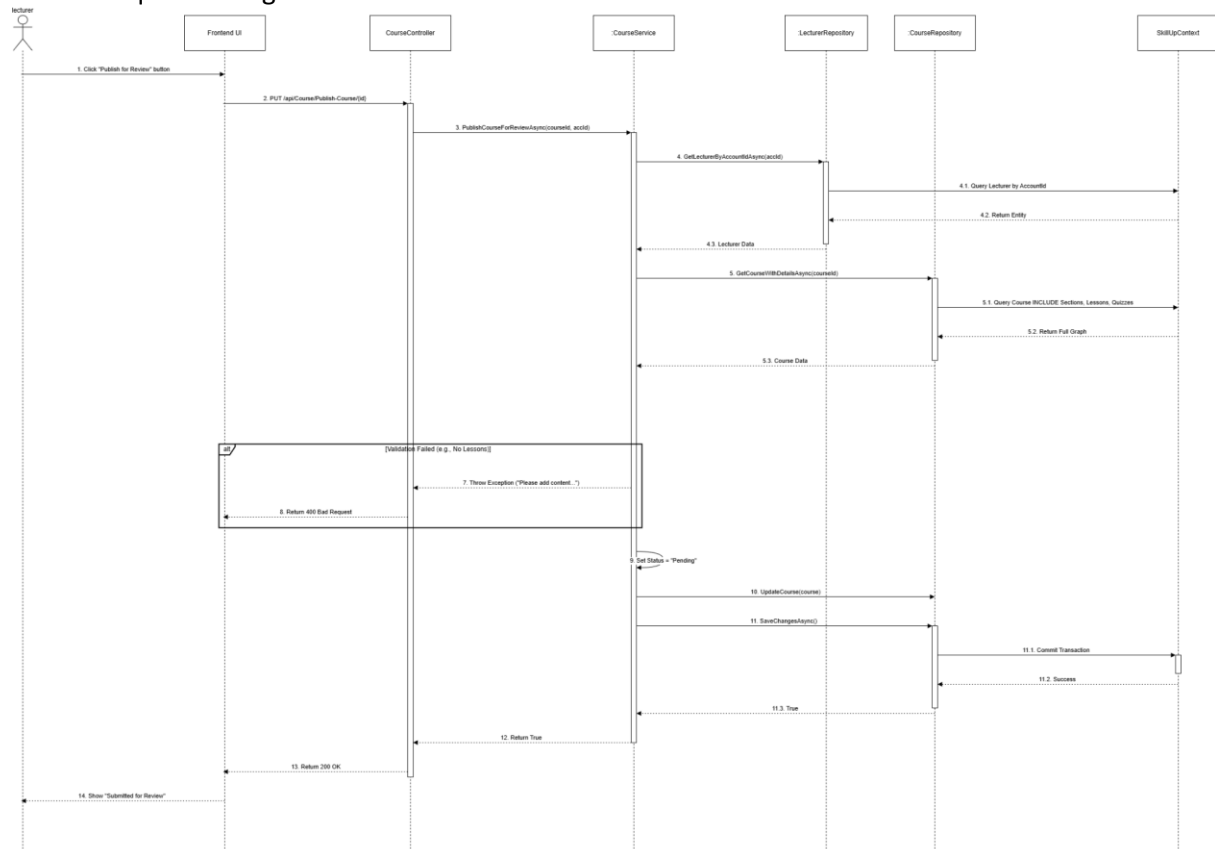
3.8 Course Feature

3.8.1 Publish Course

3.8.1.1 Class Diagram

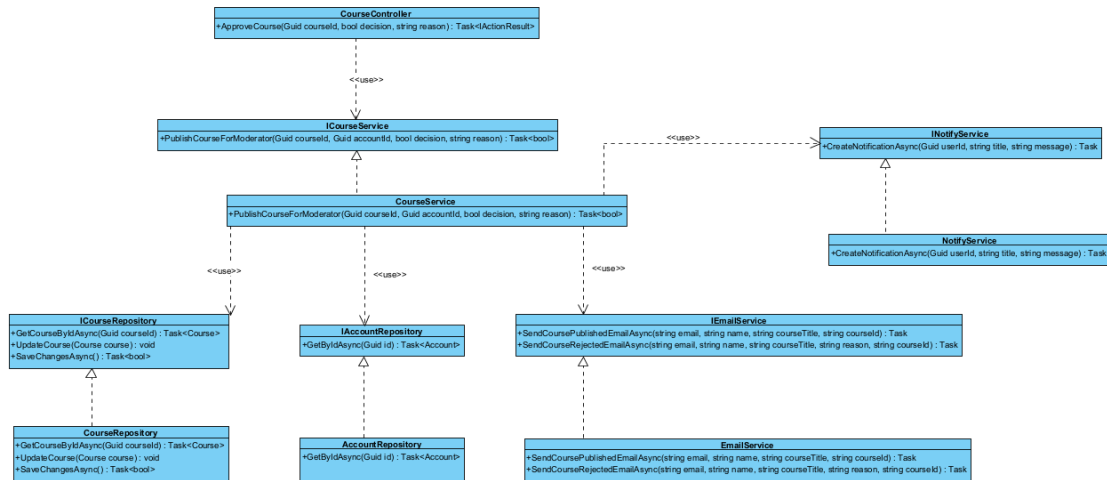


3.8.1.2 Sequence Diagram

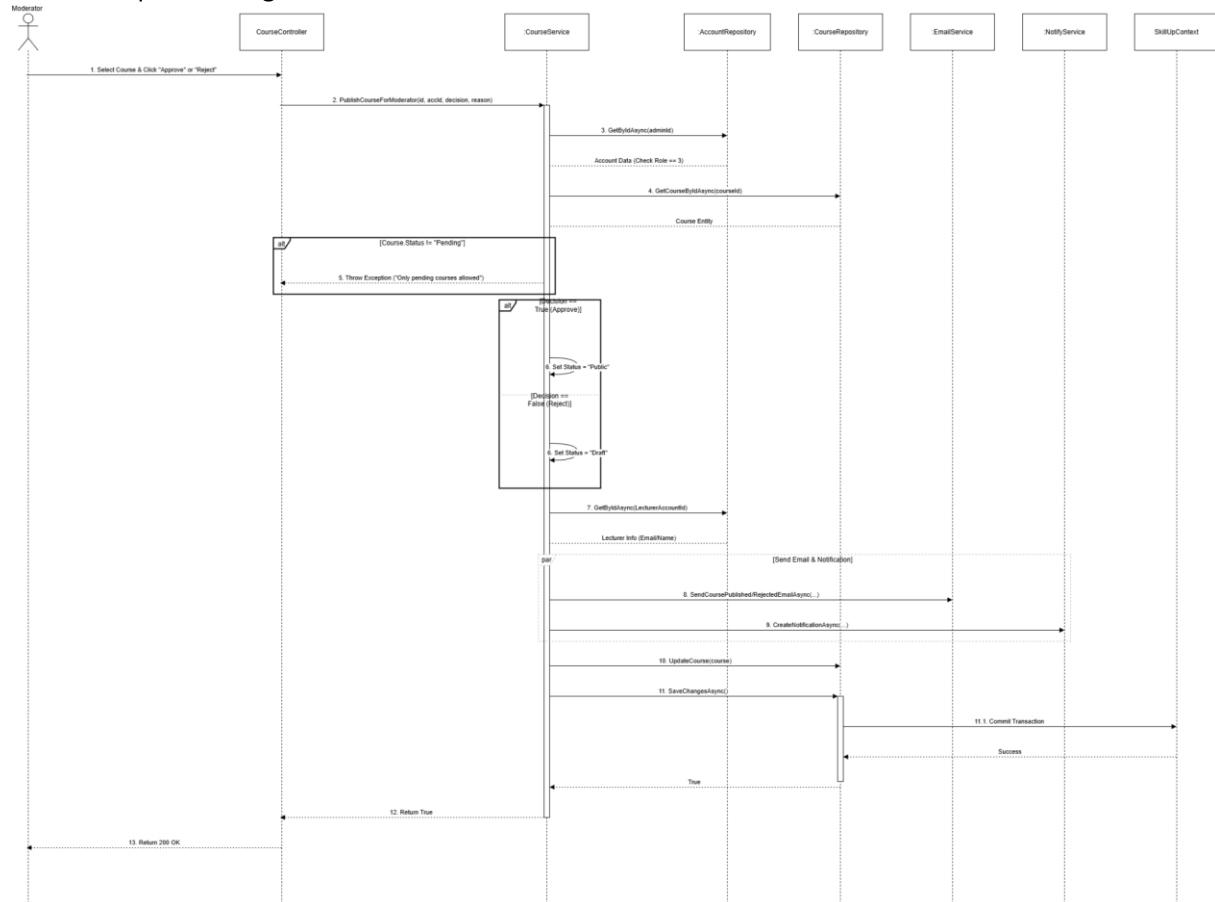


3.8.2 Approve or Reject course

3.8.1.1 Class Diagram

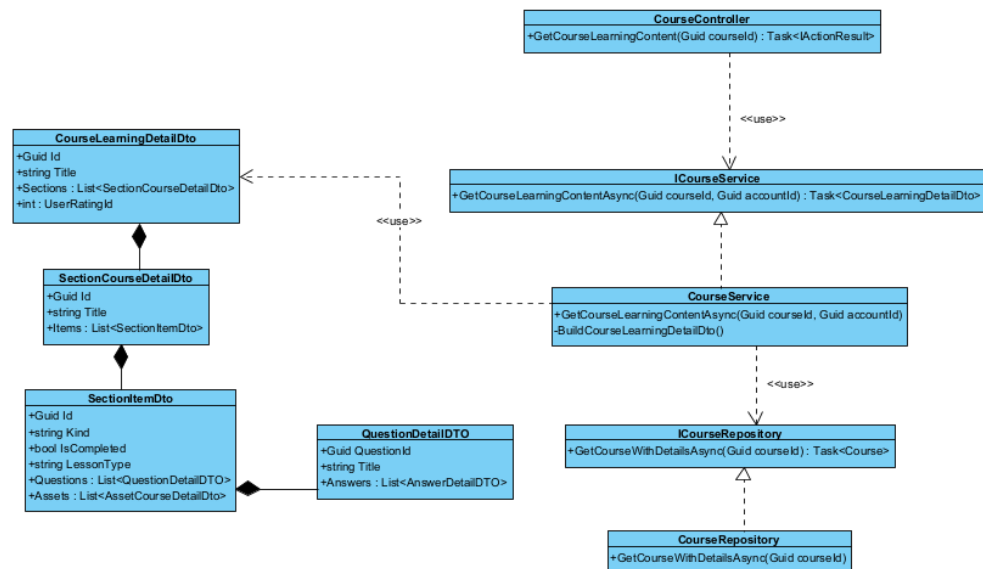


3.8.1.2 Sequence Diagram

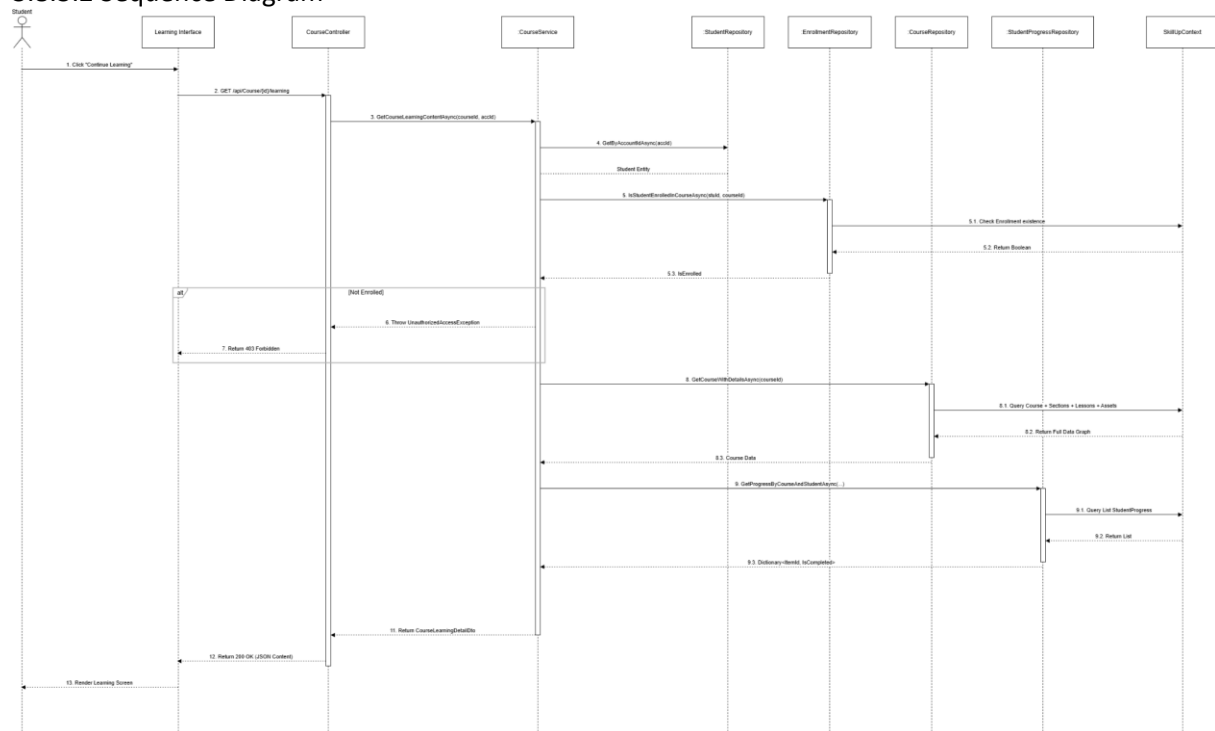


3.8.3 Access Learning Content

3.8.3.1 Class Diagram



3.8.3.2 Sequence Diagram



V. Software Testing Documentation

1. Scope of Testing

1.1 Target of Test (features & functions)

The tests will cover the OLP-SkillUp features below (primary targets):

- **Course Management:** Course creation, editing, publishing (draft/published states), curriculum organization with sections and lessons, course categories and subcategories, course enrollment and progress tracking.
- **Assessment & Quizzes:** Quiz creation with multiple question types, timer and passing percentage configuration, quiz attempts and scoring, student progress tracking.
- **Student Progress & Enrollment:** Course enrollment, lesson completion tracking, progress percentage calculation, learning history and recently accessed courses.
- **Rating & Reviews:** Course ratings (star-based), written reviews and feedback, rating aggregation and average calculation, student-only rating permissions.
- **Lecturer Application System:** CV submission for becoming a lecturer, application status tracking (Pending/Approved/Rejected), admin review and approval workflow, lecturer profile with bio and expertise.
- **Payments:** Shopping cart functionality, course pricing (free/paid), voucher/coupon system with discount codes, transaction processing, payment method integration, order history.
- **User Management:** Registration, login, profile management, role-based access (Student, Lecturer, Content Moderator), password reset with OTP, token-based authentication (JWT + Refresh Token).

1.2 Non-functional requirements

The following non-functional areas will be validated at a basic/operational level:

- **Compatibility:** RESTful API architecture for cross-platform support, CORS configuration for frontend-backend communication, support for multiple browsers (modern browsers via Vite/React), mobile-responsive design with Tailwind CSS.
- **Security (basic):** JWT-based authentication with token expiration, role-based authorization (Student, Lecturer, Content Moderator, Admin), refresh token mechanism for session management, OTP verification for password reset, input validation and sanitization, secure file upload restrictions.
- **Reliability:** correct error handling and stable recovery for common failure cases.
- **Documentation:** correctness and completeness of user guides and API docs.

2. Test Strategy

2.1 Testing Types

No	Type of Tests	Description
1	Unit testing	Executed during development iterations to verify the smallest units within the source code. Focus on core features (e.g., authentication logic, enrollment rules, voucher validation, progress calculation), external dependencies are mocked.
2	Integration testing	Executed during development iterations to validate interactions between modules and with external services. Covers data exchange, API communication, payment gateway integration, file uploads, and error handling in the system.
3	System testing	System testing is performed in the final integration phase and validates the behavior of the complete, fully integrated SkillUp platform across all user roles and business workflows.

2.2 Test Levels

Type of Tests	Test Level		
	Unit	Integration	System
Function Testing	X		X
Integration Testing		X	X
Performance Testing		X	X
API Testing	X	X	

2.3 Supporting Tools

Purpose	Tool	Vendor/In-house	Version
Test Result	Google sheets(excel)	Google	Latest
Front-end test	Chrome Developer/ Microsoft Edge	Google/Microsoft	Latest
Manage Bugs	Google sheets(excel)	Google	Latest

3. Test Plan

3.1 Human Resources

Worker/Doer	Role	Specific Responsibilities/Comments
QuatBT, HaiTT, BinhTB, TuanCM, HienVD	Team member	- Create a test plan. - Create test cases and perform unit tests. - Create test cases and perform integration tests. - Create test cases and perform system tests. - Create test cases, checklists and perform acceptance tests. - Summaries test reports.
TuanCM	Team member	- Fix bugs front-end
QuatBT, HaiTT, BinhTB, HienVD	Team member	- Fix bugs back-end

3.2 Test Environment

Purpose	Tool	Provider	Version
Test API	Postman	Postman.Inc	Lastest
Test UT	NUnit	.NET Foundation	3.14.0
Test IT	Chrome Developer/ Microsoft Edge	Google/Microsoft	Lastest
Test ST	Chrome Developer/ Microsoft Edge	Google/Microsoft	Latest

3.3 Test Milestones

Milestone Task	Start Date	End Date
Test Planning	21/9/2025	12/12/2025
Unit Testing Execution	3/11/2025	12/12/2025
Integration Testing Execution	17/11/2025	12/12/2025
System Testing Execution	7/12/2025	12/12/2025
API Testing Execution	2/10/2025	12/12/2025
Test Report Preparation	11/12/2025	12/12/2025
Bug Fixing and Retesting	6/11/2025	12/12/2025

4. Test Cases

4.1 Unit Test

The unit testing phase focuses on verifying the core methods mentioned above.

For more full detail test cases: [Report5.1 Unit Test.xlsx](#)

4.2 Integration Test

Integration testing is performed by combining individual modules and examining them as a complete group. This phase aims to ensure that the interconnected components interact properly and that the system as a whole operates according to expectations.

For more full detail test cases: [Report5.2 Integration Test.xlsx](#)

4.3 System Test

- This testing approach focuses on observing and assessing how a complete system operates. It integrates various management processes to achieve the outcomes users need.
- Test cases serve as blueprints for system testing. They specify the actions performed within the system, the data provided as input, and the anticipated results at each stage of the test.

For more full detail test cases: [Report5.3 System Test.xlsx](#)

5. Test Reports

5.1 Unit test report

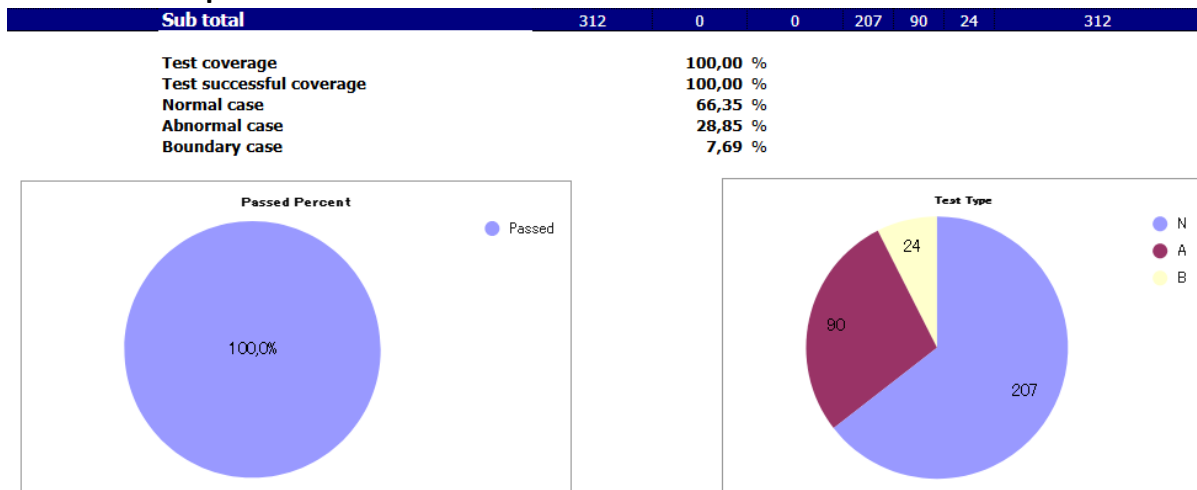


Figure 5.1: Unit test report

5.2 Integration test report

TEST STATISTICS

Project Name	Online Learning Platform – SkillUp	Creator		TuanCM
Project Code	OLP-SkillUp	Reviewer/Approver		
Document Code	OLP-SkillUp_System_Test_Report	Issue Date		3/11/2025
Notes				

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Register	3	0	0	0	3
2	Login by account	3	0	0	0	3
3	Login by google	1	0	0	0	1
4	Forgot password	2	0	0	0	2
5	Logout	3	0	0	0	3
6	View profile	1	0	0	0	1
7	Update profile	2	0	0	0	2
8	Change password	4	0	0	0	4
9	View purchase history	2	0	0	0	2
10	View cart	2	0	0	0	2
11	Add to cart	3	0	0	0	3
12	Remove from cart	2	0	0	0	2
13	Add course	2	0	0	0	2
14	Hide course	1	0	0	0	1
15	Update course - general	9	0	0	0	9
16	View section	2	0	0	0	2
17	Add section	4	0	0	0	4
18	Edit section	3	0	0	0	3
19	Delete section	1	0	0	0	1
20	Create lesson	6	0	0	0	6
21	Create question	4	0	0	0	4
22	Add from bank	2	0	0	0	2
23	View question	2	0	0	0	2
24	Edit question	3	0	0	0	3
25	Delete question	2	0	0	0	2
26	Enroll course	3	0	0	0	3
27	Learning course	3	0	0	0	3
28	Progress tracking	1	0	0	0	1
29	Checkout	5	0	0	0	5
30	Apply voucher	5	0	0	0	5
31	Take quiz	6	0	0	0	6
32	View result	6	0	0	0	6
33	Rating	2	0	0	0	2
34	Email notification	4	0	0	0	4
Sub total		104	0	0	0	104

Test coverage 100,00 %
Test successful coverage 100,00 %

Figure 5.2: Integration test report

5.3 System test report

TEST STATISTICS

Project Name	Online Learning Platform – SkillUp	Creator		TuanCM
Project Code	OLP-SkillUp	Reviewer/Approver		
Document Code	OLP-SkillUp_System_Test_Report	Issue Date		7/12/2025
Notes				

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Authentication	15	0	0	0	15
2	Workflow of System Moderator	15	0	0	0	15
3	Workflow of Content Moderator	32	0	0	0	32
4	Workflow of Lecturer	63	0	0	0	63
5	Workflow of Student	33	0	0	0	33
6	Workflow of Admin	16	0	0	0	16
7	AI assistant	5	0	0	0	5
Sub total		174	0	0	0	174

Test coverage 100,00 %
Test successful coverage 100,00 %

Figure 5.3: System test report

VI. Release Package & User Guides

2. Installation Guides

2.1 System Requirements

- .NET 8 SDK & Runtime (for backend API).
- ReactJS 19+(for frontend).
- Python 3.11.9 (for backend API)
- SQL Server 2019+ (database).
- Qdrant (vector database)

3. User Manual

3.1 Overview

The SkillUp platform revolutionizes online education by connecting students seeking knowledge with expert instructors in a seamless, automated ecosystem.

After account creation, students immediately browse courses by category, preview free lessons to evaluate quality, purchase through secure payment gateways, and gain instant access to structured learning materials including videos, text content, quizzes, and downloadable resources. Progress tracking is fully automated—completion percentages, quiz scores, and learning streaks provide clear motivation without manual intervention.

Lecturers submit applications with CVs for approval, then access powerful course creation tools that automatically handle video, content formatting, quiz building, and payment processing. They create promotional vouchers, monitor student analytics, respond to questions, and track revenue from a centralized dashboard—eliminating the need for separate hosting, payment systems, or marketing infrastructure.

Content moderators review lecturer applications for credibility, approve courses before publication to ensure quality standards, moderate community forum posts to prevent inappropriate content, and resolve support tickets efficiently.

This guide focuses on core workflows that drive platform success, organized by user role and critical tasks to ensure efficient operation.